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Flo-Therm™ G4 Heater Electrical Safety

Functional Design Specification / Software Design Specification

Line 4 Bursom Rd - UK

29th April 2014

Version 1.0

Project No.5620

Notices

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Revision History

Date	Section	Description	Reference
29th April 2014	Whole Document	First Draft	V1.0

Items to complete / to be checked

Item	Section	Description
P&ID	2.1.2	Update P&ID to finalised document.
Ethernet IP Addresses	2.2.4	Update Ethernet IP Addresses once assigned.
SCADA Screens	3.1	Update screen image with finalised 'Runtime' image.
I/O Architecture	4.2.2	Validate I/O architecture with final hardware design.
Add-On Instructions	4.3.2.1 Appendix A	Confirm finalised Add-On Instructions
User-Defined Tags	4.3.2.2 Appendix A	Confirm finalised User-Defined Tags
Routines	4.3.4	Update Routine lists as per finalised program
Risk assessment & System Design	4.3.5.2	Add supplementary information
Define Module and Safety Zone (Tags)	4.3.5.3	Add supplementary information
Configure the Guard I/O Modules	Appendix A	Add finalised module configuration
Safe I/O Test Sheets	Appendix B	Add all finalised Safe I/O Test Sheets
Flo-Mech Sequencing Test Sheet	Appendix B	Add finalised Flo-Mech Sequencing Test Sheet
Flo-Mech Truth Table Test Sheet	Appendix B	Add finalised Flo-Mech Truth Table Test Sheet

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Section 1

Introduction

1. Introduction

This Functional Design Specification/Software Design Specification (FDS/SDS) has been written to facilitate the electrical safety design and operation of the Flo-Therm™ Heater G4, which is to be installed on the existing Line 4 in Bursom Rd.

This document will describe the electrical safety design/functionality of the Flo-Therm™ Heater G4 only. General Heater Control and Optimisation are not covered in any detail in this document.

1.1 Project Information

1.1.1 Location Details

The project is to install a replacement burner and heat exchanger in the current location of Line 4's existing unit to feed an existing PC36 Fryer.

The environmental specification for the factory area is:
- External Average Temp 2 – 20°C
- External Average Humidity 74 – 89%
- Seasonal spikes do occur either way for both.

Power supply 415 VAC 3 Phase 50 Hz

1.1.2 Language

All displayed text on Operator Interface Terminals (OIT's) and equipment nameplates will be in English language only. All project documents will be in English.

1.2 Related Documents

All systems supplied will comply with the following Frito-Lay specifications:

- Electrical Engineering Specifications March 2012
- L4 Bursom Burner User Requirement Specification V3.0

1.3 Terminology Used

Throughout this Functional/Software Design Specification, a number of terms will be used.

Term	Description
OIT	Operator Interface Terminals The OIT's are the main control points for the line and are located adjacent to the fryer and the slicers. These contain computers running a SCADA application complete with touch screen monitor.
PLC	Programmable Logic Controller The PLC monitors the process variables and influences the control variables on the system.
I/O	Input and output cards fitted to the PLC, which gather and transmit systems signals.
SCADA	Supervisory Control And Data Acquisition
Wonderware Intouch	The SCADA package utilised in this system
Ethernet	The network used for TCP/IP communications between PLC's, OIT's and peripheral devices (such as printers and modems).
TCP/IP	Transmission Control Protocol/Internet Protocol An Internet protocol that provides for the reliable delivery of data streams from one host to another.
IP Address	Internet Protocol (Address). Every machine on a TCP/IP Ethernet network has a unique identifying number. A typical IP address looks like this: 192.168.1.29
CIP	Common Industrial Protocol An industrial communication protocol used by Logix 5000-based automation systems on Ethernet, ControlNet, and DeviceNet communication networks.
CIP Safety	Common Industrial Protocol – Safety Protocol SIL 3 rated version of CIP.
1oo2	1 Out Of 2 Generic term used in Safety Systems – typically indicates a dual channel architecture where each channel can execute the safety function by itself.

Table 1-1 Glossary of Terms

1.4 Tagging Convention

The tagging convention consists of 5 Components.

Tag Component	Description
Component 1	Line Numbers (L4 – Bursom Line 4)
Component 2	Letters relating to the device type *(up to three letters)*
Component 3	Production Area Code *(two digits)*
Component 4	Device number *(two digits)*
Component 5	I/O suffix *(1-3 letters, appears on I/O lists only)*

The example shown below is a tag for Valve L4V3501…

L4	V	35	01
LINE NUMBER	DEVICE IDENTIFIER	AREA IDENTIFIER	DEVICE NUMBER

Standard Device Identifiers are…

Device Identifier	Description	Device Identifier	Description
M	Motor	**ES**	Emergency Stop Actuator
VC	Vibrator Controller	**ZS**	Zone Stop Actuator / Switch
V	Electromagnetic Valve	**TS**	Thermostat
CV	Control Valve	**H**	Heat Exchanger
MV	Manual Valve	**FS**	Flow Switch
LS	Level Switch	**TT**	Temperature Transmitter
LT	Level Transmitter	**WC**	Weigh Cell
LIS	Limit Switch	**WT**	Weigh Cell Transmitter
FM	Flow Meter	**XL**	Indicator Lamp
FT	Flow Transmitter	**PB**	Pushbutton
PEC	Photoelectric Cell		

Standard Area Identifiers are…

Area Identifier	Description	Area Identifier	Description
15	Potato Infeed	**28**	Sizer / Grader Unit (future use)
16	Potato Washer	**29**	Wet Inspection
17	Destoner	**30**	Starch Recovery
18	Dewooder	**32**	Slicer Feed / Slicing
19	Reception Hopper	**33**	Flume On Belt
20	Peeler Feed	**34**	Potato Slice Speed Washer
21	Peeler 1	**35**	Heat Exchanger
22	Peeler 2	**36**	Oil Circulation & Fryer Drives
23	Peeler 3	**38**	Fryer Take Out Drives
24	Peeler 4	**44**	Optical Sorter
27	Peeler Discharge	**45**	Seasoning Infeed

Standard Device Suffixes are…

Device Suffix	Description
OPN	Valve Open Limit Switch.
CLS	Valve Closed Limit Switch.
PWD	Motor/Vibrator Controller Power Output
K	Motor/Vibrator Controller Drive Run Output
KA	Motor/Vibrator Controller Running Input
QA	Motor/Vibrator Controller Trip Input
ISO	Motor/Vibrator Controller Isolated Input
SP	Motor/Vibrator Controller Speed Output

Section 2

System Overview

2 System Overview

2.1 Process Overview

2.1.1 Process Description

The Flo-Therm™ G4 Heater is made up of two gas streams (combustion air, pollution gas) connected to a fire tube and an oil to gas heat exchanger which transfers heat to the frying oil.
A pre-heat exchanger connected to the exhaust stack is used to pre-heat the combustion air stream.
Other main component includes: (i) Weishaupt WK70 dual fuel burner, (ii) a flame tube, (iii) a pollution fan motor assembly, (iv) a combustion air fan motor assembly and the main heater tube. Instrumentation fitted is use for control, monitoring and maintenance.

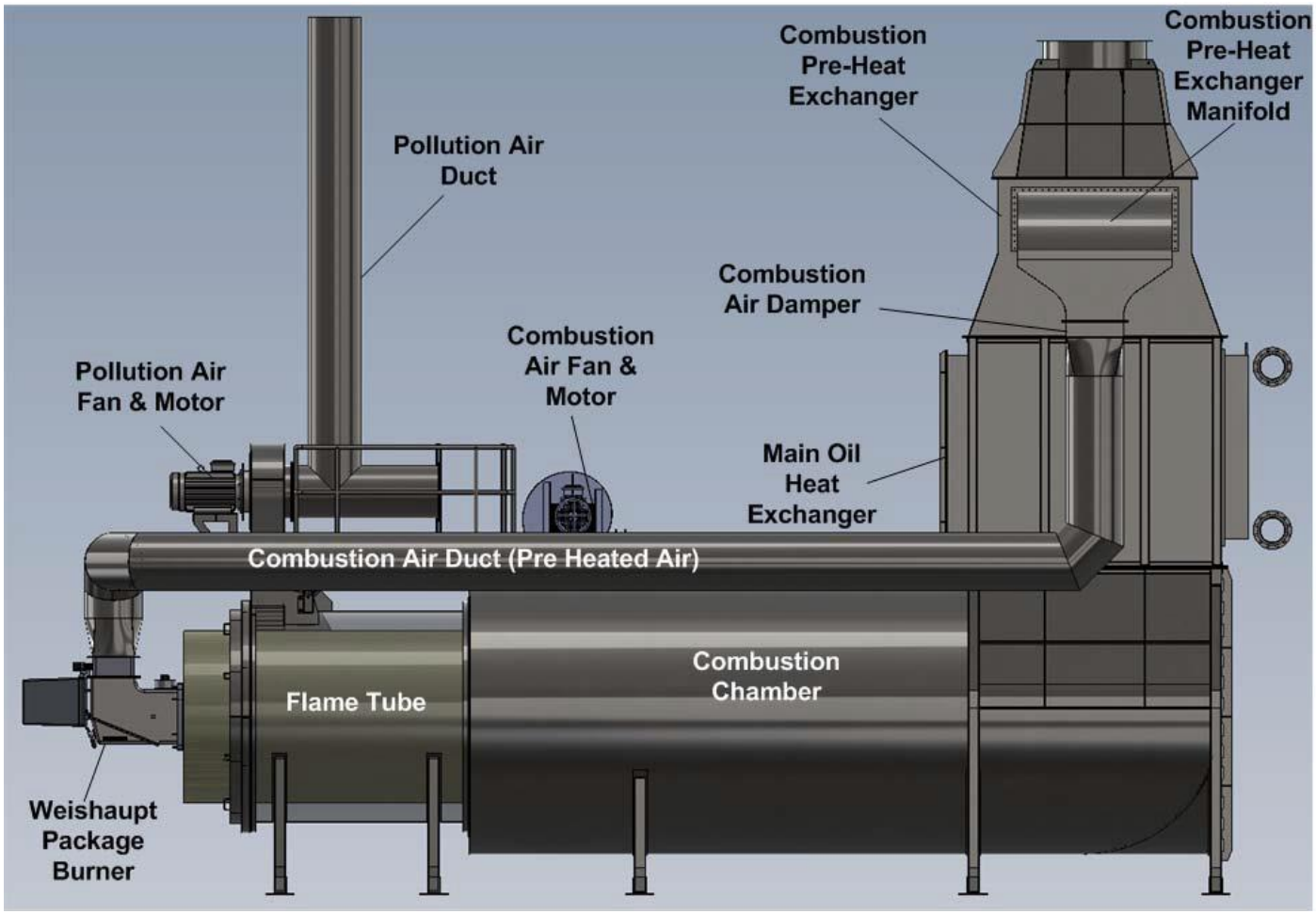


Figure 2.1-1 Flo-Therm™ G4 Heater

2.1.1.1 Combustion Chamber Assembly

The combustion chamber assembly comprises two sections. The burner fires into the flame tube where the flame is established prior to the pollution gases being introduced for incineration. The second section is a holding/mixing vessel (Combustion Chamber) for the gases and forms a support for the heat exchanger.

In the first section there is a flame tube the size of which is matched to the burner characteristics. This tube is lined with a very high quality high temperature insulation material. The flame tube is mounted concentrically in a larger tube (Combustion Chamber), which is lined with a multi-layered insulating materials made to minimise heat losses. All the layers are pinned to welded studs on the inner casing of the heater body. The combustion chamber is of the same construction but reinforced externally to carry the heat exchanger assembly. All machines are supported on and reinforced by a R.H.S. base frame designed to support the machine with suitable ground clearance and loading.

A channel running along the length of the burner allows drainage of excessive dew formation which maintains the insulation properties within spec.

2.1.1.2 Main Oil Heat Exchanger

The hot gas to oil cross flow heat exchanger is designed as a one-piece unit to eliminate external manifolds and to economise on oil content. The heat exchanger comprises a bank of stainless steel tubes in a u-tube configuration. The bottom of the heat exchanger uses a double u-tube configuration to control bulk to skin temperature hence helping to reduce FFA formation. The also minimises the system resistance and load on the main oil pump.
The headers are manufactured from stainless, as are the pass partition plates. The partition plates are fitted with a number of drain holes where the diameter has been maximise to speed up drain down time. The diameter is 12 mm.

Multi-inlet and outlet manifolds are fitted to obtain even flow characteristics in each pass. Locating and sealing grooves in the mild steel doors eliminate oil bypassing. One end of the tube bundle is fixed and fitted with one large sealing door. Each of the u-tubes is able to freely expand during the heating and cooling phases of operation.

2.1.1.3 Combustion Air Pre-Heat Exchanger

The air-to-air heat exchanger is designed to recover some heat from the flue gas and pre-heat the combustion air feeding the Weishaupt WK50. The incoming combustion air is heated in a single pass through the tubes. All of the Pre-Heat exchanger tubes are made of mild steel and fitted with internal turbulators to increase the efficiency of the heat exchanger. Turbulators can be removed from the tubes if the efficiency was to drop significantly or the pressure drop across the Pre-Heat Exchanger was to increase significantly.

The headers of the heat exchangers have been designed specifically to lower the overall pressure drop. The heat exchanger is insulated with ceramic fibre to reduce surface heat losses.

2.1.1.4 Burner

The heater is fitted with a Weishaupt WK50 package burner, with flame detection and burner management WFM equipment. Control of the burner modulation is via servomotors. The burner is bolted on to the flame tube and can be removed without interfering with the flame tube lining.
On commissioning both servo air / gas will be set-up with the WFM controller to give adequate heat input throughout the range of Delta T. In Burner Optimisation mode a secondary air valve is used to control the burner excess air (See later in FDS). Also the servo motor will be set to give optimum control of the air damper and to minimise CO formation. Under no circumstances should any of these settings be changed a serious damage to the equipment could result. The WFM burner controller is pre-programmed with all of the normal start up and shut down sequences for the single fuel burner. The engineering information can be read from the ABE interface, but parameters are locked in service level menus and cannot be adjusted by the user.

2.1.1.5 Instrumentation and Electrics

The control and instrumentation system has been designed to be robust and reliable in what can be a hostile environment, particularly due to temperature and moisture. All control panels are designed to avoid ingress of dirt or moisture, and control systems are designed to check the operation of external sensors.
All instrumentations used for the Burner Optimisation are self-purging to avoid any blockages.

2.1.1.6 Combustion Air Damper

The combustion air control valve damper **CV3501** is positioned after to the combustion fan inlet and after the combustion pre-heat outlet. It is used to control the excess air under Burner Optimisation enable mode. Whilst Burner Optimisation is disabled the damper is fully opened.

2.1.1.7 Pollution Air Fire Damper

The pollution air fire damper **V3503** is positioned in the ducting between the fryer and the Flo-Therm G4. The damper is designed to isolate the heater unit from the fryer in the event of any abnormal condition and prevents migration of heat or fire from one machine to another. The damper is set to spring close in the event of power or air loss and has limit switches to monitor position.

2.1.1.8 Fresh Air Damper

This damper **V3501** is fitted with a spring return actuator designed to spring closed in the event of a power failure. The normal function of the damper is to admit ambient air for cooling purposes at the end of production or during a boil out. The damper is hydraulically actuated allowing compressed air free operation and also allowing actuation under emergency power.

2.1.1.9 Combustion Air Fan

The combustion air fan **M3501** runs continuously whilst in operation, providing the correct air flow for combustion of the gas fuel. Excess combustion air also cools the burner chamber tube and allows cool air to dissipate the heat accumulated in the flame tube once the burner has been shut down. The fan speed is controlled by an inverter with a range of set points to achieve the programmed speed needed for a particular duty point in the burner range.

The fan will continue to run, by means of a pre-set timer situated within the PLC, for approximately two hours after the burner is shut-down. This is known as after-run.

2.1.1.10 Combustion Air Flow

As the combustion air fan is essential to the safe running and efficiency of the Flo-Therm G4, a pressure switch **PS3506** situated in the burner casing monitors continuously the pressure.
Should the fan fail to deliver the correct amount of air into the flame tube, the burner will automatically cease firing and a fault will be generated. The hard wired burner controls check for combustion air-flow as part of the burner start sequence. The switch forms part of the burner interlock chain.

The combustion air mass flow used for the Burner Optimisation is also measured via a flow transmitter **FT3503**, and pressure **PT3503** and temperature **TT3503** for correction purposes. Two solenoid air valves within the flow transmitter deliver compressed air to the flow device to periodically clean the pitot tubes.

2.1.1.11 Pollution Air Fan

This fan **M3502** is used to draw the polluted frying vapours from the fryer for incineration in the heater, and runs continuously whilst the heater is in operation. The pollution fan also cools the burner flame to yield the required 820^{0}C for the heat exchange process. The pollution fan speed is controlled by an inverter.

Like the combustion air fan, the pollution air fan will continue to run, by means of a pre-set timer situated within the electrical panel, for approximately two hours after the burner unit has been stopped.

The fan is fitted with a differential pressure switch to monitor for flow.

2.1.1.12 Pollution Air Flow

As the pollution air fan is essential to the safe running and efficiency of the Flo-Therm G4, a differential pressure **PS3501** device is installed across the inlet/discharge. During the start-up sequence the PLC checks for no-flow prior to start and flow once the fan is running. Should the switch trip during production the burner flame is extinguished and the system raises an alarm and carries out a controlled cool down. Purge valves **V3512** and **V3513** are used to purge the feed pipes with compressed air.

The pollution air mass flow used for the Burner Optimisation is also measured via a flow transmitter **FT3502**, and pressure **PT3504** and temperature **TT3508** for correction purposes. Purge valves within the flow unit are used to ensure self-cleaning of the flow device.
Humidity and mixing ratio (g/kg) is also measured via the humidity device **RH3501**.

2.1.1.13 Flame Sensor

The flame sensor **UV3501** is situated within the burner head and is a Ultra Violet (UV) scanner probe. If flame formation does not take place within a set safety period after ignition, the burner will stop automatically (Lockout) and the fuel supply valve will close. This safety aspect is controlled via the Weishaupt WFM unit. If, for any reason, the

flame sensor cannot detect a flame in operation, the burner will automatically shut-down and enter the cool down mode.

2.1.1.14 Compressed Air Flow

If the compressed air supply falls below 4 bar, the burner shuts down and a fault is generated. If the vent valves on the heat exchanger open due to air pressure failure the main oil pump will be stopped by the hard wired interlocks. Pressure switch **PS3507** monitors the mains pressure.

2.1.1.15 Oil Flow Meter

A flow meter **FT3601** is located in the existing site oil pipe work, with an electrical signal to the control panel, indicating the flow of cooking oil passing through the Flo-Therm G4 system. The oil flow meter is one of the parameter used to calculate the burner efficiency.

Should the amount of oil flowing through the system fall below a pre-set minimum, the burner will automatically cease to fire and a fault will be generated. The flow rate trip point is set at 60% of normal maximum flow. For PC21 the trip is 118 m^3/hr for normal max flow of 193 m^3/hr. The flow input is fed into the PLC. The low oil flow alarm is known as OLFD (oil flow low deviation).

2.1.1.16 Pump Contact

The main oil pump inverter drive is interlocked with the G4 heater panel. The two vent valves **V3505, V2506** and vent line flow switch **FS3501** will prevent the oil pump form running if they are not in the correct conditions (valves closed and flow switch = no flow condition on flow switch). In addition there is also a temperature interlock with the fryer stack to prevent the pump form being stopped at elevated temperatures or being re-started at elevated temperatures. The temperature interlock is known as stack inhibit (STOT).

2.1.1.17 Pollution Vapour Heater Temperature

A duplex thermocouple **TT3509A/B** is located in the pollution fan outlet duct to detect any excess temperature in pollution gases. (E.g. fire in header).

Should the pollution gases rise above a set temperature the burner will automatically cease firing and a fault will be generated. The temperature signals are fed into the PLC and the alarm is known as Pollution Air Over Temperature (PAOT)

This fault is one of the heater fire conditions.

2.1.1.18 Combustion Chamber Temperature

A duplex thermocouple **TT3501A/B** is positioned on the bend section of the Flo-Therm, with the probe of the device positioned directly in the flow of hot gases leading to the heat exchanger.

Should the temperature of the gases passing through the combustion chamber rise excessively, a high limit alarm will automatically cause the burner to cease firing and a fault will be generated. The temperature signals are fed into the PLC and the alarm is known as Combustion Chamber Over Temperature (CCOT).

If Optimisation is enabled then the CCOT = 845DegC. If the temperature increases above 845DegC, then the pollution fan speed will increased to a safe level (820DegC). If CCOT > 870DegC, the burner will stop.

2.1.1.19 Oil Outlet Temperature

A duplex thermocouple **TT3507A/B** is positioned in the outlet manifold of the heat exchanger.
Should the temperature of the oil exceed 205DegC (or 98DegC when water is being used), the burner will automatically cease firing and a fault will be generated.

2.1.1.20 HX Oil Safety Temperature

A duplex thermocouple **TT3511A/B** is positioned in the outlet manifold of the heat exchanger.
Should the temperature of the oil exceed 205DegC (or 98DegC when water is being used), the burner will automatically cease firing and a fault will be generated.

2.1.1.21 Stack Temperature

A duplex thermocouple **TT3502A/B** is positioned on the flue gas discharge section of the Flo-Therm G4, with the probe section protruding directly into the internal flow of exhausted gases flowing from main heat exchanger (i.e. between the main and pre-heat exchangers). Should the temperature rise above the safety limit, the burner will automatically cease to fire and a fault will be generated. The temperature signals are fed into the PLC and the alarm is known as Stack Temperature Over Temperature (STOT).

2.1.1.22 Pump Inhibit

If the stack temperature between main and pre-heat exchangers **TT3502A/B** is above 130˚C and the system is in operation (i.e. production or boil out) the main oil pump cannot be stopped by the user. Secondly, if the main oil pump has stopped by any method and the stack temp **TT3502A/B** is above 130˚C the pump cannot be restarted. This is to prevent thermal shock or oil overheating and is known as stack inhibit. When the temperature drops below the 130°C, the pump can be restarted.

2.1.1.23 Stack Vent Damper

A fail safe open damper, **V3502,** is fitted which will allow the fryer hood to be vented directly to atmosphere. The damper is closed for frying, open for boil out. In the event of a power failure the damper opens to vent the fryer. Should there be a fire the damper is closed directly by the fire suppression system change over relays. The damper is electro-hydraulically actuated requiring a 6 amp 24VDC supply. In the event of a power failure, followed by a fire, the damper is also directly closed by the fire suppression system providing that the fire suppression system is healthy (battery backup).

2.1.1.24 Heat Exchanger Vent Valves

The heat exchanger has a pair of venting valves (**V3505**, **V3506**) with limit switches. The main oil pump inverter has a hard wired logic relay system to prevent the inverter starting the main oil pump if the vent valves are not closed. In addition the valve limits will stop the main oil pump if the closed limits are not made in production. The valves are cycled at start up, a successful cycling of the valves will authorise a main oil pump start up. The condition of the valve seals is monitored by a flow switch LIS3502, which will also stop the main oil pump if flow is detected.

2.1.1.25 Flame Tube Drain Valve

Two drain valves **V3504** and **V3518** are connected to the flame tube and energized on a timely basis (out of sync) to allow any oil or water residue to be drained and minimise oil deposit build up.

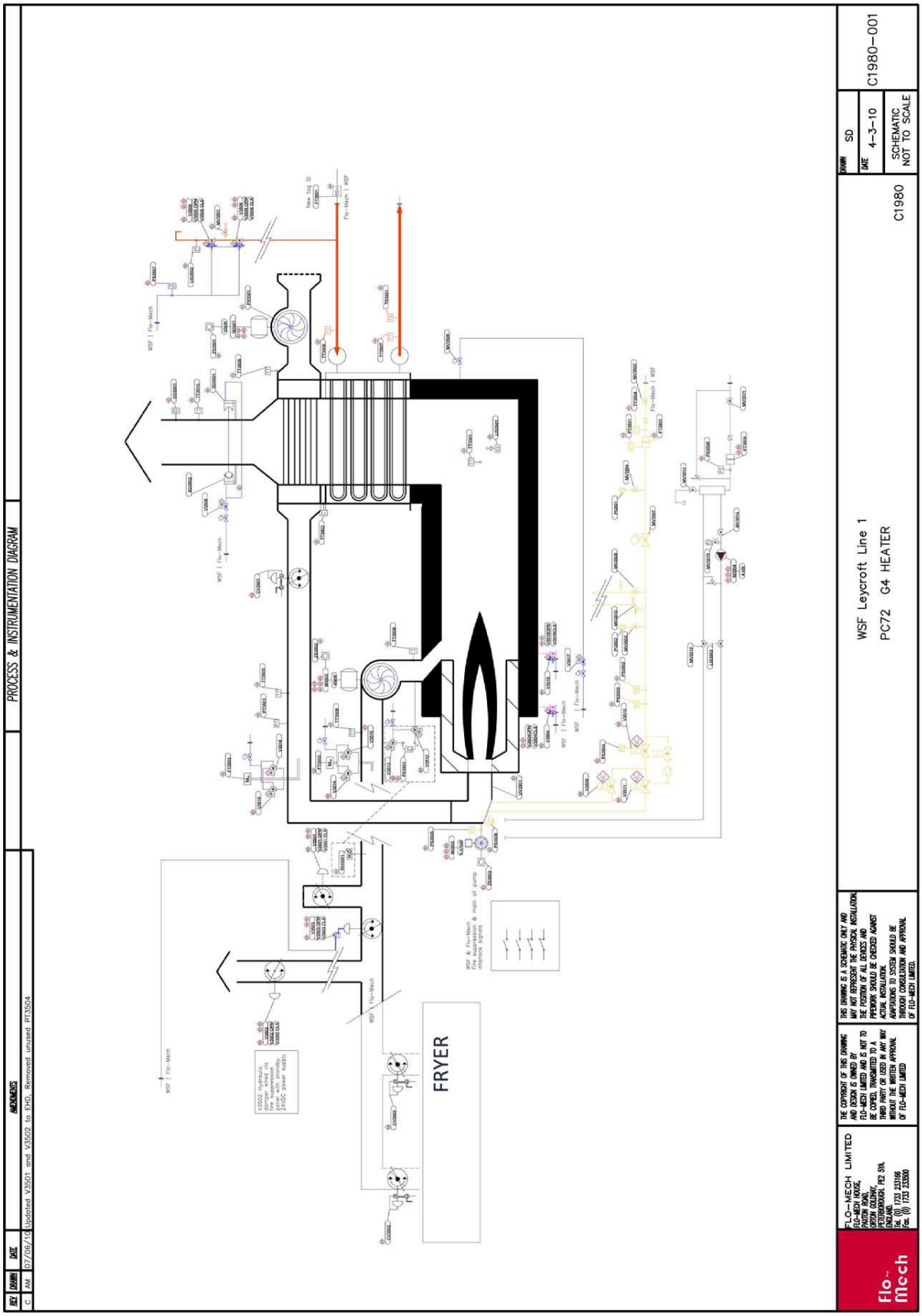
2.1.1.26 Smoke Detector

A Smoke transmitter (**SD3501**) and detector (**SD3502**) is fitted to identify any abnormal combustion conditions (i.e. a fire in the unit due to heat exchanger tube leak or pollution debris). If the alarm condition (80%) is reached the burner will be turned off and the oil circulation pump will stop. The smoke detector has a 30 second delay before alarm after smoke has been detected. The smoke detector lenses are kept clean by an air wash system, fed by a solenoid valve **V3508**. The solenoid valve is energised whenever the main e-stop circuit is healthy.

2.1.1.27 Gas and Oil Train Valve

The Weishaupt single fuel burner WK50 has been fitted with a gas train. Pressure switches, non-return valves, vent valves, motorised valve, governor have been fitted to ensure adequate regulation and control of the gas flow to the burner. The gas flow rate is measured with a flow transmitter **FT3501**. The gas train has also been fitted with a pressure and temperature probe to be used in the calculation of the burner efficiency and to correct the gas flow rate to calibration levels of 1013 mbar and 0°C.

2.1.2 P&ID


PROCESS & INSTRUMENTATION DIAGRAM
AMENDMENTS
REV DRAWN DATE
C AM 07/06/10 Updated V3501 and V3502 to EHD. Removed unused PT3504
FRYER
WSF Leycroft Line 1
PC72 G4 HEATER
C1980
DRAWN SD
DATE 4-3-10
SCHEMATIC NOT TO SCALE
C1980-001
FLO-MECH LIMITED
FLO-MECH HOUSE,
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2.2 Control System Overview

2.2.1 Panel Overview

A dedicated heater control panel will be supplied as part of the Flo-Therm™ G4 Heater installation.

The PLC for the control system is to be an Allen Bradley Compact GuardLogix® (1768-L43S).

Allen Bradley Powerflex 753 Ethernet drives with 'Safe Torque Off' will be utilised.

Local control and data acquisition will be via a panel mounted Allen Bradley PVP400.

2.2.2 PLC Overiew

The Flo-Therm™ G4 Heater control system will encompass an Allen Bradley Compact GuardLogix® 1768-L43S PLC.

Figure 2.2-1 Example Allen Bradley Compact GuardLogix Controller

The Compact GuardLogix® controller is a full-function Logix processor that also provides safety control.
The GuardLogix system is a dual processor solution that uses a primary controller and a safety partner to achieve SIL 3, PLe. A major benefit of this system is that it is a single project, with safety and standard control together.
The GuardLogix system is a 1oo2 safety architecture that provides unmatched safety and ease of use.

Just like other Logix processors during development, safety and standard have the same rules. Online editing, forcing, even multiple users are all allowed. Once the project is tested and ready for final validation, you set the Safety Task to a SIL 3 integrity level, which is then enforced by the GuardLogix controller. When safety memory is locked and protected, the safety logic cannot be modified.
On the standard side of the GuardLogix controller, all functions operate like a regular Logix controller, motion, drive, sequential, even process. Thus, online editing, forcing, and other activities are all allowed, while safety is securely isolated.

With this level of integration, safety memory can be read by standard logic and external devices, such as HMIs or other controllers, eliminating the need to condition safety memory for use elsewhere.

I/O modules used for the Flo-Therm™ G4 Heater control system will separated into two categories: -

- ***Safe I/O*** will utilise POINT Guard I/O™.
  These modules communicate by using CIP Safety protocol over EtherNet/IP.

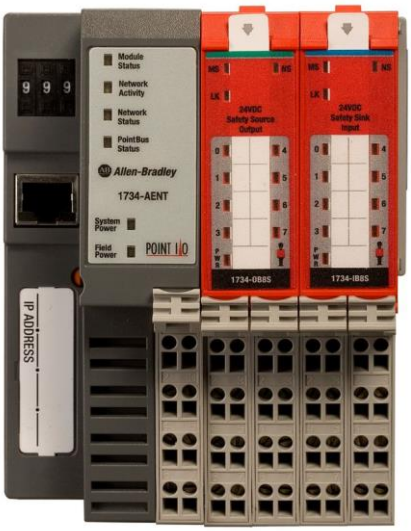


Figure 2.2-2 Example POINT Guard I/O™.

- ***Non Safe I/O*** will utilise standard 1769 Compact I/O modules.

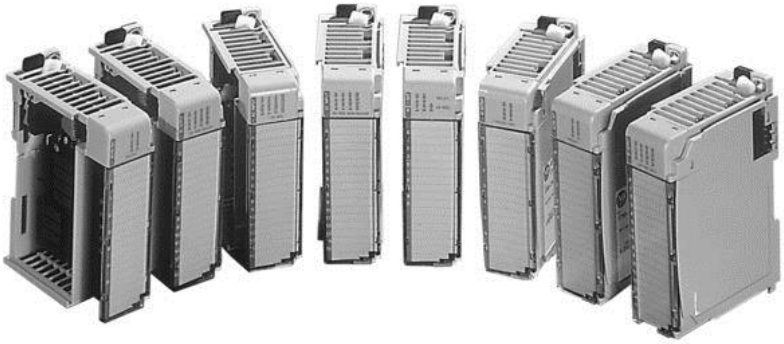


Figure 2.2-3 Example 1769 Compact I/O Modules

PLC logic generated associated with all the Safe I/O devices will be contained within the dedicated Safety Task of the PLC, all other Non-Safe I/O devices will have their associated logic contained within the standard Main Task of the PLC.

2.2.3 Operator Interface Terminals

The dedicated heater control panel will consist of a local Allen Bradley PVP 400 OIT. This will be used for local heater control and data acquisition only.

Figure 2.2-4 PanelVIew Plus 400 HMI

Existing OIT's are distributed around the factory; these will remain as is.

Where necessary existing L4 Heater SCADA screens will be amended.
Additionally newly developed SCADA screens will be added to the current existing factory SCADA application.

2.2.4 Ethernet Networks

Two Ethernet networks are associated with the Flo-Therm™ G4 Heater system.

2.2.4.1 Factory Ethernet Network

The Flo-Therm™ G4 Heater control system (Allen Bradley Compact GuardLogix PLC) will be connected to the existing corporate Ethernet network. This will then allow communication to remote PLC's (e.g. Fryer), SCADA I/O Data Servers, Historian Databases, etc.

Device	Ethernet IP Address
Line 4 G4 Heater PLC (Ethernet Module #1)	XXX.XXX.XXX.XXX

2.2.4.2 I/O Ethernet Network

The Flo-Therm™ G4 Heater control panel will utilise an internal 'standalone' Ethernet network; this will not be connected to the corporate Ethernet network.

This internal network will handle necessary I/O communications between the Flo-Therm™ G4 Heater control system (Allen Bradley Compact GuardLogix PLC), the Point Guard IO modules, the internal Allen Bradley Powerflex drives as well as the local Allen Bradley PVP400 OIT.

Device	Ethernet IP Address
Line 4 G4 Heater PLC (Ethernet Module #2)	192.168.1.XXX
Line 4 G4 Heater PVP400 OIT	192.168.1.XXX
POINT I/O Ethernet/IP adaptor	192.168.1.XXX
M3501 Combustion Fan PowerFlex 525	192.168.1.XXX
M3502 Pollution Fan PowerFlex 525	192.168.1.XXX

Section 3

System Functions And Facilities

3 System Functions And Facilities

The Flo-Therm™ G4 Heater is made up of two gas streams (combustion air, pollution gas) connected to a fire tube and an oil to gas heat exchanger which transfers heat to the frying oil.
A pre-heat exchanger connected to the exhaust stack is used to pre-heat the combustion air stream.
Other main component includes: (i) Weishaupt WK70 dual fuel burner, (ii) a flame tube, (iii) a pollution fan motor assembly, (iv) a combustion air fan motor assembly and the main heater tube. Instrumentation fitted is use for control, monitoring and maintenance.

3.1 Operation Area's

The system comprises of one discrete unit of Operation.

Unit Operation	Area Description
Heater System	Burner/Heat Exchanger which transfers heat to the frying oil.

Table 3-1 Unit Operations

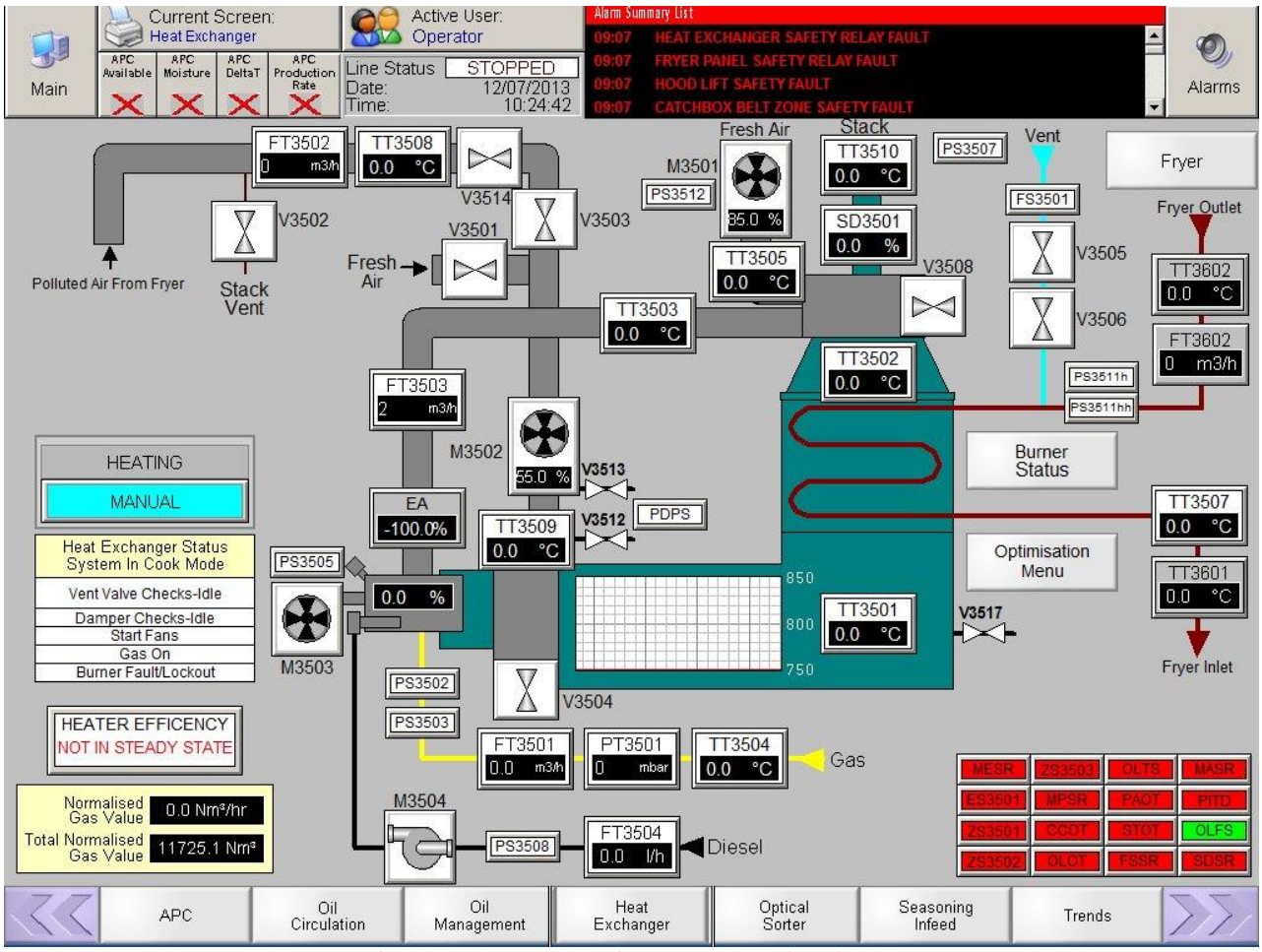


Figure 3.1-1 Heat Exchanger Operational Area

3.2 Heater Control

All 'Automatic' requests to operate the Heater will come from the Fryer and Oil Management Operation Units. These requests are processed in the Fryer PLC and sent over to the Heater PLC via an Ethernet message.
Please refer to the specific FDS for further details on the operation of these units.

Automatic Requests that are sent to the Heater PLC from the Fryer PLC (via the Fryer and Oil Management Operation Units) are as follows: -

3.2.1 Boil Out Mode

The Boil Out Mode request is sent to the Heater PLC from the Fryer PLC.
The Boil Out Mode request is only then executed by the Heater PLC if the local Cook/Clean Mode Castell Switch (SS3703) is in the 'Clean' position.
On the Boil Out Mode request being executed by the Heater PLC the Heater System will operate in a 'Boil Out' mode of operation.

3.2.2 Preheat Mode

The Preheat Mode request is sent to the Heater PLC from the Fryer PLC.
The Preheat Mode request is only then executed by the Heater PLC if the local Cook/Clean Mode Castell Switch (SS3703) is in the 'Cook' position and the Water Pipe is in a safe position.
On the Preheat Mode request being executed by the Heater PLC the Heater System will operate in a 'Preheat' mode of operation.

3.2.3 Fry Mode

The Fry Mode request is sent to the Heater PLC from the Fryer PLC.
The Fry Mode request is only then executed by the Heater PLC if the local Cook/Clean Mode Castell Switch (SS3703) is in the 'Cook' position and the Water Pipe is in a safe position.
On the Fry Mode request being executed by the Heater PLC the Heater System will operate in a 'Fry' mode of operation.

3.2.4 Main Oil Pump Run (Vent Valve Check)

When the Fryer Mail Oil Pump (M3604) is requested to run a request to check the Vent Valves is sent to Heater PLC from the Fryer PLC.
On the Vent Valve request being executed by the Heater PLC the Oil Vent Valves (V3505 & V3506) are checked for open and close functionality. Only when the Vent Valves are proved to be fully operational will an interlock be sent back to the Fryer PLC to let the Fryer Mail Oil Pump (M3604) to be started.

Note, the Main Oil Pump (M3604) is inhibited from starting until the Stack Temperature falls below 130°C. This in turn will also hold off the Vent Valve check until the Stack Temperature also falls below 130°C.

Additionally the Main Oil Pump (M3604) will be generally inhibited from stopping until the Stack Temperature falls below 130°C.

3.2.5 Burner Run

When the Fryer requires the oil (or water) to be heated a Burner Run request is sent to the Heater PLC from the Fryer PLC.
On the Burner Run request being executed by the Heater PLC a number of test are carried out: -

1. The Fresh Air Damper (V3501), Stack Damper (V3502), Fire Damper (V3503) and the CIP Damper Valve (V3514) will be checked for open and close functionality.
2. Once the Damper test is passed the Combustion Air Fan (M3501), Pollution Fan (M3502) and the Flame Sensing Cooling Fan (M3503) are requested to start.
3. On the Combustion Air Fan (M3501), Pollution Fan (M3502) and the Flame Sensing Cooling Fan (M3503) all running (and all other associated Burner faults are clear) a 'Burner Run' request to the Weishaupt Burner controller will be made.
4. The Weishaupt Burner controller will then go through a number of additional tests, and assuming OK will light the burner. This is indicated back to the Heater PLC by the 'Gas On' signal.

Temperature control of the oil (or water) is handled in the Fryer PLC by controlling the flow of gas to the burner into the heat exchanger. This analogue signal is sent directly from the Fryer PLC to the Weishaupt Burner controller.

In general the burner will operate in one of three modes, Boil Out, Preheat or Fry.
In Boil Out and Preheat modes, the analogue signal sent to the Weishaupt Burner controller to control the gas flow is limited (generally set @ 50%).

In all three modes of operation the Combustion Air Fan (M3501), Pollution Fan (M3502) and the Flame Sensing Cooling Fan (M3503) will run.
The position of the Air Dampers and the Vent Valves will be dependent on the mode of operation and additionally the fault status of the Heater System.

The following truth table details the required states of the Heater System devices: -

Fault	Device	ID	Alarm	Main Oil Pump Shutdown	Burner Shutdown	Combustion Air Fan Running	Pollution Fan Running	Burner Cooling Fan Running	Fuel Oil Pump (Burner) Running	Fresh Air Damper	Fire Damper	Stack Vent Damper	CIP Damper	Vent Valve Position
						M3501	M3502	M3503	M3504	V3501	V3503	V3502	V3504	[illegible]
BURNER ALARMS														
High gas pressure	TBC	HGP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Low gas pressure	TBC	LGP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Valve proving (VFM Only)	TBC	N/A	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Combustion air pressure (VFM Only)	TBC	N/A	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Cooling Fan Pressure Switch	TBC	FSP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Burner Lockout	TBC	BFR	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
OPERATIONAL ALARMS														
Complete Power Loss (Incl Gen.	TBC		NA	Y	Y	N	N	N	N	Closed	Closed	Open	Open	Open
Combustion Chamber Temp (>870°C)	TBC	CCO	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Pollution Air Pressure	TBC	PDP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Combustion Air Fan Trip	TBC		Y	N	Y	N	Y	Y	N	Open	Closed	Open	Open	Closed
Pollution Fan Trip	TBC		Y	N	Y	Y	N	Y	N	Closed	Closed	Open	Open	Closed
Burner Cooling Fan Trip	TBC		Y	N	Y	Y	Y	N	N	Open	Closed	Open	Open	Closed
Fuel Oil Pump Trip	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Pollution Header Temperature	TBC	PAO	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Stack Temperature	TBC	STO	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Oil Flow (Vent Line)	TBC	VLFS	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Vent Valve 1 Failure	TBC	VV1	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Vent Valve 2 Failure	TBC	VV2	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Low Oil /Water flow	TBC	OLF	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Hi Water Temperature (PLC Setpoint	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Hi Oil Temperature	TBC	OLO	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Hi Hi Oil Temperature	TBC	OLT	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Main Oil Pump Failure (By Others)	TBC		Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Fire Suppression Release	TBC	FSS	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open*	Open	Open
Instrument Air Pressure Low (Failure)	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Complete Loss of Instrument Air	TBC		Y	Y	Y	Y	Y	Y	N	Closed	Closed	Open	Open	Open
Smoke Detection	TBC	SDS	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Strainer Differential Pressure (Hi Hi)	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Strainer Differential Pressure (Hi)	TBC		Y	N	N	Y	Y	Y	N	Closed	Open	Closed	Open	Closed
Zone Stop Combustion Air	TBC	CFZ	Y	N	Y	N	Y	Y	N	Open	Closed	Open	Open	Closed
Zone Stop Pollution Air	TBC	PFZ	Y	N	Y	Y	N	Y	N	Open	Closed	Open	Open	Closed
Zone Stop Burner Cooling Fan	TBC	FSZ	Y	N	Y	Y	Y	N	N	Open	Closed	Open	Open	Closed
Zone Stop Burner Fuel Oil Pump	TBC	OPZ	Y	N	Y	Y	Y	Y	Y	Open	Closed	Open	Open	Closed
Main Emergency Stop	TBC	MES	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Fuel Oil Supply Pressure Switch	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Manway Access Safety Relay	TBC	MAS	Y	N	Y	N	N	Y	N	Closed	Closed	Open	Open	Closed
OPTIMISATION ALARMS														
Combustion Flow High/Low Flow Alarm	TBC		Y	N	N	Y	Y	Y	Y	Closed	Open	Closed	Open	Closed
Pollution Flow High/Low Flow Alarm	TBC		Y	N	N	Y	Y	Y	Y	Closed	Open	Closed	Open	Closed
Combustion Chamber Temp (>850 and	TBC		Y	N	N	Y	Y	Y	Y	Closed	Open	Closed	Open	Closed
OPERATIONAL MODES														
Frying Mode			N	N	N	Y	Y	Y	Y	Closed	Open	Closed	Open	Closed
Boil Out/CIP Mode			N	N	N	Y	Y	Y	Y	Open	Closed	Open	Closed	Closed
Pre Heat Mode			N	N	N	Y	Y	Y	Y	Open	Open	Closed	Open	Closed
Shut Down Mode (Circ Pump Stopped)			N	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open

3.2.6 Burner Shut Down

When the Fryer no longer requires the oil (or water) to be heated the Burner Run request is removed to the Heater PLC (from the Fryer PLC).
On the Burner Run request being removed in the Heater PLC, the 'Burner Run' request to the Weishaupt Burner controller will be removed and the burner will shut down immediately and go into a 'Cool Down' mode.
In 'Cool Down' mode the Combustion Air Fan (M3501), Pollution Fan (M3502) and the Flame Sensing Cooling Fan (M3503) will continue to run for a 120 minutes.

Note, the Main Oil Pump (M3604) will be inhibited from stopping until the Stack Temperature falls below 130°C.

Section 4

Software Design

4 Software Design

4.1 PLC

4.1.1 PLC Overview

The Flo-Therm™ G4 Heater control system will encompass an Allen Bradley Compact GuardLogix® 1768-L43S PLC.

The Compact GuardLogix® controller is a full-function Logix processor that also provides safety control.
The GuardLogix system is a dual processor solution that uses a primary controller and a safety partner to achieve SIL 3, PLe. A major benefit of this system is that it is a single project, with safety and standard control together.
The GuardLogix system is a 1oo2 safety architecture that provides unmatched safety and ease of use.

Just like other Logix processors during development, safety and standard have the same rules. Online editing, forcing, even multiple users are all allowed. Once the project is tested and ready for final validation, you set the Safety Task to a SIL 3 integrity level, which is then enforced by the GuardLogix controller. When safety memory is locked and protected, the safety logic cannot be modified.
On the standard side of the GuardLogix controller, all functions operate like a regular Logix controller, motion, drive, sequential, even process. Thus, online editing, forcing, and other activities are all allowed, while safety is securely isolated.

With this level of integration, safety memory can be read by standard logic and external devices, such as HMIs or other controllers, eliminating the need to condition safety memory for use elsewhere.

I/O modules used for the Flo-Therm™ G4 Heater control system will separated into two categories: -

- ***Safe I/O*** will utilise POINT Guard I/O™.
  These modules communicate by using CIP Safety protocol over EtherNet/IP.

- ***Non-Safe I/O*** will utilise standard 1769 Compact I/O modules located in the local chassis of the controller.

Logic generated associated with all the Safe I/O devices will be contained within the dedicated Safety Task of the PLC, all other Non-Safe I/O devices will have their associated logic contained within the standard Main Task of the PLC.

4.2 I/O

4.2.1 I/O Overview

The Flo-Therm™ G4 Heater control system will consist of ‘Safe’ and ‘Non-safe’ I/O.
The allocation of each is detailed below: -

4.2.1.1 Non-Safe Digital Inputs

Device/Component Tag	Description
FT3501	NATURAL GAS FLOW PULSE
FT3504	FUEL OIL FLOW PULSE
FRR	FAN REQUEST SIGNAL
GASR	GAS ON SIGNAL
OILR	FUEL OIL ON SIGNAL
BFR	BURNER FAULT SIGNAL (LOCKOUT)
S1	RESET BUTTON

4.2.1.2 Non-Safe Digital Outputs

Device/Component Tag	Description
FSR	FUEL SELECTION RELAY
RR1/RR2	RESET RELAY
S1	PANEL CONTROL ON LAMP
V3508	AIR WASH VALVE
V3512	POLLUTION FAN SWITCH PURGE VALVE 1
V3513	POLLUTION FAN SWITCH PURGE VALVE 2
V3517	SIGHT GLASS AIR WASH VALVE

4.2.1.3 Non-Safe Analogue Inputs

Device/Component Tag	Description
TT3503	COMBUSTION WARM AIR STREAM TEMPERATURE
TT3504	NATURAL GAS TEMPERATURE
TT3505	COMBUSTION COLD AIR STREAM TEMPERATURE
TT3508	POLLUTION VAPOUR TEMPERATURE
TT3510	EXHAUST TEMPERATURE
FT3501	NATURAL GAS FLOW
FT3502	POLLUTION DIFFERENTIAL PRESSURE
FT3503	COMBUSTION AIR DIFFERENTIAL PRESSURE
FT3504	FUEL OIL FLOW
PT3501	NATURAL GAS PRESSURE

4.2.1.4 Non-Safe Analogue Outputs

Device/Component Tag	Description
FT3602	MAIN OIL FLOWMETER RE-TRANSMITION TO FRYER
CV3502	STACK PROCESS DAMPER 1
CV3503	STACK PROCESS DAMPER 2

4.2.1.5 Safe Digital Inputs

Device/Component Tag	Description
M3501K	COMBUSTION AIR FAN RUNNING
M3501KA	COMBUSTION AIR FAN TRIPPED
M3501KQ	COMBUSTION AIR FAN ISOLATED
M3502K	POLLUTION FAN RUNNING
M3502KA	POLLUTION FAN TRIPPED
M3502Q	POLLUTION FAN ISOLATED

M3503K	FLAME SENSOR COOLING FAN RUNNING
M3503KA	FLAME SENSOR COOLING FAN TRIPPED
M3503Q	FLAME SENSOR COOLING FAN ISOLATED
M3504K	FUEL OIL PUMP RUNNING
M3504KA	FUEL OIL PUMP TRIPPED
M3504Q	FUEL OIL PUMP ISOLATED
PS3511	OIL STRAINER PRESSURE HIGH
PS3511	OIL STRAINER PRESSURE HIGH / HIGH
PS3507	COMPRESSED AIR HEALTHY
PS3508	FUEL OIL PRESSURE SWITCH
ES3501A	PANEL E-STOP CHANNEL 1
ES3502A	PANEL E-STOP CHANNEL 2
ZS3501A	COMBUSTION AIR FAN ZONE STOP CHANNEL 1
ZS3501B	COMBUSTION AIR FAN ZONE STOP CHANNEL 2
ZS3502A	POLLUTION FAN ZONE STOP CHANNEL 1
ZS3502B	POLLUTION FAN ZONE STOP CHANNEL 2
ZS3503A	FLAME SENSOR COOLING FAN ZONE STOP CHANNEL 1
ZS3503B	FLAME SENSOR COOLING FAN ZONE STOP CHANNEL 2
SS3501	MANWAY ACCESS CHANNEL 1
SS3501	MANWAY ACCESS CHANNEL 2
SD3501	SMOKE DETECTOR HEALTHY
PS3501	POLLUTION DIFFERENTIAL PRESSURE HEALTHY
PS3505	FLAME SENSOR COOLING PRESSURE HEALTHY
FS3501	VENT VALVE LINE FLOW SWITCH HEALTHY
PS3502	HIGH GAS PRESSURE FAULT
PS3503	LOW GAS PRESSURE FAULT
M3604	MAIN OIL PUMP RUNNING (REMOTE)
MPSR	MAIN OIL PUMP SAFETY RELAY HEALTHY (REMOTE)
RB	FIRE SUPPRESSION NOT ACTIVE (REMOTE)
SS3703	WATER PIPE IN SAFE POSITION (REMOTE)
SS3703	COOK SELECTED AND WATER PIPE IN SAFE POSITION (REMOTE)
V3501OPN	FRESH AIR DAMPER OPEN
V3501CLS	FRESH AIR DAMPER CLOSED
V3502OPN	STACK DAMPER OPEN
V3502CLS	STACK DAMPER CLOSED
V3503OPN	FIRE DAMPER OPEN
V3503CLS	FIRE DAMPER CLOSED
V3504OPN	POLLUTANT DRAIN VALVE 1 OPEN
V3504CLS	POLLUTANT DRAIN VALVE 1 CLOSED
V3505OPN	VENT VALVE 2 OPEN

V3505CLS	VENT VALVE 2 CLOSED
V3506OPN	VENT VALVE 1 OPEN
V3506CLS	VENT VALVE 1 CLOSED
V3514OPN	CIP DAMPER OPEN
V3514CLS	CIP DAMPER CLOSED
V3518OPN	POLLUTANT DRAIN VALVE 2 OPEN
V3518CLS	POLLUTANT DRAIN VALVE 2 CLOSED

4.2.1.6 Safe Digital Outputs

Device/Component Tag	Description
M3501K1	COMBUSTION AIR FAN RUN
M3502K1	POLLUTION FAN RUN
M3503K	FLAME SENSOR COOLING FAN
M3504K	FUEL OIL PUMP RUN
M3604K	MAIN OIL PUMP RUN SIGNAL (REMOTE)
V3501	FRESH AIR DAMPER VALVE
V3502	STACK DAMPER
V3503	FIRE DAMPER VALVE
V3504	POLLUTANT DRAIN VALVE 1
V3505	VENT VALVE 1
V3506	VENT VALVE 2
V3514	CIP DAMPER VALVE
V3518	POLLUTANT DRAIN VALVE 2
BRR	BURNER RUN RELAY 1
BRR	BURNER RUN RELAY 2

4.2.1.7 Safe Analogue Inputs

Device/Component Tag	Description
FT3602	MAIN OIL FLOWMETER
TT3501A	COMBUSTION CHAMBER TEMPERATURE TRANSMITTER 1
TT3501B	COMBUSTION CHAMBER TEMPERATURE TRANSMITTER 2
TT3502A	STACK TEMPERATURE TRANSMITTER 1
TT3502B	STACK TEMPERATURE TRANSMITTER 2
TT3507A	OIL OUTLET TEMPERATURE TRANSMITTER 1
TT3507B	OIL OUTLET TEMPERATURE TRANSMITTER 2
TT3511A	HX OIL SAEFTY TEMPERATURE TRANSMITTER 1
TT3511B	HX OIL SAEFTY TEMPERATURE TRANSMITTER 2
TT3509A	POLLUTION TEMPERATURE TRANSMITTER 1

TT3509B	POLLUTION TEMPERATURE TRANSMITTER 2
SD3501	SMOKE DETECTOR OBSCURITY

4.2.2 I/O Architecture

To accommodate the I/O requirements of the Flo-Therm™ G4 Heater control system the following IO architecture will be used: -

The local chassis of the Compact GuardLogix® PLC will contain the following modules…

Slot Position	I/O Module	Description
1768 Bus – Slot 2	1768-ENBT	Ethernet/IP Bridge Module
1768 Bus – Slot 1	1768-ENBT	Ethernet/IP Bridge Module
Slot 0	1768-L43S	CompactLogix L43 Safety Processor
1769 Bus – Slot 1	1769-IF8	8 Channel Analog Current/Voltage Input Module
1769 Bus – Slot 2	1769-IF8	8 Channel Analog Current/Voltage Input Module
1769 Bus – Slot 3	1769-OF4CI	4 Channel Analog Current Isolated Output Module
1769 Bus – Slot 4	1769-IQ16	16 Point 24 VDC Sinking/Sourcing Input Module
1769 Bus – Slot 5	1769-OB16	16 Point 24 VDC Sourcing Output Module

Safe I/O will be wired to a POINT Guard I/O™ system, this modular system will contain the following modules…

Slot Position	I/O Module	Description
Slot 0	1734-AENT	EtherNet/IP Twisted Pair Media I/O Adapter
Slot 1	1734-IE4S	POINT Guard I/O Safety Module - 4 Safety Analog Input Module
Slot 2	1734-IE4S	POINT Guard I/O Safety Module - 4 Safety Analog Input Module
Slot 3	1734-IE4S	POINT Guard I/O Safety Module - 4 Safety Analog Input Module
Slot 4	1734-IE4S	POINT Guard I/O Safety Module - 4 Safety Analog Input Module
Slot 5	1734-IB8S	POINT Guard I/O Safety Module - 8 Point Input Module
Slot 6	1734-IB8S	POINT Guard I/O Safety Module - 8 Point Input Module
Slot 7	1734-IB8S	POINT Guard I/O Safety Module - 8 Point Input Module
Slot 8	1734-IB8S	POINT Guard I/O Safety Module - 8 Point Input Module
Slot 9	1734-IB8S	POINT Guard I/O Safety Module - 8 Point Input Module
Slot 10	1734-IB8S	POINT Guard I/O Safety Module - 8 Point Input Module
Slot 11	1734-IB8S	POINT Guard I/O Safety Module - 8 Point Input Module
Slot 12	1734-OB8S	POINT Guard I/O Safety Module - 8 Safety Sourcing Output Module
Slot 13	1734-OB8S	POINT Guard I/O Safety Module - 8 Safety Sourcing Output Module
Slot 14	1734-OB8S	POINT Guard I/O Safety Module - 8 Safety Sourcing Output Module

4.3 PLC Program

4.3.1 PLC Program Overview

Rockwell RSLogix5000 V20 programming software will be used to generate the PLC program.

4.3.2 PLC Program Structure

PLC software will be designed in a structured manner, using a modular approach, with each functional module occupying a separate program file.

All rungs to be annotated in English with a meaningful comment, and each functional module or sub-module to be annotated with a description of the module function.

All addresses used within the PLC to be fully commented to identify items of equipment. Addresses which refer to input or output elements to have the I/O Tag number as part of the comment.

4.3.2.1 Add-On Instructions

Within the PLC Program common Add-On Instructions (AOI's) are utilised.

With Add-On Instructions, you can create new instructions for sets of commonly-used logic, provide a common interface to this logic, and provide documentation for the instruction. Add-On Instructions are intended to be used to encapsulate commonly used functions or device control.

These Add-On Instructions are then called for each time a function is then required.

The Flo-Therm™ G4 Heater control program will utilise the following Add-On Instructions: -

Add-On Instruction	Description
InputDigital	Handles all non-safe the digital inputs to the program controlling conditions, faults and SCADA status.
Motor1S1D	Handles the single speed single direction motors controlling conditions, faults, manual controls and SCADA status.
MotorPowerFlexInv	Handles the inverter driven motors controlling conditions, faults, manual controls and SCADA status.
ValveLimits	Handles maintained valves with both open and closed limit switches controlling conditions, faults, manual controls and SCADA status.
ValveNoLimits	Handles maintained valves with no open or closed limit switches controlling conditions, faults, manual controls and SCADA status.

Full structure details on the individual Add-On Instructions can be found in Appendix A.

4.3.2.2 User Defined Tags

Within the PLC Program common User Defined Tags (UDT's) are utilised.
A User Defined Tag structure lets you combine multiple data types (e.g. INTS, DINTS, FLOATS and other UDT's) into one single structure.

The Flo-Therm™ G4 Heater control program will utilise the following User Defined Tags: -

UDT	Description
UDTAlarmHandler	Defined tag structure for general Alarm Handling functionality (e.g. new alarm indication, resetting, muting, etc.).
UDTAnalogInput	Defined tag structure for an Analogue Input device.
UDTAnalogOutput	Defined tag structure for an Analogue Output device.
UDTBurner	Defined tag structure for general Burner control functionality (e.g. Cook mode, etc.).
UDTCombustionManagement	Defined tag structure for general Combustion Management functionality (e.g. Gas Ratio's, device control, etc.).
UDTCombustionManagerFunctions	Defined tag structure for general Combustion Manager Function functionality (e.g. Energy Balance calculations for Optimisation, etc.).
UDTDigitalInput	Defined tag structure for a Digital Input device.
UDTEng	Defined tag structure for Stack Damper Lookup Table.
UDTFlowCalc	Defined tag structure for Flow Rate calculations.
UDTGenericMultiConstantCalc	Defined tag structure for Energy Balance calculations.
UDTHeatEff	Defined tag structure for general Burner Efficiency functionality.
UDTInterfaceHXToFryer	Defined tag structure for tags sent to Fryer PLC.
UDTLookupTable	Defined tag structure for a generic Lookup Table.
UDTMisc	Defined tag structure for miscellaneous control (e.g. flash bits, always on flag etc.).
UDTMotorDOL	Defined tag structure for a One Speed One Direction (DOL) type motor.
UDTPIDControl	Defined tag structure for generic PID Loop Controllers.
UDTPLCStatus	Defined tag structure for general PLC Status (e.g. CPU mode, battery status, etc.).
UDTPowerFlexINV	Defined tag structure for an Inverter type motor.
UDTRecipe	Defined tag structure for general Burner functionality based on selected recipe in Fryer.
UDTScadaAnalogs	Defined tag structure for analogue tags to be used in SCADA application.
UDTScadaMisc	Defined tag structure for miscellaneous tags to be used in SCADA application.
UDTScadaStatus	Defined tag structure for general status tags to be used in SCADA application.
UDTValve	Defined tag structure for an Analogue Input device.

Full structure details on the individual User Defined Tags can be found in Appendix A.

4.3.2.3 Tasks

A Task contains programs, each with its own routines and program-scoped tags. Once a task is triggered, all the Programs assigned to the Task execute in the order in which they are listed (in the Task properties).

A Task can be configured as either continuous, periodic, or event driven.

The Flo-Therm™ G4 Heater control program will utilise the following Tasks: -

Task	Type	Description
Main Task	Continuous	Contains all the non-safety programs/routines for the Flo-Therm™ G4 Heater control.
Safety Task	Periodic	Contains all the safety programs/routines for the Flo-Therm™ G4 Heater control.

4.3.3 Programs

A Program organizes groups of routines. They are used to isolate logic and data from other Programs.
All Programs can share common data by the use of 'Controller' Scoped tags; however isolated data that is only accessible from that Program can be handled by the use of 'Program' Scoped tags.

Each Program contains a number of Routines specific to that Program. Additionally each Program has a Main Routine that is the first routine to execute within a program.

The 'Main Task' will utilise the following Programs: -

Program	Description
MainProgram	Contains all the core 'Non-Safe' routines for the general control of the Flo-Therm™ G4 Heater.
IOMapping	Contains all the routines for the handling of the Flo-ThermTM G4 Heater Non-Safe I/O.

The 'Safety Task' will utilise the following Programs: -

Program	Description
SafetyProgram	Contains all the core routines for the general 'Safety' control of the Flo-Therm™ G4 Heater.

4.3.4 Routines

Routines contain the executable code (logic).

The Main Routine (within the Program) will be the first Routine (within the Program) to be executed.
The Main Routine also generally calls for the other Routines (within the Program) to be executed.

The 'MainProgram' Program will contain the following Routines: -

Routine	Description
Main	Calls all the core 'Non-Safe' routines for the general control of the Flo-Therm™ G4 Heater.
[Add Routines]	[Add Routines as per final PLC program]

The 'IOMapping' Program will contain the following Routines: -

Routine	Description
Main	Calls all the routines for the handling of the Flo-ThermTM G4 Heater Non-Safe I/O.
[Add Routines]	[Add Routines as per final PLC program]

The 'SafetyProgram' Program will contain the following Routines: -

Routine	Description
Main	Calls all the core routines for the general 'Safety' control of the Flo-Therm™ G4 Heater.
S001_Analogue_In	Handles safety instructions for Safety Analogue Inputs.
S002_Digital_In	Handles safety instructions for Safety Digital Inputs.
S002_Digital_Out	Handles safety instructions for Safety Digital Outputs.
[Add Routines]	[Add Routines as per final PLC program]

4.3.5 Safety Control Overview

The Flo-Therm™ G4 Heater control system will encompass an Allen Bradley Compact GuardLogix® 1768-L43S PLC.

The Compact GuardLogix® controller is a full-function Logix processor that also provides safety control.
The GuardLogix system is a dual processor solution that uses a primary controller and a safety partner to achieve SIL 3, PLe. A major benefit of this system is that it is a single project, with safety and standard control together.
The GuardLogix system is a 1oo2 safety architecture that provides unmatched safety and ease of use.

Just like other Logix processors during development, safety and standard have the same rules. Online editing, forcing, even multiple users are all allowed. Once the project is tested and ready for final validation, you set the Safety Task to a SIL 3 integrity level, which is then enforced by the GuardLogix controller. When safety memory is locked and protected, the safety logic cannot be modified.
On the standard side of the GuardLogix controller, all functions operate like a regular Logix controller, motion, drive, sequential, even process. Thus, online editing, forcing, and other activities are all allowed, while safety is securely isolated.

With this level of integration, safety memory can be read by standard logic and external devices, such as HMIs or other controllers, eliminating the need to condition safety memory for use elsewhere.

4.3.5.1 Differentiate between Standard and Safety logic

Because it is a Logix-series controller, both standard (non-safety-related) and safety-related components can be used in the GuardLogix control system.

The Compact GuardLogix® 1768-L43S controller provide the same functionality as other 1768-L43 CompactLogix controllers. You can perform standard automation control from standard tasks within a GuardLogix project.

What differentiates the GuardLogix controller from the standard controller is that it can provide a SIL 3-capable safety task.

However, a logical and visible distinction is required between the standard and safety-related portions of the application. RSLogix 5000 software provides this differentiation via the safety task, safety programs, safety routines, safety tags, and safety I/O modules.
You can implement both SIL 2 and SIL 3 levels of safety control with the safety task of the GuardLogix controller.

4.3.5.1.1 Safety Task

The creation of a GuardLogix project automatically creates a single safety task. The safety task has these additional characteristics:

- GuardLogix controllers are the only controllers that support the safety task.
- The safety task cannot be deleted.
- GuardLogix controllers support a single safety task.
- Within the safety task, you can use multiple safety programs composed of multiple safety routines.
- You cannot schedule or execute standard routines from within the safety task.

The safety task is a periodic timed task with a user-selectable task priority and watchdog. In most cases, it is the controller's top priority and the user-defined program watchdog must be set to accommodate fluctuations in the execution of the safety task.

Both the safety task period and the safety task watchdog are specified.
The safety task period is the period at which the safety task executes.
The safety task watchdog is the maximum time allowed from the start of safety task scheduled execution to its completion.

The safety task period is limited to a maximum of 500 ms and cannot be modified online. The safety task period is set to ensure that the safety task has enough time to finish before it is triggered again.
Safety task watchdog timeout, a non-recoverable safety fault in the GuardLogix controller, occurs if the safety task is triggered while it is still executing from the previous trigger.

The safety task executes in the same manner as standard periodic tasks, with the following exceptions:

- The safety task does not begin executing until the primary controller and safety partner have established their control partnership and the coordinated system time (CST) is synchronized. However, standard tasks begin executing as soon as the controller transitions to Run mode.
- Although the configurable range of the requested packet interval (RPI) for safety inputs and safety consumed tags is 6…500 ms, safety input tags and safety-consumed tags are updated only at the beginning of safety task execution. This means that even though the I/O RPI can be faster than the safety task period, the data does not change during safety task execution. The data is read only once at the beginning of the safety task execution.
- Safety input values are frozen at the start of safety task execution. As a result, timer-related instructions, such as TON and TOF, will not update during a single safety task execution. They will keep accurate time from one task execution to another, but the accumulated time will not change during safety task execution.
- For standard tags that are mapped to safety tags, the standard tag values are copied into safety memory at the start of the safety task and do not change during safety task execution.
- Safety output tag (output and produced) values are updated at the conclusion of safety task execution.
- The safety task responds to mode changes (for example, Run to Program or Program to Run) at timed intervals. As a result, the safety task may take more than one task period, but always less than two, to make a mode transition.

4.3.5.1.2 Safety Programs

A safety program has all the attributes of a standard program, except that it can be scheduled only in the safety task.
Additionally:

- A safety program may also define 'Program Scoped' safety tags.
- A safety program may be scheduled or unscheduled.
- A safety program can contain only safety components.
- All of the routines in a safety program are safety routines.
- A safety program cannot contain standard routines or standard tags.

4.3.5.1.3 Safety Routines

Safety routines have all the attributes of standard routines, except that they can exist only in safety programs.
One safety routine may be designated as the main routine. Another safety routine may be designated as the fault routine.
Only safety-certified instructions may be used in safety routines.

Please note, in order to preserve SIL 3, you must make sure that the safety logic does not attempt to read or write standard tags.

4.3.5.1.4 Safety Tags

The GuardLogix control system supports the use of both standard and safety tags in the same project. However, RSLogix5000 software operationally differentiates standard tags from safety tags.
Safety tags have all the attributes of standard tags with the addition of mechanisms to provide SIL 3 data integrity.

RSLogix 5000 software prevents the direct creation of invalid tags in a safety program. In the event that invalid tags are imported, they cannot be verified.
In safety programs RSLogix 5000 software also prevents the following:

- The creation of a safety tag using a structured type (user-defined, Add-On defined, predefined) when they contain one or more members of the REAL data type, including nested structures.
- The modification of a user-defined, or Add-On defined type that would cause an invalid data type to be included when the user-defined or Add-On defined type is already referenced directly or indirectly by a safety tag.
- Invalid tags created by using with the New Tag or Tag Properties dialog box.

Tags classified as safety tags are either controller-scoped or program-scoped.
Controller-scoped safety tags can be read by either standard or safety logic or other communication devices, but can only be written to by safety logic or another GuardLogix safety controller.
Program-scoped safety tags are accessible only by local safety routines. These are routines that reside within the safety program.
Tags associated with Safety I/O and produced or consumed safety data must be controller-scoped safety tags.

Any controller-scoped safety tag is readable by any standard routine, but the update rate is based on the execution of the safety task. This means that safety tags are updated at the safety task periodic rate, which is different from standard tag behaviour.

Controller-scoped standard tags can be mapped into safety tags, providing you with a mechanism to synchronize standard and safety actions.

Please note, when using standard data in a safety routine, you are responsible for providing a reliable means of ensuring that the data is used in an appropriate manner. Using standard data in a safety tag does not make it safety data. You must not directly control a safety output with standard tag data.

4.3.5.1.5 Safety Instructions

Only safety-certified instructions may be used in safety routines. For full details of the available safety-certified instructions please refer to the appropriate Rockwell Software documentation.

With RSLogix 5000 software you can create safety Add-On Instructions.
Safety Add-On Instructions let you encapsulate commonly-used safety logic into a single instruction, making it modular and easier to reuse. Safety Add-On Instructions use the instruction signature of high-integrity Add-On Instructions and also a SIL 3 safety instruction signature for use in safety related functions up to and including SIL 3.

4.3.5.2 Safety Control Development

There are a number of steps to be undertaken prior to the generation of logic routines in the Compact GuardLogix® controller.

These are briefly detailed in the following sections:

4.3.5.2.1 Risk Assessment & System Design

[Detail supplementary information with regards to Risk Assessments, Performance Levels, etc.]

4.3.5.2.2 Define Module and Safety Zone (Tags)

[Detail supplementary information with regards to clarification of safe modules/zones.]

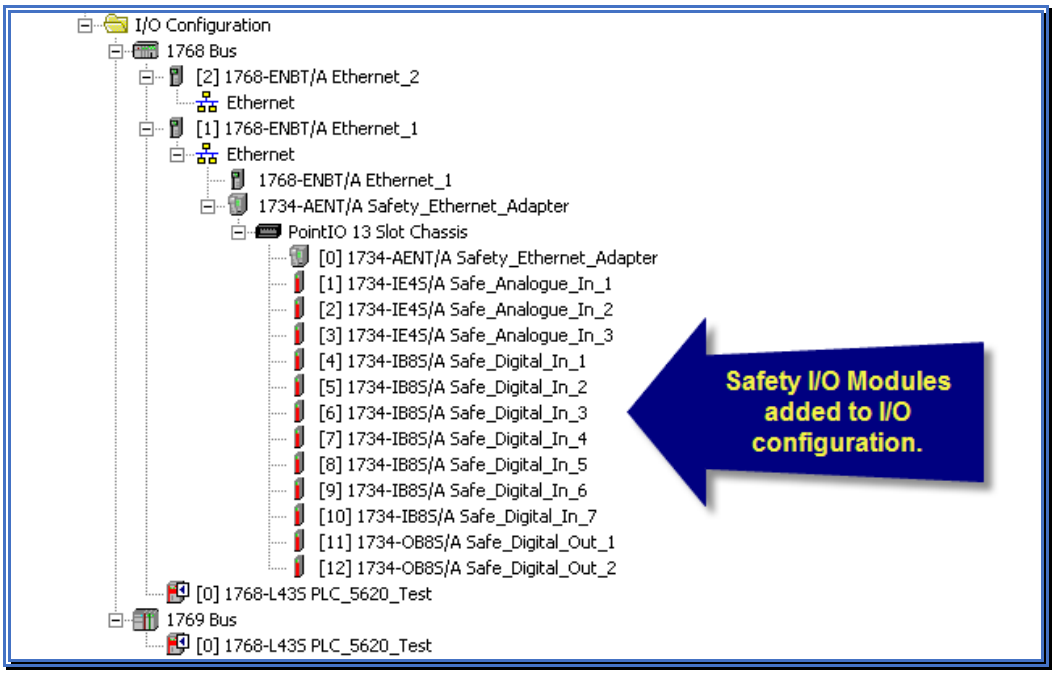
4.3.5.2.3 Configure the Guard I/O Modules

CIP Safety I/O modules can be connected to safety input and output devices, allowing these devices to be monitored and controlled by the GuardLogix controller.

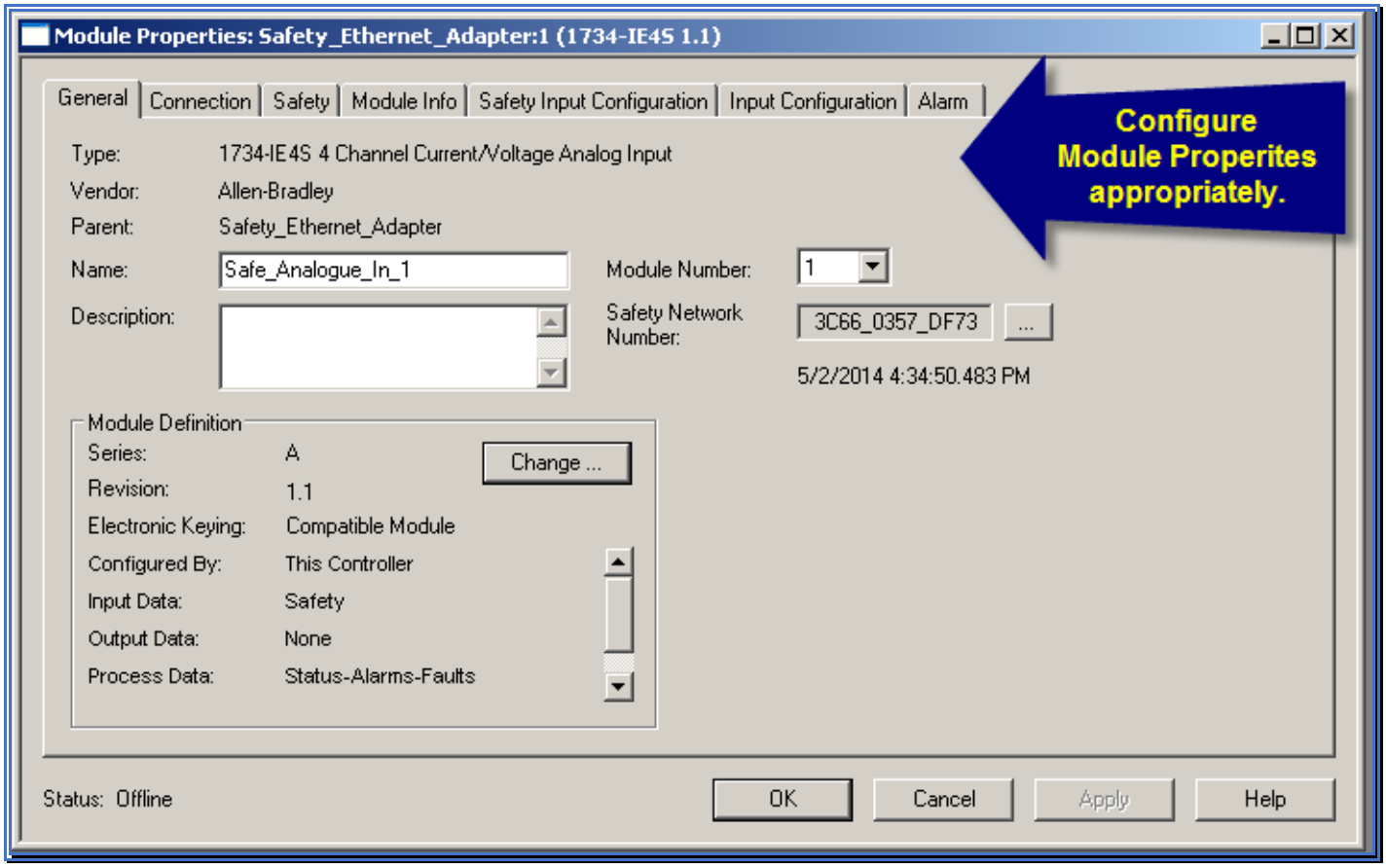
The Flo-Therm™ G4 Heater control system uses POINT Guard I/O safety modules; these are connected on an EtherNet/IP network via a POINT I/O Ethernet/IP adaptor (1734-AENT).

Each Guard I/O Module that is to be used is configured using the RSLogix5000 software.

All Safety I/O modules are added to PLC I/O configuration tree:



For each Safety I/O module configure the 'Module Properties' appropriately.



For further details on the final configuration of the modules please refer to Appendix A.

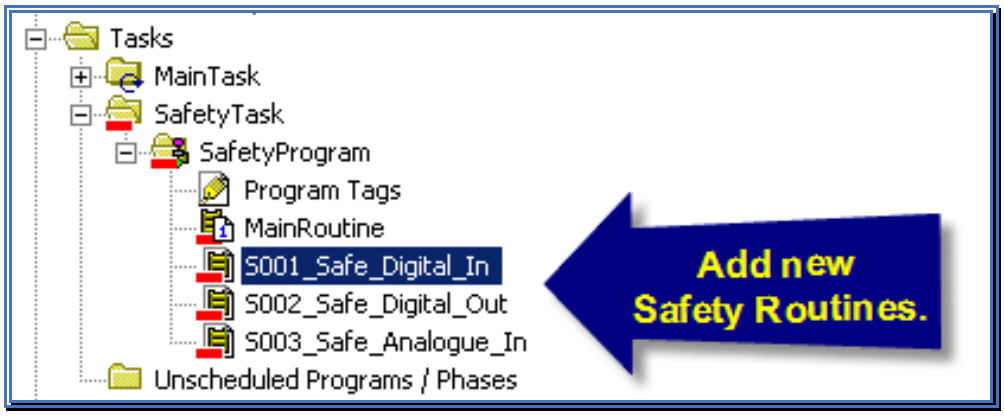
Once the above steps have been finalised the Safety logic Programs/Routines can be generated.

4.3.5.3 Safety Logic Generation

To separate the safety logic for digital inputs, digital outputs and analogue inputs individual routines are generated within the SafetyProgram:

The 'SafetyProgram' Program will contain the following Routines: -

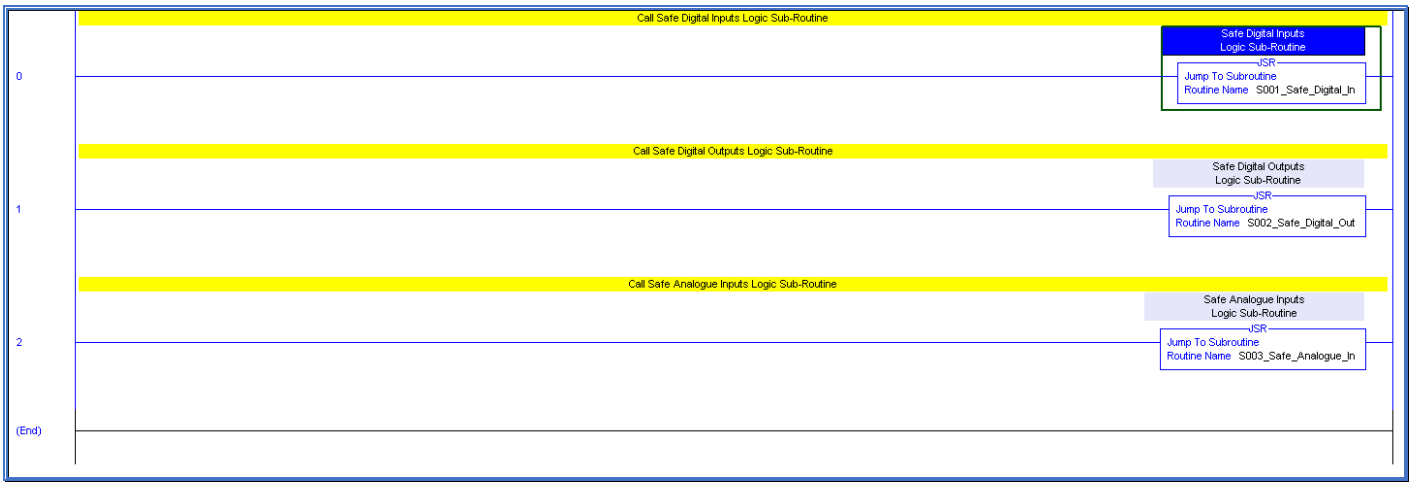
Routine	Description
Main	Calls all the 'Safety' Routines with the SafetyProgram.
S001_Analogue_In	Handles safety instructions for Safety Analogue Inputs.
S002_Digital_In	Handles safety instructions for Safety Digital Inputs.
S002_Digital_Out	Handles safety instructions for Safety Digital Outputs.



Each safety routine is briefly detailed:

4.3.5.3.1 MainRoutine Routine

Simply calls all other routines within the SafetyProgram sequentially using standard Jump To Subroutine (JSR) instructions:



4.3.5.3.2 S001_Safe_Digital_In Routine

Handles all logic for each Safety Digital Input device used on the Flo-Therm™ G4 Heater control system.

Dependent on the SIL rating requirement for the module/zone specific safety instructions will be utilised.
In general single-channel digital inputs will use SIL 2 category safety instructions, dual-channel digital inputs will use SIL 3 category safety instruction.

Single-channel digital inputs (for SIL 2) will use a Safe Examine If Closed (XIC) instructions.

Dual-channel digital inputs (for SIL 3) will use Dual-Channel Input Stop (DCS) instructions.

The Dual-Channel Input Stop instruction monitors dual-input safety devices whose main function is to stop a machine/process safely, for example, an E-stop or safety switch.
The instruction can only energize its output flag when both safety inputs, Channel A and Channel B, are in the active state as determined by the Input type parameter, and the correct reset actions are carried out.

The following devices are handled in the S001_Safe_Digital_In routine:

Device/Component Tag	Instruction Type	Description
M3501K	XIC	COMBUSTION AIR FAN RUNNING
M3501KA	XIC	COMBUSTION AIR FAN TRIPPED
M3501KQ	XIC	COMBUSTION AIR FAN ISOLATED
M3502K	XIC	POLLUTION FAN RUNNING
M3502KA	XIC	POLLUTION FAN TRIPPED
M3502Q	XIC	POLLUTION FAN ISOLATED
M3503K	XIC	FLAME SENSOR COOLING FAN RUNNING
M3503KA	XIC	FLAME SENSOR COOLING FAN TRIPPED
M3503Q	XIC	FLAME SENSOR COOLING FAN ISOLATED
M3504K	XIC	FUEL OIL PUMP RUNNING
M3504KA	XIC	FUEL OIL PUMP TRIPPED
M3504Q	XIC	FUEL OIL PUMP ISOLATED
PS3511	XIC	OIL STRAINER PRESSURE HIGH
PS3511	XIC	OIL STRAINER PRESSURE HIGH / HIGH
PS3507	XIC	COMPRESSED AIR HEALTHY
PS3508	XIC	FUEL OIL PRESSURE SWITCH
ES3501A	DCS	PANEL E-STOP CHANNEL 1
ES3502A	DCS	PANEL E-STOP CHANNEL 2
ZS3501A	DCS	COMBUSTION AIR FAN ZONE STOP CHANNEL 1
ZS3501B	DCS	COMBUSTION AIR FAN ZONE STOP CHANNEL 2
ZS3502A	DCS	POLLUTION FAN ZONE STOP CHANNEL 1
ZS3502B	DCS	POLLUTION FAN ZONE STOP CHANNEL 2
ZS3503A	DCS	FLAME SENSOR COOLING FAN ZONE STOP CHANNEL 1
ZS3503B	DCS	FLAME SENSOR COOLING FAN ZONE STOP CHANNEL 2
SS3501	DCS	MANWAY ACCESS CHANNEL 1
SS3501	DCS	MANWAY ACCESS CHANNEL 2
SD3501	XIC	SMOKE DETECTOR HEALTHY
PS3501	XIC	POLLUTION DIFFERENTIAL PRESSURE HEALTHY

PS3505	XIC	FLAME SENSOR COOLING PRESSURE HEALTHY
FS3501	XIC	VENT VALVE LINE FLOW SWITCH HEALTHY
PS3502	XIC	HIGH GAS PRESSURE FAULT
PS3503	XIC	LOW GAS PRESSURE FAULT
M3604	XIC	MAIN OIL PUMP RUNNING (REMOTE)
MPSR	XIC	MAIN OIL PUMP SAFETY RELAY HEALTHY (REMOTE)
RB	XIC	FIRE SUPPRESSION NOT ACTIVE (REMOTE)
SS3703	XIC	WATER PIPE IN SAFE POSITION (REMOTE)
SS3703	XIC	COOK SELECTED AND WATER PIPE IN SAFE POSITION (REMOTE)
V3501OPN	XIC	FRESH AIR DAMPER OPEN
V3501CLS	XIC	FRESH AIR DAMPER CLOSED
V3502OPN	XIC	STACK DAMPER OPEN
V3502CLS	XIC	STACK DAMPER CLOSED
V3503OPN	XIC	FIRE DAMPER OPEN
V3503CLS	XIC	FIRE DAMPER CLOSED
V3504OPN	XIC	POLLUTANT DRAIN VALVE 1 OPEN
V3504CLS	XIC	POLLUTANT DRAIN VALVE 1 CLOSED
V3505OPN	XIC	VENT VALVE 2 OPEN
V3505CLS	XIC	VENT VALVE 2 CLOSED
V3506OPN	XIC	VENT VALVE 1 OPEN
V3506CLS	XIC	VENT VALVE 1 CLOSED
V3514OPN	XIC	CIP DAMPER OPEN
V3514CLS	XIC	CIP DAMPER CLOSED
V3518OPN	XIC	POLLUTANT DRAIN VALVE 2 OPEN
V3518CLS	XIC	POLLUTANT DRAIN VALVE 2 CLOSED

4.3.5.3.3 S002_Safe_Digital_Out Routine

Handles all logic for each Safety Digital Output device used on the Flo-Therm™ G4 Heater control system.

Dependent on the SIL rating requirement for the module/zone specific safety instructions will be utilised.
In general both single-channel digital inputs and dual-channel digital inputs will both use SIL 3 category digital output safety instructions.

For all safe digital outputs, Configurable Redundant Output (CROUT) instructions will be used.

The Configurable Redundant Output instruction controls and monitors redundant outputs.
The reaction time for output feedback is configurable. The instruction supports positive and negative feedback signals

The following devices are handled in the S002_Safe_Digital_Out routine:

Device/Component Tag	Instruction Type	Description
M3501K1	CROUT	COMBUSTION AIR FAN RUN
M3502K1	CROUT	POLLUTION FAN RUN

M3503K	CROUT	FLAME SENSOR COOLING FAN
M3504K	CROUT	FUEL OIL PUMP RUN
M3604K	CROUT	MAIN OIL PUMP RUN SIGNAL (REMOTE)
V3501	CROUT	FRESH AIR DAMPER VALVE
V3502	CROUT	STACK DAMPER
V3503	CROUT	FIRE DAMPER VALVE
V3504	CROUT	POLLUTANT DRAIN VALVE 1
V3505	CROUT	VENT VALVE 1
V3506	CROUT	VENT VALVE 2
V3514	CROUT	CIP DAMPER VALVE
V3518	CROUT	POLLUTANT DRAIN VALVE 2
BRR	CROUT	BURNER RUN RELAY 1
BRR	CROUT	BURNER RUN RELAY 2

4.3.5.3.4 S003_Safe_Analogue_In Routine

Handles all logic for each Safety Analogue Input device used on the Flo-Therm™ G4 Heater control system.

For all single-channel and dual-channel safe analogue inputs, Dual-Channel Analog Input (DCA) instructions will be used.

The Dual-channel Analog Input instruction monitors two analogue input channels originating from an analogue input module. The instruction can only energize its output flag when both analogue inputs (Channel A and Channel B) are within the Tolerance and the High and Low Limit settings, and the correct reset actions have been performed.

In the case of a single-channel type device, the Dual-channel Analog Input instruction will have Channel A and Channel B mapped to the same physical input address.

The following devices are handled in the S003_Safe_Analogue_In routine:

Device/Component Tag	Instruction Type	Description
FT3602	DCA	MAIN OIL FLOWMETER
TT3501A	DCA	COMBUSTION CHAMBER TEMPERATURE TRANSMITTER 1
TT3501B	DCA	COMBUSTION CHAMBER TEMPERATURE TRANSMITTER 2
TT3502A	DCA	STACK TEMPERATURE TRANSMITTER 1
TT3502B	DCA	STACK TEMPERATURE TRANSMITTER 2
TT3507A	DCA	OIL OUTLET TEMPERATURE TRANSMITTER 1
TT3507B	DCA	OIL OUTLET TEMPERATURE TRANSMITTER 2
TT3511A	DCA	HX OIL SAEFTY TEMPERATURE TRANSMITTER 1
TT3511B	DCA	HX OIL SAEFTY TEMPERATURE TRANSMITTER 2
TT3509A	DCA	POLLUTION TEMPERATURE TRANSMITTER 1

4.4 Logic Validation

To ensure that the finalised logic achieves the requirements of the Flo-Therm™ G4 Heater control system a series of tests will be carried out and validated.

After the Flo-Therm™ G4 Heater control system has been fully I/O tested, specific safety and operational testing can be carried out.

Initially each Safe I/O device will undertake an appropriate test; some devices may undergo multiple tests. To supplement these tests individual test sheets for each test will be generated and completed. In general the test sheet will consist of the following information:

- Test name/reference.
- Test description.
- Tag name of device being tested.
- Test procedure.
- Expected result of test.
- Test result.
- Test comments.
- Sign off / test date

In addition to the Safe I/O test, the general Flo-Therm™ G4 Heater control system will undergo a series of general operational and safety tests (as specified by Flo-Mech).

Again these tests will be fully documented and validated.

The complete set of test sheets can be found in Appendix B.

Section 5

Appendix A

5 Appendix A

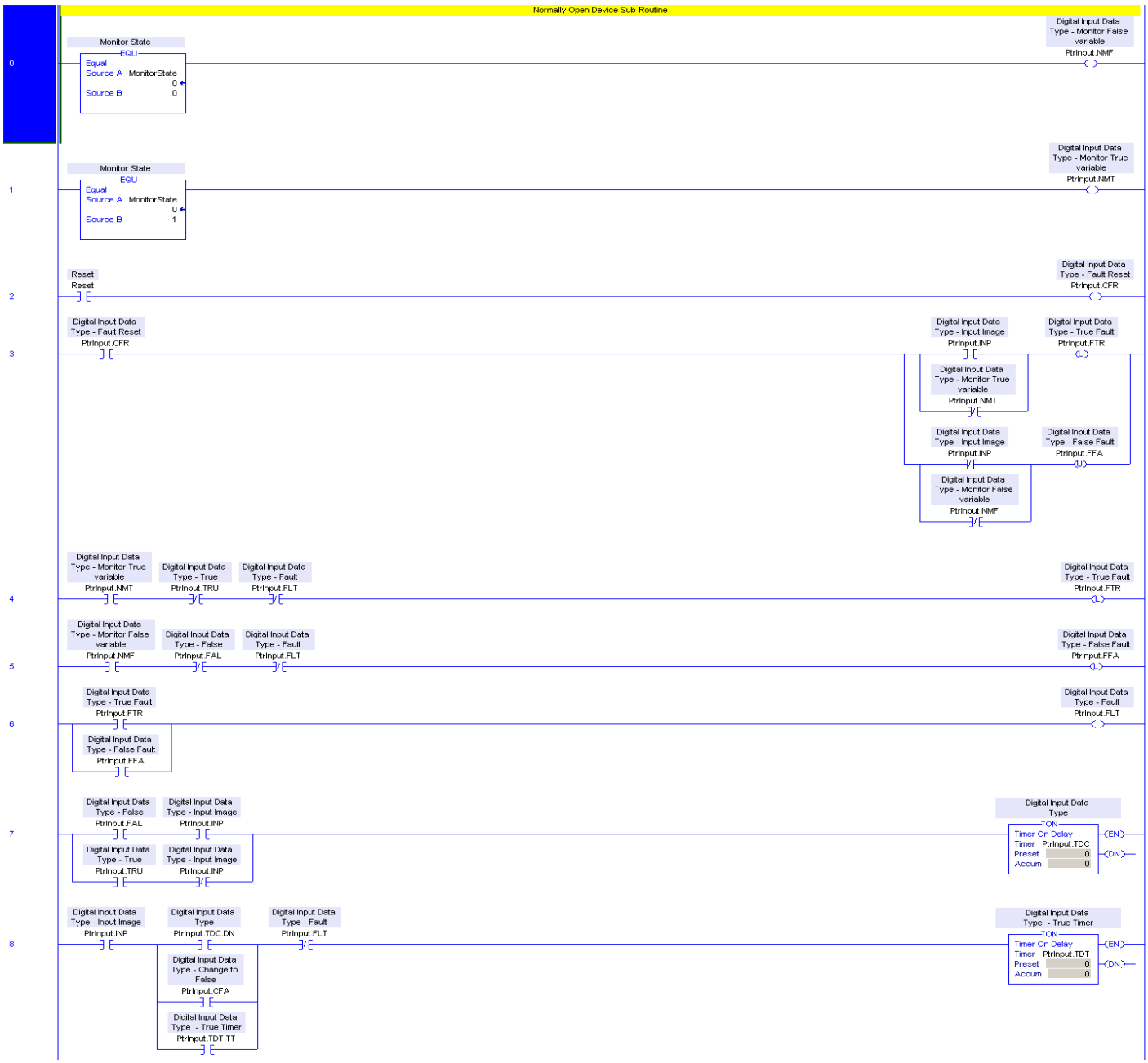
5.1 Add-On Instructions

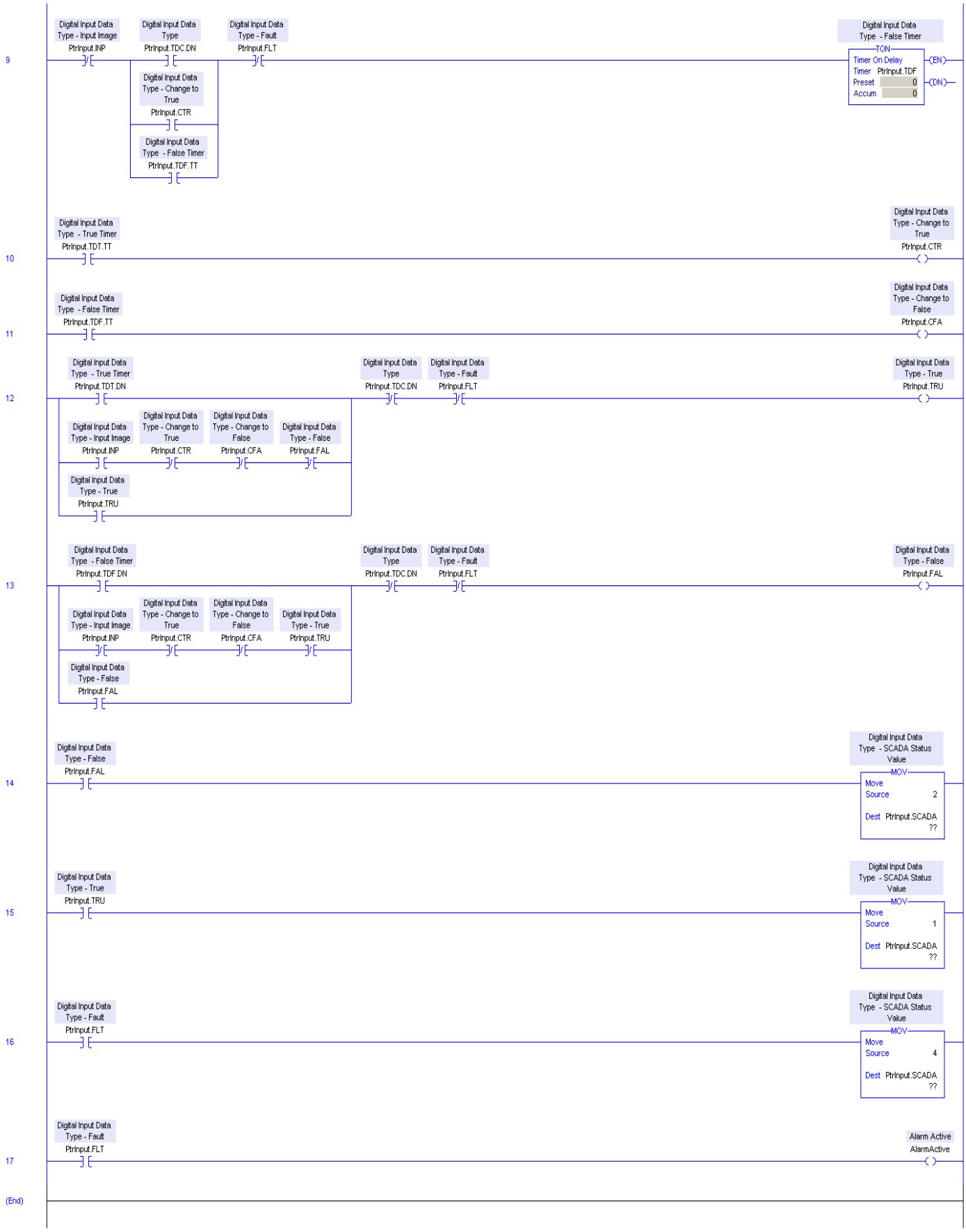
5.1.1 InputDigital

5.1.1.1 Parameters And Local Tags

Name	Usage	Default	Force Mask	Style	Data Type	Description	Constant
AlarmActive	Output	0		Decimal	BOOL	Alarm Active	☐
EnableIn	Input	1		Decimal	BOOL	Enable Input - System Defined Parameter	☐
EnableOut	Output	0		Decimal	BOOL	Enable Output - System Defined Parameter	☐
⊞ MonitorState	Input	0		Decimal	SINT	Monitor State	☐
⊞ PtrInput	InOut	{...}	{...}		UDTDigitalInput	Digital Input Data Type	☐
Reset	Input	0		Decimal	BOOL	Reset	☐

5.1.1.2 Logic



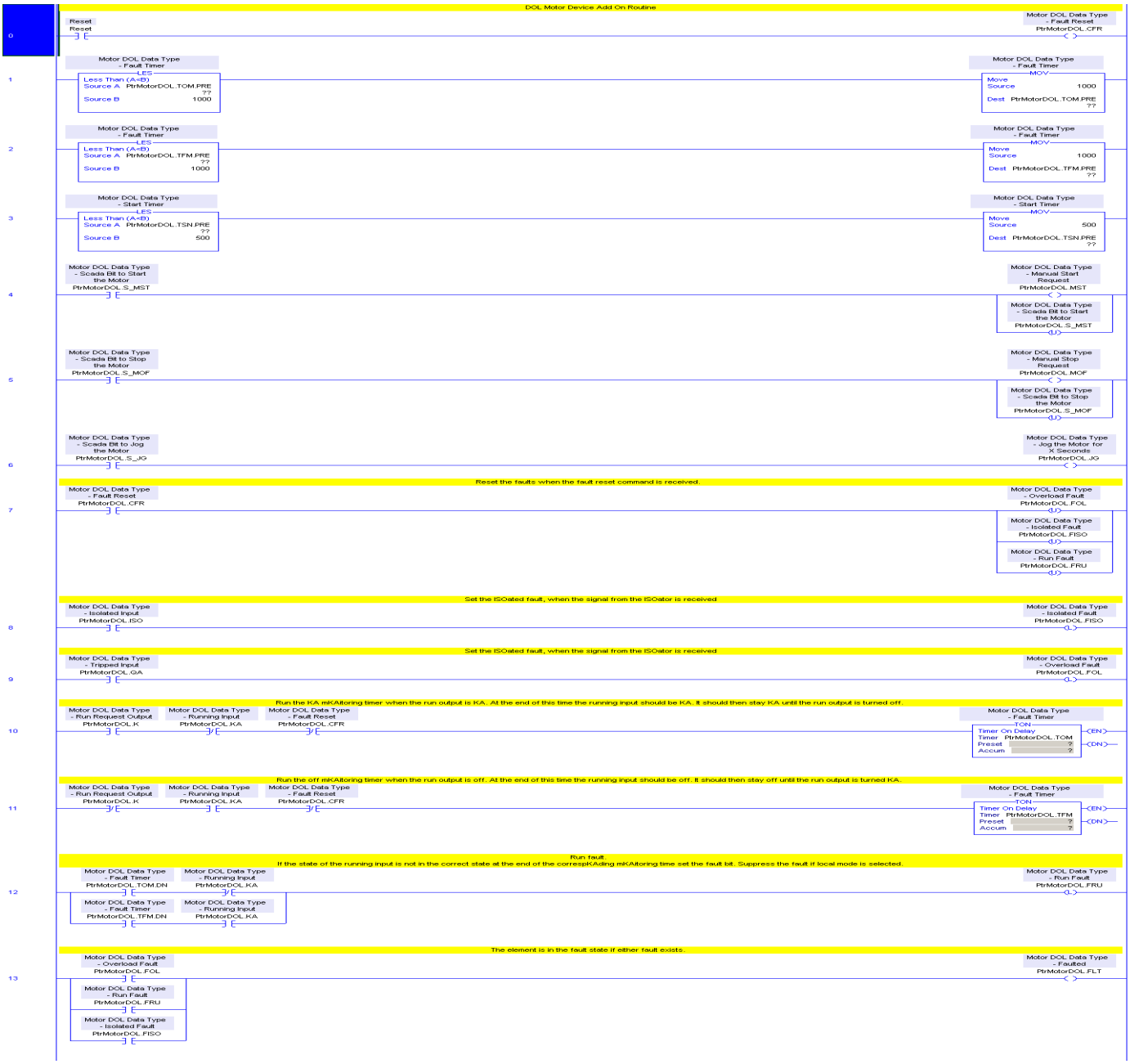

9
Digital Input Data Type - Input Image PtrInput.INP
Digital Input Data Type PtrInput.TDC.DN
Digital Input Data Type - Fault PtrInput.FLT
Digital Input Data Type - Change to True PtrInput.CTR
Digital Input Data Type - False Timer PtrInput.TDF.TT
Digital Input Data Type - False Timer
TON
Timer On Delay
Timer PtrInput.TDF
Preset 0
Accum 0
EN
DN
10
Digital Input Data Type - True Timer PtrInput.TDT.TT
Digital Input Data Type - Change to True PtrInput.CTR
11
Digital Input Data Type - False Timer PtrInput.TDF.TT
Digital Input Data Type - Change to False PtrInput.CFA
12
Digital Input Data Type - True Timer PtrInput.TDT.DN
Digital Input Data Type PtrInput.TDC.DN
Digital Input Data Type - Fault PtrInput.FLT
Digital Input Data Type - True PtrInput.TRU
Digital Input Data Type - Input Image PtrInput.INP
Digital Input Data Type - Change to True PtrInput.CTR
Digital Input Data Type - Change to False PtrInput.CFA
Digital Input Data Type - False PtrInput.FAL
Digital Input Data Type - True PtrInput.TRU
13
Digital Input Data Type - False Timer PtrInput.TDF.DN
Digital Input Data Type PtrInput.TDC.DN
Digital Input Data Type - Fault PtrInput.FLT
Digital Input Data Type - False PtrInput.FAL
Digital Input Data Type - Input Image PtrInput.INP
Digital Input Data Type - Change to True PtrInput.CTR
Digital Input Data Type - Change to False PtrInput.CFA
Digital Input Data Type - True PtrInput.TRU
Digital Input Data Type - False PtrInput.FAL
14
Digital Input Data Type - False PtrInput.FAL
Digital Input Data Type - SCADA Status Value
MOV
Move
Source 2
Dest PtrInput.SCADA ??
15
Digital Input Data Type - True PtrInput.TRU
Digital Input Data Type - SCADA Status Value
MOV
Move
Source 1
Dest PtrInput.SCADA ??
16
Digital Input Data Type - Fault PtrInput.FLT
Digital Input Data Type - SCADA Status Value
MOV
Move
Source 4
Dest PtrInput.SCADA ??
17
Digital Input Data Type - Fault PtrInput.FLT
Alarm Active AlarmActive
(End)

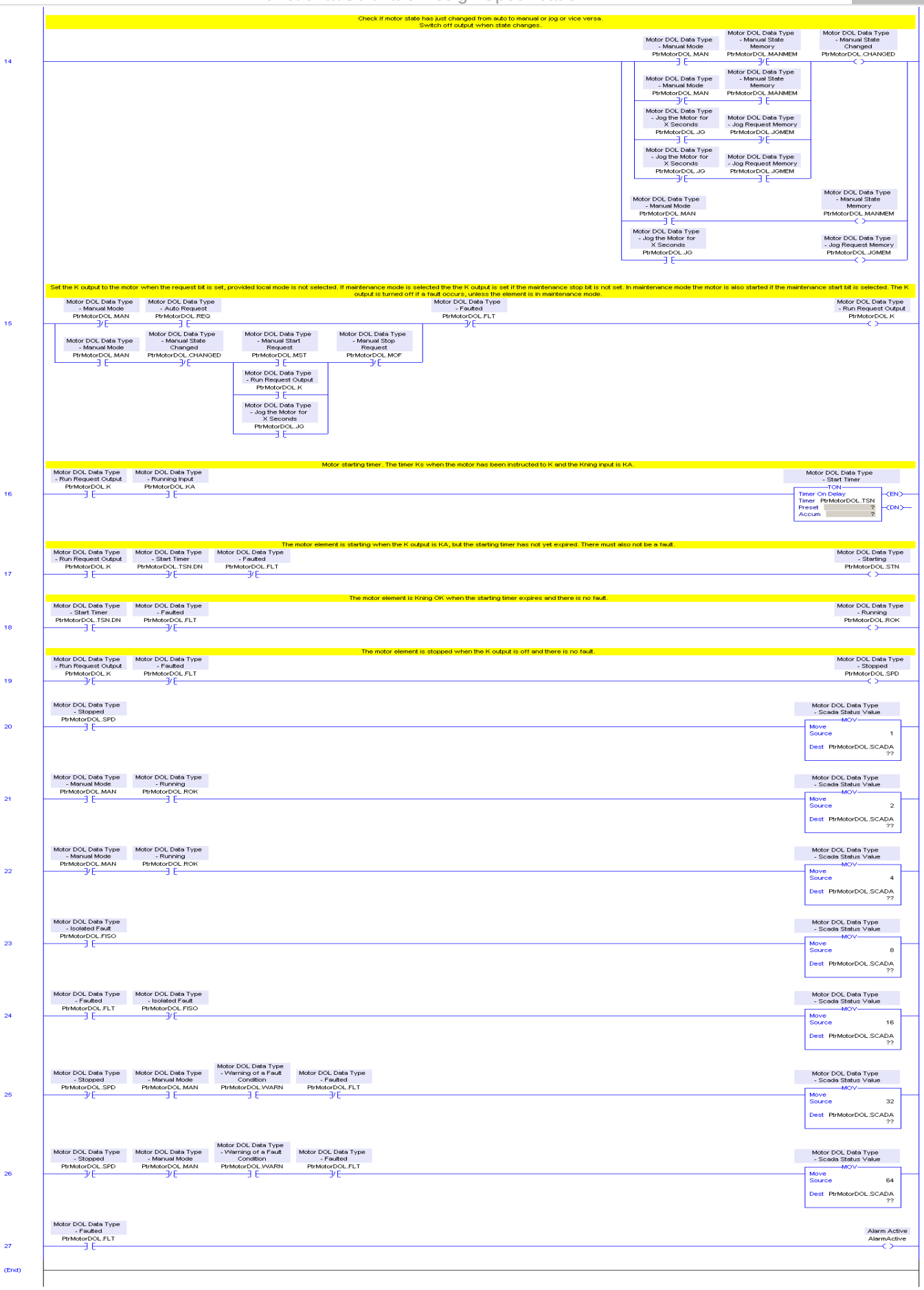
5.1.2 Motor1S1D

5.1.2.1 Parameters And Local Tags

Name	Usage	Default	Force Mask	Style	Data Type	Description	Constant
AlarmActive	Output	0		Decimal	BOOL	Alarm Active	☐
EnableIn	Input	1		Decimal	BOOL	Enable Input - System Defined Parameter	☐
EnableOut	Output	0		Decimal	BOOL	Enable Output - System Defined Parameter	☐
⊞ PtrMotorDOL	InOut	{...}	{...}		UDTMotorDOL	Motor DOL Data Type	☐
Reset	Input	0		Decimal	BOOL	Reset	☐

5.1.2.2 Logic



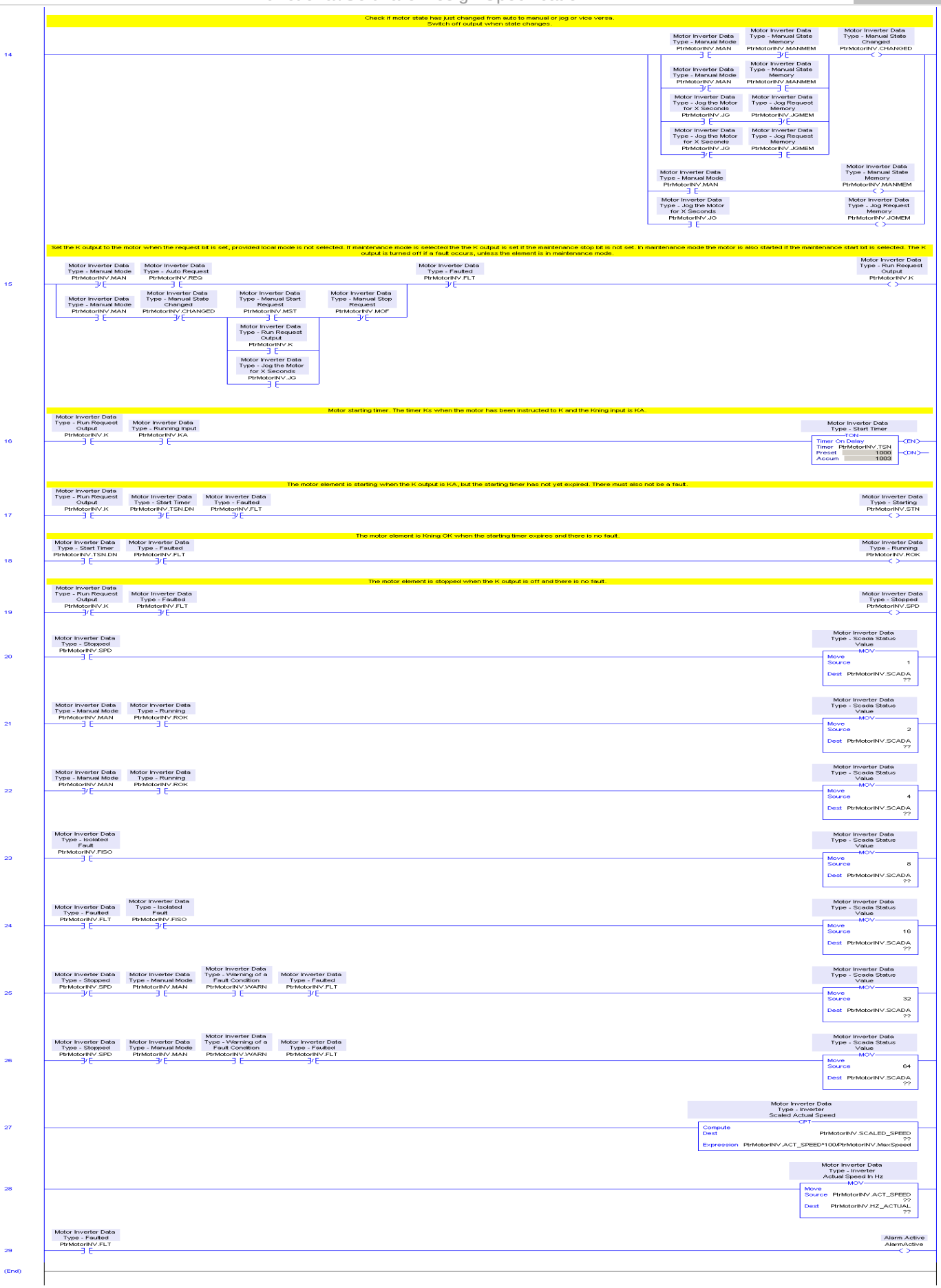

Check if motor state has just changed from auto to manual or jog or vice versa.
Switch off output when state changes.
Motor DOL Data Type - Manual Mode PtrMotorDOL.MAN
Motor DOL Data Type - Manual State Memory PtrMotorDOL.MANMEM
Motor DOL Data Type - Manual State Changed PtrMotorDOL.CHANGED
Motor DOL Data Type - Jog the Motor for X Seconds PtrMotorDOL.JG
Motor DOL Data Type - Jog Request Memory PtrMotorDOL.JGMEM
Set the K output to the motor when the request bit is set, provided local mode is not selected. If maintenance mode is selected the the K output is set if the maintenance stop bit is not set. In maintenance mode the motor is also started if the maintenance start bit is selected. The K output is turned off if a fault occurs, unless the element is in maintenance mode.
Motor DOL Data Type - Auto Request PtrMotorDOL.REQ
Motor DOL Data Type - Faulted PtrMotorDOL.FLT
Motor DOL Data Type - Run Request Output PtrMotorDOL.K
Motor DOL Data Type - Manual State Changed PtrMotorDOL.CHANGED
Motor DOL Data Type - Manual Start Request PtrMotorDOL.MST
Motor DOL Data Type - Manual Stop Request PtrMotorDOL.MOF
Motor DOL Data Type - Jog the Motor for X Seconds PtrMotorDOL.JG
Motor starting timer. The timer Ks when the motor has been instructed to K and the Kning input is KA.
Motor DOL Data Type - Running Input PtrMotorDOL.KA
Motor DOL Data Type - Start Timer
TON
Timer On Delay
Timer PtrMotorDOL.TSN
Preset ?
Accum ?
EN
DN
The motor element is starting when the K output is KA, but the starting timer has not yet expired. There must also not be a fault.
Motor DOL Data Type - Start Timer PtrMotorDOL.TSN.DN
Motor DOL Data Type - Starting PtrMotorDOL.STN
The motor element is Kning OK when the starting timer expires and there is no fault.
Motor DOL Data Type - Running PtrMotorDOL.ROK
The motor element is stopped when the K output is off and there is no fault.
Motor DOL Data Type - Stopped PtrMotorDOL.SPD
Motor DOL Data Type - Isolated Fault PtrMotorDOL.FISO
Motor DOL Data Type - Warning of a Fault Condition PtrMotorDOL.WARN
Motor DOL Data Type - Scada Status Value
MOV
Move
Source 1
Source 2
Source 4
Source 8
Source 16
Source 32
Source 64
Dest PtrMotorDOL.SCADA ??
Alarm Active AlarmActive
(End)

5.1.3 MotorPowerFlexInv

5.1.3.1 Parameters And Local Tags

Name	Usage	Default	Force Mask	Style	Data Type	Description	Constant
AlarmActive	Output	0		Decimal	BOOL	Alarm Active	☐
EnableIn	Input	1		Decimal	BOOL	Enable Input - System Defined Parameter	☐
EnableOut	Output	0		Decimal	BOOL	Enable Output - System Defined Parameter	☐
⊞ PtrMotorINV	InOut	{...}	{...}		UDTPowerflexINV	Motor Inverter Data Type	☐
Reset	Input	0		Decimal	BOOL	Reset	☐

5.1.3.2 Logic

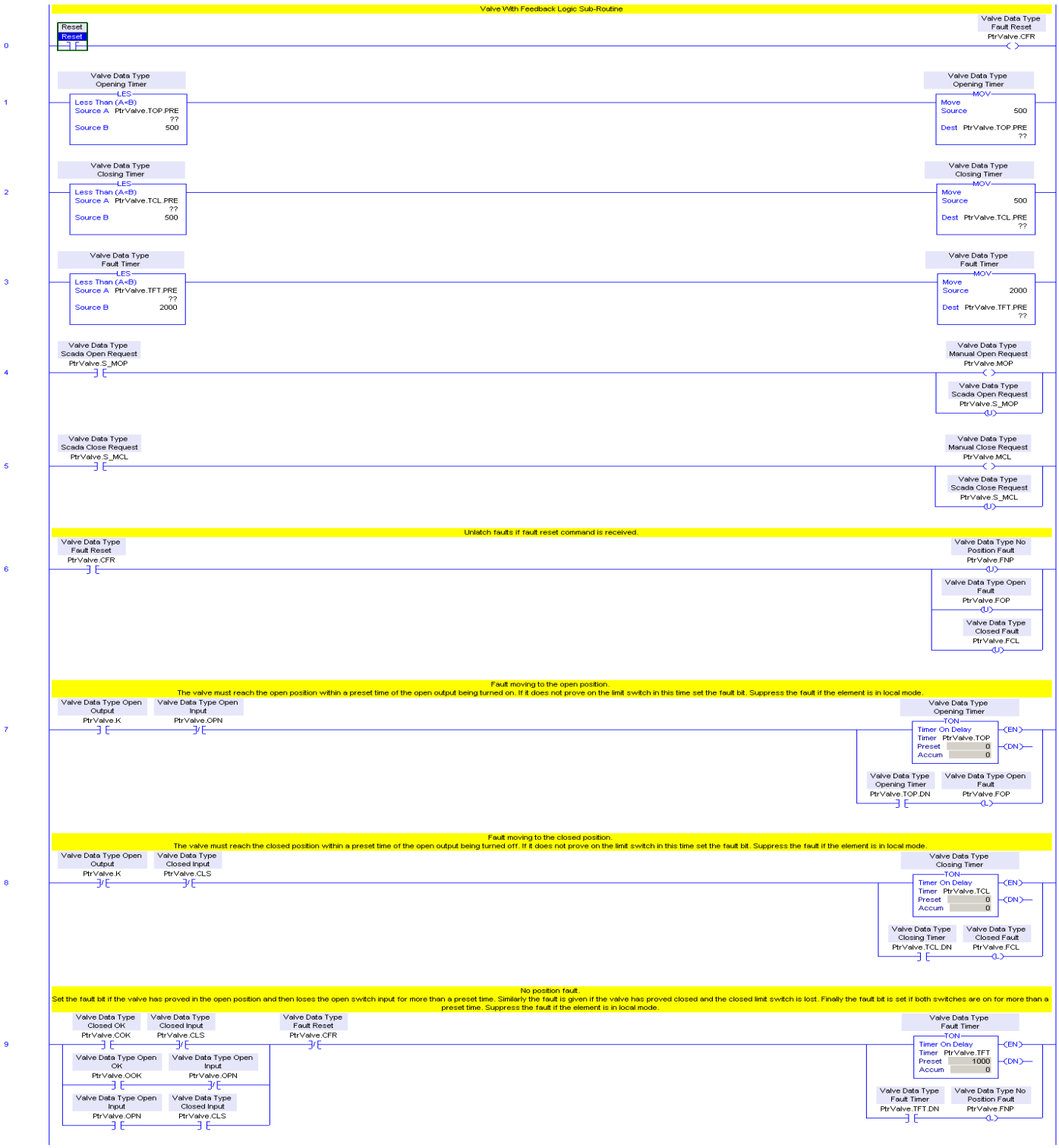

Check if motor state has just changed from auto to manual or jog or vice versa.
Switch off output when state changes.
14
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Manual State Memory PtrMotorINV.MANMEM
Motor Inverter Data Type - Manual State Changed PtrMotorINV.CHANGED
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Manual State Memory PtrMotorINV.MANMEM
Motor Inverter Data Type - Jog the Motor for X Seconds PtrMotorINV.JG
Motor Inverter Data Type - Jog Request Memory PtrMotorINV.JGMEM
Motor Inverter Data Type - Jog the Motor for X Seconds PtrMotorINV.JG
Motor Inverter Data Type - Jog Request Memory PtrMotorINV.JGMEM
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Manual State Memory PtrMotorINV.MANMEM
Motor Inverter Data Type - Jog the Motor for X Seconds PtrMotorINV.JG
Motor Inverter Data Type - Jog Request Memory PtrMotorINV.JGMEM
Set the K output to the motor when the request bit is set, provided local mode is not selected. If maintenance mode is selected the the K output is set if the maintenance stop bit is not set. In maintenance mode the motor is also started if the maintenance start bit is selected. The K output is turned off if a fault occurs, unless the element is in maintenance mode.
15
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Auto Request PtrMotorINV.REQ
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Motor Inverter Data Type - Run Request Output PtrMotorINV.K
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Manual State Changed PtrMotorINV.CHANGED
Motor Inverter Data Type - Manual Start Request PtrMotorINV.MST
Motor Inverter Data Type - Manual Stop Request PtrMotorINV.MOF
Motor Inverter Data Type - Run Request Output PtrMotorINV.K
Motor Inverter Data Type - Jog the Motor for X Seconds PtrMotorINV.JG
Motor starting timer. The timer Ks when the motor has been instructed to K and the Kning input is KA.
16
Motor Inverter Data Type - Run Request Output PtrMotorINV.K
Motor Inverter Data Type - Running Input PtrMotorINV.KA
Motor Inverter Data Type - Start Timer
TON
Timer On Delay
Timer PtrMotorINV.TSN
Preset 1000
Accum 1003
EN
DN
The motor element is starting when the K output is KA, but the starting timer has not yet expired. There must also not be a fault.
17
Motor Inverter Data Type - Run Request Output PtrMotorINV.K
Motor Inverter Data Type - Start Timer PtrMotorINV.TSN.DN
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Motor Inverter Data Type - Starting PtrMotorINV.STN
The motor element is Kning OK when the starting timer expires and there is no fault.
18
Motor Inverter Data Type - Start Timer PtrMotorINV.TSN.DN
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Motor Inverter Data Type - Running PtrMotorINV.ROK
The motor element is stopped when the K output is off and there is no fault.
19
Motor Inverter Data Type - Run Request Output PtrMotorINV.K
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Motor Inverter Data Type - Stopped PtrMotorINV.SPD
20
Motor Inverter Data Type - Stopped PtrMotorINV.SPD
Motor Inverter Data Type - Scada Status Value
MOV
Move
Source 1
Dest PtrMotorINV.SCADA ??
21
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Running PtrMotorINV.ROK
Motor Inverter Data Type - Scada Status Value
MOV
Move
Source 2
Dest PtrMotorINV.SCADA ??
22
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Running PtrMotorINV.ROK
Motor Inverter Data Type - Scada Status Value
MOV
Move
Source 4
Dest PtrMotorINV.SCADA ??
23
Motor Inverter Data Type - Isolated Fault PtrMotorINV.FISO
Motor Inverter Data Type - Scada Status Value
MOV
Move
Source 8
Dest PtrMotorINV.SCADA ??
24
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Motor Inverter Data Type - Isolated Fault PtrMotorINV.FISO
Motor Inverter Data Type - Scada Status Value
MOV
Move
Source 16
Dest PtrMotorINV.SCADA ??
25
Motor Inverter Data Type - Stopped PtrMotorINV.SPD
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Warning or a Fault Condition PtrMotorINV.WARN
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Motor Inverter Data Type - Scada Status Value
MOV
Move
Source 32
Dest PtrMotorINV.SCADA ??
26
Motor Inverter Data Type - Stopped PtrMotorINV.SPD
Motor Inverter Data Type - Manual Mode PtrMotorINV.MAN
Motor Inverter Data Type - Warning or a Fault Condition PtrMotorINV.WARN
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Motor Inverter Data Type - Scada Status Value
MOV
Move
Source 64
Dest PtrMotorINV.SCADA ??
27
Motor Inverter Data Type - Inverter Scaled Actual Speed
CPT
Compute
Dest PtrMotorINV.SCALED_SPEED ??
Expression PtrMotorINV.ACT_SPEED*100/PtrMotorINV.MaxSpeed
28
Motor Inverter Data Type - Inverter Actual Speed In Hz
MOV
Move
Source PtrMotorINV.ACT_SPEED ??
Dest PtrMotorINV.HZ_ACTUAL ??
29
Motor Inverter Data Type - Faulted PtrMotorINV.FLT
Alarm Active AlarmActive
(End)

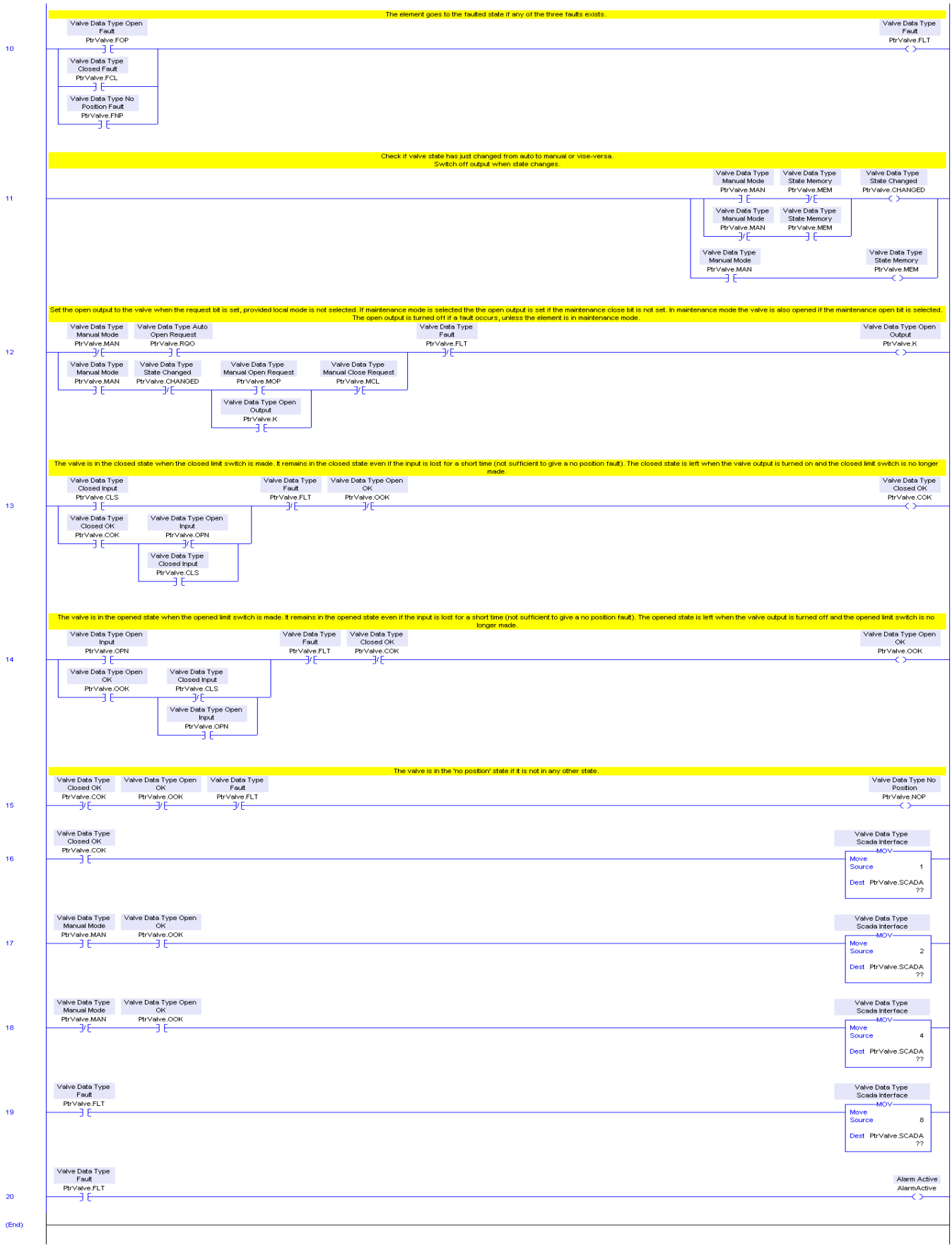
5.1.4 ValveLimits

5.1.4.1 Parameters And Local Tags

Name	Usage	Default	Force Mask	Style	Data Type	Description	Constant
AlarmActive	Output	0		Decimal	BOOL	Alarm Active	☐
EnableIn	Input	1		Decimal	BOOL	Enable Input - System Defined Parameter	☐
EnableOut	Output	0		Decimal	BOOL	Enable Output - System Defined Parameter	☐
⊞ PtrValve	InOut	{...}	{...}		UDTValve	Valve Data Type	☐
Reset	Input	0		Decimal	BOOL	Reset	☐

5.1.4.2 Logic



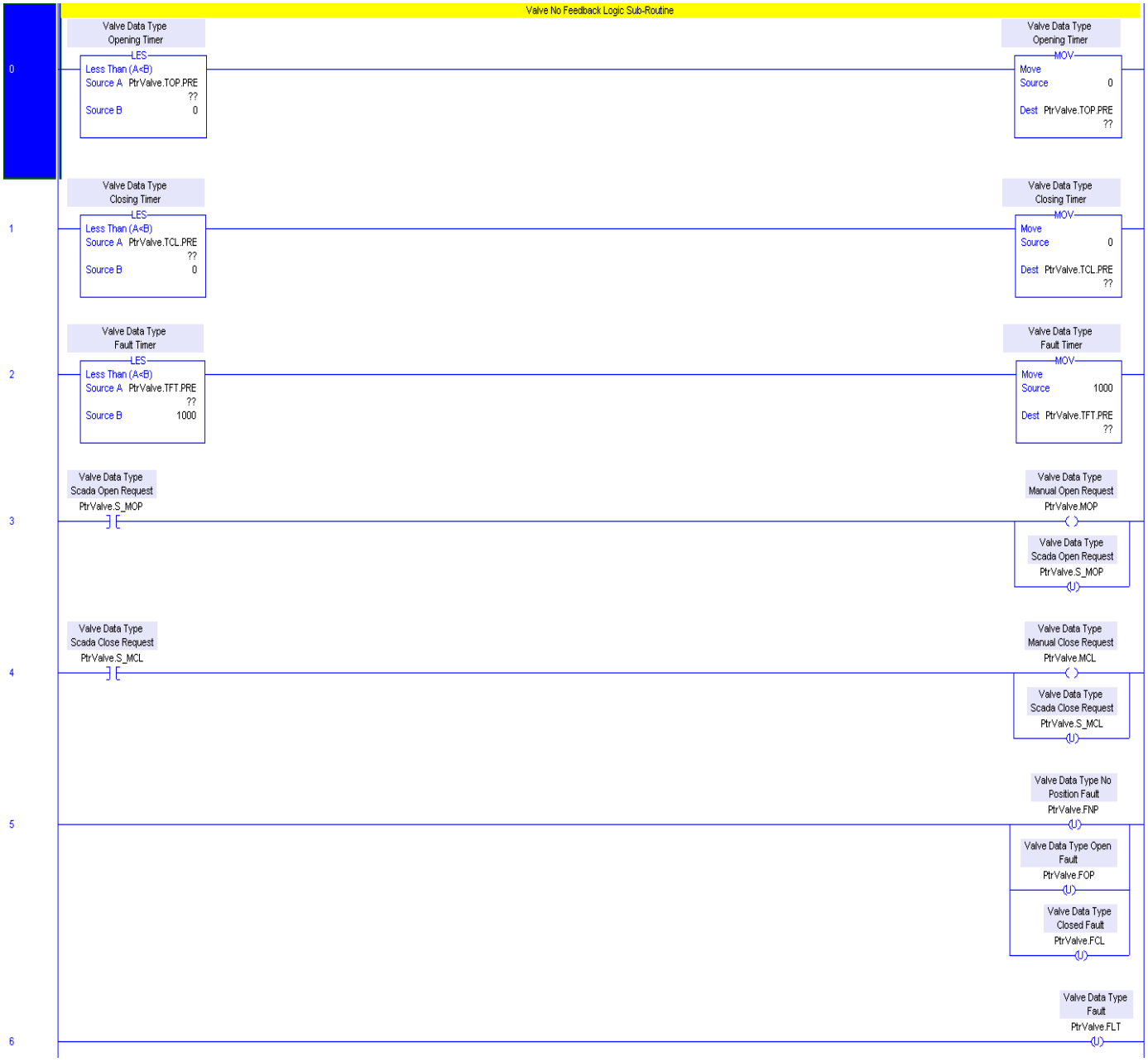

The element goes to the faulted state if any of the three faults exists.
Valve Data Type Open Fault PtrValve.FOP
Valve Data Type Closed Fault PtrValve.FCL
Valve Data Type No Position Fault PtrValve.FNP
Valve Data Type Fault PtrValve.FLT
Check if valve state has just changed from auto to manual or vise-versa.
Switch off output when state changes.
Valve Data Type Manual Mode PtrValve.MAN
Valve Data Type State Memory PtrValve.MEM
Valve Data Type State Changed PtrValve.CHANGED
Set the open output to the valve when the request bit is set, provided local mode is not selected. If maintenance mode is selected the the open output is set if the maintenance close bit is not set. In maintenance mode the valve is also opened if the maintenance open bit is selected. The open output is turned off if a fault occurs, unless the element is in maintenance mode.
Valve Data Type Auto Open Request PtrValve.RQO
Valve Data Type Manual Open Request PtrValve.MOP
Valve Data Type Manual Close Request PtrValve.MCL
Valve Data Type Open Output PtrValve.K
The valve is in the closed state when the closed limit switch is made. It remains in the closed state even if the input is lost for a short time (not sufficient to give a no position fault). The closed state is left when the valve output is turned on and the closed limit switch is no longer made.
Valve Data Type Closed Input PtrValve.CLS
Valve Data Type Closed OK PtrValve.COK
Valve Data Type Open Input PtrValve.OPN
Valve Data Type Open OK PtrValve.OOK
The valve is in the opened state when the opened limit switch is made. It remains in the opened state even if the input is lost for a short time (not sufficient to give a no position fault). The opened state is left when the valve output is turned off and the opened limit switch is no longer made.
The valve is in the 'no position' state if it is not in any other state.
Valve Data Type No Position PtrValve.NOP
Valve Data Type Scada Interface
MOV
Move Source 1 Dest PtrValve.SCADA ??
Move Source 2 Dest PtrValve.SCADA ??
Move Source 4 Dest PtrValve.SCADA ??
Move Source 8 Dest PtrValve.SCADA ??
Alarm Active AlarmActive
(End)

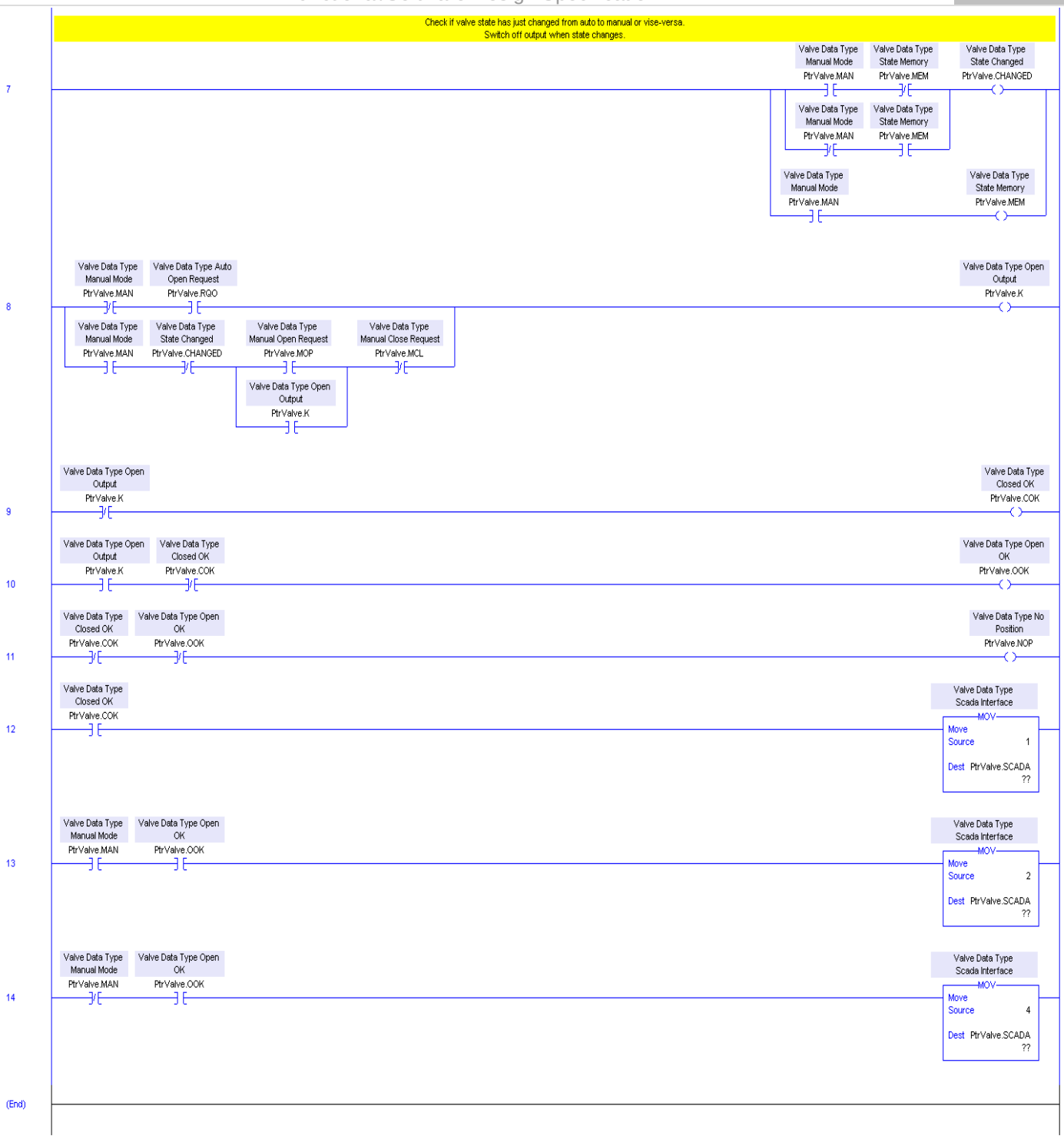
5.1.5 ValveNoLimits

5.1.5.1 Parameters And Local Tags

Name	Usage	Default	Force Mask	Style	Data Type	Description	Constant
EnableIn	Input	1		Decimal	BOOL	Enable Input - System Defined Parameter	☐
EnableOut	Output	0		Decimal	BOOL	Enable Output - System Defined Parameter	☐
⊞ PtrValve	InOut	{...}	{...}		UDTValve	Valve Data Type	☐

5.1.5.2 Logic




Check if valve state has just changed from auto to manual or vise-versa.
Switch off output when state changes.
7
Valve Data Type Manual Mode PtrValve.MAN
Valve Data Type State Memory PtrValve.MEM
Valve Data Type State Changed PtrValve.CHANGED
8
Valve Data Type Manual Mode PtrValve.MAN
Valve Data Type Auto Open Request PtrValve.RQO
Valve Data Type State Changed PtrValve.CHANGED
Valve Data Type Manual Open Request PtrValve.MOP
Valve Data Type Manual Close Request PtrValve.MCL
Valve Data Type Open Output PtrValve.K
9
Valve Data Type Open Output PtrValve.K
Valve Data Type Closed OK PtrValve.COK
10
Valve Data Type Open Output PtrValve.K
Valve Data Type Closed OK PtrValve.COK
Valve Data Type Open OK PtrValve.OOK
11
Valve Data Type Closed OK PtrValve.COK
Valve Data Type Open OK PtrValve.OOK
Valve Data Type No Position PtrValve.NOP
12
Valve Data Type Closed OK PtrValve.COK
Valve Data Type Scada Interface
MOV
Move
Source 1
Dest PtrValve.SCADA ??
13
Valve Data Type Manual Mode PtrValve.MAN
Valve Data Type Open OK PtrValve.OOK
Valve Data Type Scada Interface
MOV
Move
Source 2
Dest PtrValve.SCADA ??
14
Valve Data Type Manual Mode PtrValve.MAN
Valve Data Type Open OK PtrValve.OOK
Valve Data Type Scada Interface
MOV
Move
Source 4
Dest PtrValve.SCADA ??
(End)

5.2 User-Defined Tags

5.2.1 UDTAlarmHandler

Members: Data Type Size: 304 byte(s)

Name	Data Type	Style	Description	External Access
Current	DINT[32]	Decimal		Read/Write
Last	DINT[32]	Decimal		Read/Write
⊞ Mute_Timer	TIMER			Read/Write
⊞ Half_Sec_Timer	TIMER			Read/Write
Alarm_Pointer	DINT	Decimal		Read/Write
DINT_All_ON	DINT	Decimal		Read/Write
Alarm_Check1	DINT	Decimal		Read/Write
Alarm_Check2	DINT	Decimal		Read/Write
Alarm_Check3	DINT	Decimal		Read/Write
New_Fault_Detected	BOOL	Decimal		Read/Write
A_Fault_Exists	BOOL	Decimal		Read/Write
Alarm_Mute	BOOL	Decimal		Read/Write
Fault_Reset	BOOL	Decimal		Read/Write
Beacon_Req	BOOL	Decimal		Read/Write
Horn_Req	BOOL	Decimal		Read/Write
Start_Warning	BOOL	Decimal		Read/Write
Flash_Half_Sec	BOOL	Decimal		Read/Write
Flash_Half_Sec_Mem	BOOL	Decimal		Read/Write

5.2.2 UDTAnalogInput

Members: Data Type Size: 28 byte(s)

Name	Data Type	Style	Description	External Access
REAL	REAL	Float	- Scaled Value	Read/Write
EWMA	REAL	Float	- FilteredValue	Read/Write
EWMA_RATIO	REAL	Float	- Filter Ratio	Read/Write
MAX_Value	REAL	Float	- Max Value	Read/Write
MIN_Value	REAL	Float	- Min Value	Read/Write
Scale	REAL	Float	- the scale factor	Read/Write
Actual	REAL	Float	- Actual Faceplate Value	Read/Write

5.2.3 UDTAnalogOutput

Members: Data Type Size: 200 byte(s)

Name	Data Type	Style	Description	External Access
OUTPUT	DINT	Decimal	- Output Value	Read/Write
CV	DINT	Decimal	- Control Variable	Read/Write
⊞ PID	PID		- PID Block	Read/Write
MAN	BOOL	Decimal	- Manual Mode	Read/Write
SCADA_INPUT_	REAL	Float	- Scada Input Position	Read/Write

5.2.4 UDTBurner

Name	Data Type	Style	Description	External Access
AutoRequest	BOOL	Decimal		Read/Write
BoilOutMode	BOOL	Decimal		Read/Write
BoilOutInhibit	BOOL	Decimal		Read/Write
BoilOutTemperatureSP	REAL	Float		Read/Write
DamperChecks	BOOL	Decimal		Read/Write
DamperChecksComplete	BOOL	Decimal		Read/Write
DamperChecksFail	BOOL	Decimal		Read/Write
⊞ DamperChecksOpenTimer	TIMER		Check from Closed to Open Positions	Read/Write
⊞ DamperChecksCloseTimer	TIMER		Check from Open to Closed Positions	Read/Write
⊞ DamperCheckInitialOpenTimer	TIMER		Force dampers into a known position	Read/Write
Enable	BOOL	Decimal		Read/Write
Enable2	BOOL	Decimal		Read/Write
FansRunOnTime	REAL	Float		Read/Write
FreshAirDamperRequest	BOOL	Decimal		Read/Write
FireDamperRequest	BOOL	Decimal		Read/Write
FuelOil_Selected	BOOL	Decimal		Read/Write
FryMode	BOOL	Decimal		Read/Write
FryTemperatureSP	REAL	Float		Read/Write
Gas_Selected	BOOL	Decimal		Read/Write
HighBoilOutTemperature	BOOL	Decimal		Read/Write
⊞ InAutoOffTimer	TIMER			Read/Write
ManualRequest	BOOL	Decimal		Read/Write
OilSetpoint	REAL	Float		Read/Write
PreHeatMode	BOOL	Decimal		Read/Write
PreHeatTemperatureSP	REAL	Float		Read/Write
⊞ PS3508_Rise_Timer	TIMER			Read/Write
⊞ PS3508_Fall_Timer	TIMER			Read/Write
Run	BOOL	Decimal		Read/Write
Shutdown	BOOL	Decimal		Read/Write
ShutdownFault	BOOL	Decimal		Read/Write
StartEnable	BOOL	Decimal		Read/Write
⊞ StartupTimer	TIMER			Read/Write

	ShutdownRequest	BOOL	Decimal		Read/Write
⊞	ShutdownRequestTimer	TIMER			Read/Write
⊞	ShutdownTimer	TIMER			Read/Write
	StackDamperRequest	BOOL	Decimal		Read/Write
	StartStatus	REAL	Float		Read/Write
	Started	BOOL	Decimal		Read/Write
	Status	DINT	Decimal		Read/Write
	TemperatureInterlocks	BOOL	Decimal		Read/Write
	ValveChecks	BOOL	Decimal		Read/Write
	ValveChecksComplete	BOOL	Decimal		Read/Write
	ValveChecksFail	BOOL	Decimal		Read/Write
⊞	ValveChecksOpenTimer	TIMER			Read/Write
⊞	ValveChecksCloseTimer	TIMER			Read/Write
	VentCloseRequest	BOOL	Decimal		Read/Write
	ValvesHealthy	BOOL	Decimal		Read/Write
	VentChecks	BOOL	Decimal		Read/Write
	VentClose	BOOL	Decimal		Read/Write
	VentFault	BOOL	Decimal		Read/Write
⊞	VentOpenTimer	TIMER			Read/Write
10f 010					

5.2.5 UDTCombustionManagement

	Name	Data Type	Style	Description	External Access
	Damper_StartUpRunOK	BOOL	Decimal		Read/Write
	Damper_AutoCmd	BOOL	Decimal		Read/Write
	Damper_ManualCmd	BOOL	Decimal		Read/Write
	Damper_AutoSelected	BOOL	Decimal		Read/Write
	PollFan_StartUpRunOK	BOOL	Decimal		Read/Write
	PollFan_AutoCmd	BOOL	Decimal		Read/Write
	PollFan_ManualCmd	BOOL	Decimal		Read/Write
	PollFan_AutoSelected	BOOL	Decimal		Read/Write
	PollTemp_StartUpRunOK	BOOL	Decimal		Read/Write
	PollTemp_AutoCmd	BOOL	Decimal		Read/Write
	PollTemp_ManualCmd	BOOL	Decimal		Read/Write
	PollTemp_AutoSelected	BOOL	Decimal		Read/Write
	ExcessAir_StartUpRunOK	BOOL	Decimal		Read/Write
	ExcessAir_AutoCmd	BOOL	Decimal		Read/Write
	ExcessAir_ManualCmd	BOOL	Decimal		Read/Write
	ExcessAir_AutoSelected	BOOL	Decimal		Read/Write
	LookUp_StartUpRunOK	BOOL	Decimal		Read/Write
	LookUp_AutoCmd	BOOL	Decimal		Read/Write
	LookUp_ManualCmd	BOOL	Decimal		Read/Write
	LookUp_AutoSelected	BOOL	Decimal		Read/Write
	RecircFan_StartUpRunOK	BOOL	Decimal		Read/Write
	RecircFan_AutoCmd	BOOL	Decimal		Read/Write
	RecircFan_ManualCmd	BOOL	Decimal		Read/Write
	RecircFan_AutoSelected	BOOL	Decimal		Read/Write
	EfficiencySP_StartUpRunOK	BOOL	Decimal		Read/Write
	EfficiencySP_AutoCmd	BOOL	Decimal		Read/Write
	EfficiencySP_ManualCmd	BOOL	Decimal		Read/Write
	EfficiencySP_AutoSelected	BOOL	Decimal		Read/Write

⊞ Damper_AutoTimer	TIMER			Read/Write
⊞ Damper_ManualTimer	TIMER			Read/Write
⊞ PollFan_AutoTimer	TIMER			Read/Write
⊞ PollFan_ManualTimer	TIMER			Read/Write
⊞ PollTemp_AutoTimer	TIMER			Read/Write
⊞ PollTemp_ManualTimer	TIMER			Read/Write
⊞ ExcessAir_AutoTimer	TIMER			Read/Write
⊞ ExcessAir_ManualTimer	TIMER			Read/Write
⊞ LookUp_AutoTimer	TIMER			Read/Write
⊞ LookUp_ManualTimer	TIMER			Read/Write
⊞ RecircFan_AutoTimer	TIMER			Read/Write
⊞ RecircFan_ManualTimer	TIMER			Read/Write
⊞ EfficiencySP_AutoTimer	TIMER			Read/Write
⊞ EfficiencySP_ManualTimer	TIMER			Read/Write
InletDamperPosition	REAL	Float		Read/Write
InletDamperPositionAuto	REAL	Float		Read/Write
InletDamperPositionManual	REAL	Float		Read/Write
InletDamperPositionLUT	REAL	Float		Read/Write
InletDamperCVMax	REAL	Float		Read/Write
InletDamperCVMin	REAL	Float		Read/Write
OutletDamperPosition	REAL	Float		Read/Write
OutletDamperPositionAuto	REAL	Float		Read/Write
OutletDamperPositionManual	REAL	Float		Read/Write
OutletDamperPositionLUT	REAL	Float		Read/Write
OutletDamperCVMax	REAL	Float		Read/Write
OutletDamperCVMin	REAL	Float		Read/Write
PollutionTempSetpoint	REAL	Float		Read/Write
PollutionTempSetpointAuto	REAL	Float		Read/Write
PollutionTempSetpointManual	REAL	Float		Read/Write
PollutionFanSpeed	REAL	Float		Read/Write
PollutionFanSpeedAuto	REAL	Float		Read/Write
PollutionFanSpeedManual	REAL	Float		Read/Write
CombustionFanSpeedAuto	REAL	Float		Read/Write
CombustionFanSpeedManual	REAL	Float		Read/Write
Excess_Air	REAL	Float	Excess Air	Read/Write
Excess_Air_Manual_SP	REAL	Float		Read/Write
Combustion_Air_SP	REAL	Float		Read/Write
AirValveSP_FromPID	REAL	Float	Air Valve Output Positio	Read/Write
Air_Flow_Deviation	REAL	Float		Read/Write
Gas_Ratio	REAL	Float	Gas Ratio	Read/Write
Volume_of_Gas	REAL	Float	Mass of Gas	Read/Write
Calculated_Excess_Air	REAL	Float	Calculated Excess Air	Read/Write
Recirculation_FanSpeed	REAL	Float		Read/Write
Recirculation_FanSpeedAuto	REAL	Float		Read/Write
Recirculation_FanSpeedManual	REAL	Float		Read/Write
EfficiencySP	REAL	Float		Read/Write
EfficiencySPAuto	REAL	Float		Read/Write
EfficiencySPManual	REAL	Float		Read/Write
CarbonMonoxideSetpoint	REAL	Float		Read/Write
A_Value	REAL	Float		Read/Write
Equiv_Ratio	REAL	Float		Read/Write
GasComp1	REAL	Float		Read/Write
GasComp2	REAL	Float		Read/Write
GasComp3	REAL	Float		Read/Write
GasComp4	REAL	Float		Read/Write
GasRatioMeasured	REAL	Float		Read/Write
CV3404DamperPositionRcp1	REAL	Float		Read/Write
CV3404DamperPositionRcp2	REAL	Float		Read/Write

5.2.6 UDTCombustionManagerFunctions

Name	Data Type	Style	Description	External Access
Cp_O2	REAL	Float	Specific Heat Oxygen Function Result	Read/Write
Cp_O2_Tp	REAL	Float	Specific Heat Oxygen Temperature Variable	Read/Write
Cp_O2_a	REAL	Float	Specific Heat Oxygen Constant a	Read/Write
Cp_O2_b	REAL	Float	Specific Heat Oxygen Constant b	Read/Write
Cp_O2_c	REAL	Float	Specific Heat Oxygen Constant c	Read/Write
Cp_O2_d	REAL	Float	Specific Heat Oxygen Constant d	Read/Write
Cp_O2_e	REAL	Float	Specific Heat Oxygen Constant e	Read/Write
Cp_O2_f	REAL	Float	Specific Heat Oxygen Constant f	Read/Write
Cp_O2_g	REAL	Float	Specific Heat Oxygen Constant g	Read/Write
Cp_O2_h	REAL	Float	Specific Heat Oxygen Constant h	Read/Write
Cp_O2_i	REAL	Float	Specific Heat Oxygen Constant i	Read/Write
Cp_CO2	REAL	Float	Specific Heat Carbon Dioxide Function Result	Read/Write
Cp_CO2_Tp	REAL	Float	Specific Heat Carbon Dioxide Temperature Variable	Read/Write
Cp_CO2_a	REAL	Float	Specific Heat Carbon Dioxide Constant a	Read/Write
Cp_CO2_b	REAL	Float	Specific Heat Carbon Dioxide Constant b	Read/Write
Cp_CO2_c	REAL	Float	Specific Heat Carbon Dioxide Constant c	Read/Write
Cp_CO2_d	REAL	Float	Specific Heat Carbon Dioxide Constant d	Read/Write
Cp_CO2_e	REAL	Float	Specific Heat Carbon Dioxide Constant e	Read/Write
Cp_CO2_f	REAL	Float	Specific Heat Carbon Dioxide Constant f	Read/Write
Cp_CO2_g	REAL	Float	Specific Heat Carbon Dioxide Constant g	Read/Write
Cp_CO2_h	REAL	Float	Specific Heat Carbon Dioxide Constant h	Read/Write
Cp_CO2_i	REAL	Float	Specific Heat Carbon Dioxide Constant i	Read/Write
Cp_CO2_j	REAL	Float	Specific Heat Carbon Dioxide Constant j	Read/Write
Cp_CO2_k	REAL	Float	Specific Heat Carbon Dioxide Constant k	Read/Write
Cp_N2	REAL	Float	Specific Heat Nitrogen Function Result	Read/Write
Cp_N2_Tp	REAL	Float	Specific Heat Nitrogen Temperature Variable	Read/Write
Cp_N2_a	REAL	Float	Specific Heat Nitrogen Constant a	Read/Write
Cp_N2_b	REAL	Float	Specific Heat Nitrogen Constant b	Read/Write
Cp_N2_c	REAL	Float	Specific Heat Nitrogen Constant c	Read/Write
Cp_N2_d	REAL	Float	Specific Heat Nitrogen Constant d	Read/Write
Cp_N2_e	REAL	Float	Specific Heat Nitrogen Constant e	Read/Write
Cp_N2_f	REAL	Float	Specific Heat Nitrogen Constant f	Read/Write
Cp_N2_g	REAL	Float	Specific Heat Nitrogen Constant g	Read/Write
Cp_N2_h	REAL	Float	Specific Heat Nitrogen Constant h	Read/Write
Cp_N2_i	REAL	Float	Specific Heat Nitrogen Constant i	Read/Write
Cp_N2_j	REAL	Float	Specific Heat Nitrogen Constant j	Read/Write
Cp_N2_k	REAL	Float	Specific Heat Nitrogen Constant k	Read/Write
Cp_H2O	REAL	Float	Specific Heat Water Function Result	Read/Write
Cp_H2O_Tp	REAL	Float	Specific Heat Water Temperature Variable	Read/Write
Cp_H2O_a	REAL	Float	Specific Heat Water Constant a	Read/Write
Cp_H2O_b	REAL	Float	Specific Heat Water Constant b	Read/Write
Cp_H2O_c	REAL	Float	Specific Heat Water Constant c	Read/Write
Cp_H2O_d	REAL	Float	Specific Heat Water Constant d	Read/Write
Cp_H2O_e	REAL	Float	Specific Heat Water Constant e	Read/Write
Cp_H2O_f	REAL	Float	Specific Heat Water Constant f	Read/Write
Cp_H2O_g	REAL	Float	Specific Heat Water Constant g	Read/Write
Cp_H2O_h	REAL	Float	Specific Heat Water Constant h	Read/Write
Cp_H2O_i	REAL	Float	Specific Heat Water Constant i	Read/Write
Cp_H2O_j	REAL	Float	Specific Heat Water Constant j	Read/Write
Cp_H2O_k	REAL	Float	Specific Heat Water Constant k	Read/Write

	Cp_AIR	REAL	Float	Specific Heat Air Function Result	Read/Write
	Cp_AIR_Tp	REAL	Float	Specific Heat Air Temperature Variable	Read/Write
	Cp_AIR_a	REAL	Float	Specific Heat Air Constant a	Read/Write
	Cp_AIR_b	REAL	Float	Specific Heat Air Constant b	Read/Write
	Cp_AIR_c	REAL	Float	Specific Heat Air Constant c	Read/Write
	Cp_AIR_d	REAL	Float	Specific Heat Air Constant d	Read/Write
	Cp_AIR_e	REAL	Float	Specific Heat Air Constant e	Read/Write
	Cp_AIR_f	REAL	Float	Specific Heat Air Constant f	Read/Write
	Cp_AIR_g	REAL	Float	Specific Heat Air Constant g	Read/Write
	Cp_CH4	REAL	Float	Constant Pressure Specific Heat Methane Function Result	Read/Write
	Cp_CH4_Tp	REAL	Float	Constant Pressure Specific Heat Methane Temperature Variable	Read/Write
	Cp_CH4_Tpk	REAL	Float	Constant Pressure Specific Heat Methane Temperature Variable / 100	Read/Write
	Cp_C2H4	REAL	Float	Constant Pressure Specific Heat Ethyiene Function Result	Read/Write
	Cp_C2H4_Tp	REAL	Float	Constant Pressure Specific Heat Ethyiene Temperature Variable	Read/Write
	Cp_C2H4_Tpk	REAL	Float	Constant Pressure Specific Heat Ethyiene Temperature Variable / 100	Read/Write
	Cp_C2H6	REAL	Float	Constant Pressure Specific Heat Ethane Function Result	Read/Write
	Cp_C2H6_Tp	REAL	Float	Constant Pressure Specific Heat Ethane Temperature Variable	Read/Write
	Cp_C2H6_Tpk	REAL	Float	Constant Pressure Specific Heat Ethane Temperature Variable / 100	Read/Write
	Cp_C3H8	REAL	Float	Constant Pressure Specific Heat Propane Function Result	Read/Write
	Cp_C3H8_Tp	REAL	Float	Constant Pressure Specific Heat Propane Temperature Variable	Read/Write
	Cp_C3H8_Tpk	REAL	Float	Constant Pressure Specific Heat Propane Temperature Variable / 100	Read/Write
	Cp_C4H10	REAL	Float	Constant Pressure Specific Heat Butane Function Result	Read/Write
	Cp_C4H10_Tp	REAL	Float	Constant Pressure Specific Heat Butane Temperature Variable	Read/Write
	Cp_C4H10_Tpk	REAL	Float	Constant Pressure Specific Heat Butane Temperature Variable / 100	Read/Write
	Cpm_CO2	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Function Result	Read/Write
	Cpm_CO2_Tp	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Temperature Variable	Read/Write
	Cpm_CO2_a	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Constant a	Read/Write
	Cpm_CO2_b	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Constant b	Read/Write
	Cpm_CO2_c	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Constant c	Read/Write
	Cpm_CO2_d	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Constant d	Read/Write
	Cpm_CO2_e	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Constant e	Read/Write
	Cpm_CO2_f	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Constant f	Read/Write
	Cpm_CO2_g	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Carbon Dioxide Constant g	Read/Write
	Cpm_H2O	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Water Function Result	Read/Write
	Cpm_H2O_Tp	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Water Temperature Variable	Read/Write
	Cpm_H2O_a	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Water Constant a	Read/Write
	Cpm_H2O_b	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Water Constant b	Read/Write
	Cpm_H2O_c	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Water Constant c	Read/Write
	Cpm_H2O_d	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Water Constant d	Read/Write
	Cpm_H2O_e	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Water Constant e	Read/Write
	Cpm_N2	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Function Result	Read/Write
	Cpm_N2_Tp	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Temperature Variable	Read/Write
	Cpm_N2_a	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Constant a	Read/Write
	Cpm_N2_b	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Constant b	Read/Write
	Cpm_N2_c	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Constant c	Read/Write
	Cpm_N2_d	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Constant d	Read/Write
	Cpm_N2_e	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Constant e	Read/Write
	Cpm_N2_f	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Nitrogen Constant f	Read/Write
	Cpm_O2	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Oxygen Function Result	Read/Write
	Cpm_O2_Tp	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Oxygen Temperature Variable	Read/Write
	Cpm_O2_a	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Oxygen Constant a	Read/Write
	Cpm_O2_b	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Oxygen Constant b	Read/Write
	Cpm_O2_c	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Oxygen Constant c	Read/Write
	Cpm_O2_d	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Oxygen Constant d	Read/Write
	Cpm_O2_e	REAL	Float	Molal Constant Pressure Specific Heat (Kj/kMole.k) Oxygen Constant e	Read/Write

Rho_AIR	REAL	Float	Density Air Function Result	Read/Write
Rho_AIR_Tp	REAL	Float	Density Air Temperature Variable	Read/Write
Rho_AIR_a	REAL	Float	Density Air Constant a	Read/Write
Rho_AIR_b	REAL	Float	Density Air Constant b	Read/Write
Rho_AIR_c	REAL	Float	Density Air Constant c	Read/Write
Rho_AIR_d	REAL	Float	Density Air Constant d	Read/Write
Rho_AIR_e	REAL	Float	Density Air Constant e	Read/Write
Rho_AIR_f	REAL	Float	Density Air Constant f	Read/Write
Rho_AIR_g	REAL	Float	Density Air Constant g	Read/Write
Rho_AIR_h	REAL	Float	Density Air Constant h	Read/Write
Rho_AIR_i	REAL	Float	Density Air Constant i	Read/Write
K_AIR	REAL	Float	Conductivity Air Function Result	Read/Write
K_AIR_Tp	REAL	Float	Conductivity Air Temperature Variable	Read/Write
K_AIR_a	REAL	Float	Conductivity Air Constant a	Read/Write
K_AIR_b	REAL	Float	Conductivity Air Constant b	Read/Write
K_AIR_c	REAL	Float	Conductivity Air Constant c	Read/Write
K_AIR_d	REAL	Float	Conductivity Air Constant d	Read/Write
K_AIR_e	REAL	Float	Conductivity Air Constant e	Read/Write
K_AIR_f	REAL	Float	Conductivity Air Constant f	Read/Write
K_AIR_g	REAL	Float	Conductivity Air Constant g	Read/Write
K_AIR_h	REAL	Float	Conductivity Air Constant h	Read/Write
K_AIR_i	REAL	Float	Conductivity Air Constant i	Read/Write
hfg	REAL	Float	Latent Heat of Vaporisation of Water Function Result	Read/Write
hfg_Tp	REAL	Float	Latent Heat of Vaporisation of Water Temperature Variable	Read/Write
hfg_a	REAL	Float	Latent Heat of Vaporisation of Water Constant a	Read/Write
hfg_b	REAL	Float	Latent Heat of Vaporisation of Water Constant b	Read/Write
hfg_c	REAL	Float	Latent Heat of Vaporisation of Water Constant c	Read/Write
hfg_d	REAL	Float	Latent Heat of Vaporisation of Water Constant d	Read/Write
hfg_e	REAL	Float	Latent Heat of Vaporisation of Water Constant e	Read/Write
hfg_f	REAL	Float	Latent Heat of Vaporisation of Water Constant f	Read/Write
hf_N2	REAL	Float	Enthalpy of Formation Nitrogen Function Result	Read/Write
hf_N2_Tp	REAL	Float	Enthalpy of Formation Nitrogen Temperature Variable	Read/Write
hf_N2_T	REAL	Float	Enthalpy of Formation Nitrogen Temperature Variable (Deg K)	Read/Write
hf_N2_a	REAL	Float	Enthalpy of Formation Nitrogen Temperature Variable Constant a	Read/Write
hf_N2_b	REAL	Float	Enthalpy of Formation Nitrogen Temperature Variable Constant b	Read/Write
hf_N2_c	REAL	Float	Enthalpy of Formation Nitrogen Temperature Variable Constant c	Read/Write
hf_N2_d	REAL	Float	Enthalpy of Formation Nitrogen Temperature Variable Constant d	Read/Write
hf_O2	REAL	Float	Enthalpy of Formation Oxygen Function Result	Read/Write
hf_O2_Tp	REAL	Float	Enthalpy of Formation Oxygen Temperature Variable	Read/Write
hf_O2_T	REAL	Float	Enthalpy of Formation Oxygen Temperature Variable (Deg K)	Read/Write
hf_O2_a	REAL	Float	Enthalpy of Formation Oxygen Temperature Variable Constant a	Read/Write
hf_O2_b	REAL	Float	Enthalpy of Formation Oxygen Temperature Variable Constant b	Read/Write
hf_O2_c	REAL	Float	Enthalpy of Formation Oxygen Temperature Variable Constant c	Read/Write
hf_O2_d	REAL	Float	Enthalpy of Formation Oxygen Temperature Variable Constant d	Read/Write
hf_O2_e	REAL	Float	Enthalpy of Formation Oxygen Temperature Variable Constant e	Read/Write
hf_CO2	REAL	Float	Enthalpy of Formation Carbon Dioxide Function Result	Read/Write
hf_CO2_Tp	REAL	Float	Enthalpy of Formation Carbon Dioxide Temperature Variable	Read/Write
hf_CO2_T	REAL	Float	Enthalpy of Formation Carbon Dioxide Temperature Variable (Deg K)	Read/Write
hf_CO2_a	REAL	Float	Enthalpy of Formation Carbon Dioxide Temperature Variable Constant a	Read/Write
hf_CO2_b	REAL	Float	Enthalpy of Formation Carbon Dioxide Temperature Variable Constant b	Read/Write
hf_CO2_c	REAL	Float	Enthalpy of Formation Carbon Dioxide Temperature Variable Constant c	Read/Write
hf_CO2_d	REAL	Float	Enthalpy of Formation Carbon Dioxide Temperature Variable Constant d	Read/Write
hf_CO2_e	REAL	Float	Enthalpy of Formation Carbon Dioxide Temperature Variable Constant e	Read/Write
hf_H2O	REAL	Float	Enthalpy of Formation Water Function Result	Read/Write
hf_H2O_Tp	REAL	Float	Enthalpy of Formation Water Temperature Variable	Read/Write
hf_H2O_T	REAL	Float	Enthalpy of Formation Water Temperature Variable (Deg K)	Read/Write
hf_H2O_a	REAL	Float	Enthalpy of Formation Water Temperature Variable Constant a	Read/Write
hf_H2O_b	REAL	Float	Enthalpy of Formation Water Temperature Variable Constant b	Read/Write
hf_H2O_c	REAL	Float	Enthalpy of Formation Water Temperature Variable Constant c	Read/Write
hf_H2O_d	REAL	Float	Enthalpy of Formation Water Temperature Variable Constant d	Read/Write
hf_H2O_e	REAL	Float	Enthalpy of Formation Water Temperature Variable Constant e	Read/Write
hf_H2O_f	REAL	Float	Enthalpy of Formation Water Temperature Variable Constant f	Read/Write
hf_H2O_g	REAL	Float	Enthalpy of Formation Water Temperature Variable Constant g	Read/Write

5.2.7 UDTDigitalInput

Members: Data Type Size: 44 byte(s)

Name	Data Type	Style	Description	External Access
NMT	BOOL	Decimal	- Monitor True variable	Read/Write
NMF	BOOL	Decimal	- Monitor False variable	Read/Write
CFR	BOOL	Decimal	- Fault Reset	Read/Write
Spare1	BOOL	Decimal	- Spare BOOLEN varia	Read/Write
INP	BOOL	Decimal	- Input Image	Read/Write
FTR	BOOL	Decimal	- True Fault	Read/Write
FFA	BOOL	Decimal	- False Fault	Read/Write
FLT	BOOL	Decimal	- Fault	Read/Write
CTR	BOOL	Decimal	- Change to True	Read/Write
CFA	BOOL	Decimal	- Change to False	Read/Write
TRU	BOOL	Decimal	- True	Read/Write
FAL	BOOL	Decimal	- False	Read/Write
⊞ TDC	TIMER			Read/Write
⊞ TDT	TIMER		- True Timer	Read/Write
⊞ TDF	TIMER		- False Timer	Read/Write
SCADA	DINT	Decimal	- SCADA Status Value	Read/Write

5.2.8 UDTEng

Name	Data Type	Style	Description	External Access
StackDmp1_FDT19_1	REAL	Float		Read/Write
StackDmp1_FDT20_1	REAL	Float		Read/Write
StackDmp1_FDT21_1	REAL	Float		Read/Write
StackDmp1_FDT22_1	REAL	Float		Read/Write
StackDmp1_FDT23_1	REAL	Float		Read/Write
StackDmp1_FDT24_1	REAL	Float		Read/Write
StackDmp1_FDT25_1	REAL	Float		Read/Write
StackDmp1_FDT26_1	REAL	Float		Read/Write
StackDmp1_FDT27_1	REAL	Float		Read/Write
StackDmp1_FDT19_2	REAL	Float		Read/Write
StackDmp1_FDT20_2	REAL	Float		Read/Write
StackDmp1_FDT21_2	REAL	Float		Read/Write
StackDmp1_FDT22_2	REAL	Float		Read/Write
StackDmp1_FDT23_2	REAL	Float		Read/Write
StackDmp1_FDT24_2	REAL	Float		Read/Write
StackDmp1_FDT25_2	REAL	Float		Read/Write
StackDmp1_FDT26_2	REAL	Float		Read/Write
StackDmp1_FDT27_2	REAL	Float		Read/Write
StackDmp2_FDT19_1	REAL	Float		Read/Write
StackDmp2_FDT20_1	REAL	Float		Read/Write
StackDmp2_FDT21_1	REAL	Float		Read/Write
StackDmp2_FDT22_1	REAL	Float		Read/Write
StackDmp2_FDT23_1	REAL	Float		Read/Write
StackDmp2_FDT24_1	REAL	Float		Read/Write
StackDmp2_FDT25_1	REAL	Float		Read/Write
StackDmp2_FDT26_1	REAL	Float		Read/Write
StackDmp2_FDT27_1	REAL	Float		Read/Write
StackDmp2_FDT19_2	REAL	Float		Read/Write
StackDmp2_FDT20_2	REAL	Float		Read/Write
StackDmp2_FDT21_2	REAL	Float		Read/Write
StackDmp2_FDT22_2	REAL	Float		Read/Write
StackDmp2_FDT23_2	REAL	Float		Read/Write
StackDmp2_FDT24_2	REAL	Float		Read/Write
StackDmp2_FDT25_2	REAL	Float		Read/Write
StackDmp2_FDT26_2	REAL	Float		Read/Write
StackDmp2_FDT27_2	REAL	Float		Read/Write
PalmOilSpecificHeat	REAL	Float		Read/Write
PalmOilDensity	REAL	Float		Read/Write
OliveOilSpecificHeat	REAL	Float		Read/Write
OliveOilDensity	REAL	Float		Read/Write
CornOilSpecificHeat	REAL	Float		Read/Write
CornOilDensity	REAL	Float		Read/Write
CalibratedGasTemperature	REAL	Float		Read/Write
CalibratedAtmosphericPressure	REAL	Float		Read/Write
GasCalorificValue	REAL	Float		Read/Write

5.2.9 UDTFlowCalc

Members: Data Type Size: 80 byte(s)

Name	Data Type	Style	Description	External Access
A	REAL	Float	Cross Sectional Area Of Ducting (m2)	Read/Write
Cf	REAL	Float	Correction Factor	Read/Write
d	REAL	Float	Diameter Of Ducting (m)	Read/Write
Delta_P	REAL	Float	Differential Pressure (Pa)	Read/Write
K	REAL	Float	Constant	Read/Write
M	REAL	Float	Mass Flow Rate Of Air (kg/hr)	Read/Write
Mag	REAL	Float	Magnification Factor	Read/Write
NV	REAL	Float	Normalised Volumetric Flow Rate (Nm3/hr)	Read/Write
P	REAL	Float	Absolute Pressure (Pa)	Read/Write
Patm	REAL	Float	Site Atmospheric Pressure (Pa)	Read/Write
Rho	REAL	Float	Density Of Air (kg/m3)	Read/Write
Rho_da	REAL	Float	Density Of Dry Air (kg/m3)	Read/Write
R	REAL	Float	Gas Constant (J/kg K)	Read/Write
T	REAL	Float	Air Temperature (deg K)	Read/Write
V	REAL	Float	Volumetric Air Flow (m3/hr)	Read/Write
velocity	REAL	Float	Air Velocity (m/s)	Read/Write
Spare_Real_1	REAL	Float		Read/Write
Spare_Real_2	REAL	Float		Read/Write
Spare_Real_3	REAL	Float		Read/Write
Spare_Real_4	REAL	Float		Read/Write

5.2.10 UDTGenericMultiConstantCalc

Name	Data Type	Style	Description	External Access
RESULT	REAL	Float		Read/Write
INPUT_1	REAL	Float		Read/Write
INPUT_2	REAL	Float		Read/Write
INPUT_3	REAL	Float		Read/Write
INPUT_4	REAL	Float		Read/Write
INPUT_5	REAL	Float		Read/Write
INPUT_6	REAL	Float		Read/Write
CONSTANT_1	REAL	Float		Read/Write
CONSTANT_2	REAL	Float		Read/Write
CONSTANT_3	REAL	Float		Read/Write
CONSTANT_4	REAL	Float		Read/Write
CONSTANT_5	REAL	Float		Read/Write
CONSTANT_6	REAL	Float		Read/Write
CONSTANT_7	REAL	Float		Read/Write
CONSTANT_8	REAL	Float		Read/Write
CONSTANT_9	REAL	Float		Read/Write
CONSTANT_10	REAL	Float		Read/Write
CONSTANT_11	REAL	Float		Read/Write
CONSTANT_12	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_1	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_2	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_3	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_4	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_5	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_6	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_7	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_8	REAL	Float		Read/Write
SUB_RESULT_LEVEL_1_9	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_1	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_2	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_3	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_4	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_5	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_6	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_7	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_8	REAL	Float		Read/Write
SUB_RESULT_LEVEL_2_9	REAL	Float		Read/Write

5.2.11 UDTHeatEff

Members: Data Type Size: 92 byte(s)

Name	Data Type	Style	Description	External Access
Sunflower_Oil_In_Use	BOOL	Decimal	Sunflower Oil In Use	Read/Write
DensityOfSunflowerOil	REAL	Float	Density of Sunflower Oil	Read/Write
SpecHeatOfSunflowerOil	REAL	Float	Spec Heat of Sunflower Oil	Read/Write
DensityOfPalmoilienOil	REAL	Float	Density of Palmoilien Oil	Read/Write
SpecHeatOfPalmoilienOil	REAL	Float	Spec Heat of Palmoilien Oil	Read/Write
DensityOfOil	REAL	Float	Density of Oil	Read/Write
SpecHeatOfOil	REAL	Float	Spec Heat of Oil	Read/Write
PressureCorrection	REAL	Float	Gas Pressure Correction	Read/Write
TemperatureCorrection	REAL	Float	Gas Temperature Correction	Read/Write
NormalisedGas	REAL	Float	Normalised Gas Flow	Read/Write
GasEfficiency	REAL	Float	Gas Efficiency	Read/Write
GasCalorificValue	REAL	Float	Gas Calorific Value	Read/Write
OilMassFlow	REAL	Float	Oil Mass Flow	Read/Write
OilEfficiency	REAL	Float	Oil Efficiency	Read/Write
DeltaT	REAL	Float	Delta T	Read/Write
Efficiency	REAL	Float	Efficiency	Read/Write
GasPressure	REAL	Float	Gas Pressure	Read/Write
GasTemp	REAL	Float	Gas Temperature	Read/Write
GasFlow	REAL	Float	Gas Flow	Read/Write
OilFlow	REAL	Float	Oil Flow	Read/Write
AtmosPressure	REAL	Float	Atmospheric Pressure	Read/Write
Altitude	REAL	Float	Altitude	Read/Write
OilType	INT	Decimal		Read/Write

5.2.12 UDTInterfaceHXToFryer

Members: Data Type Size: 28 byte(s)

Name	Data Type	Style	Description	External Access
WatchDog	INT	Decimal	Heater PLC watchdog value (0-99)	Read/Write
Burner_Load_Control	BOOL	Decimal	Burner Released for modulation by fryer	Read/Write
HeaterGroup_ImmStopChecks	BOOL	Decimal	Heater group Immediate stop checks	Read/Write
HeaterGroup_CriticalFaults	BOOL	Decimal	Heater group critical faults healthy	Read/Write
HeaterGroup_OKToStart	BOOL	Decimal	Heater group ok to start	Read/Write
Fans_CriticalFaults	BOOL	Decimal	Burner Fans critical faults healthy	Read/Write
Fans_OKToStart	BOOL	Decimal	Burner fans ok to start	Read/Write
Burner_CriticalFaults	BOOL	Decimal	Burner critical faults healthy	Read/Write
Burner_OKToStart	BOOL	Decimal	Burner ok to start	Read/Write
M3604_OKSTART	BOOL	Decimal	OK to start main oil circ pump	Read/Write
M3604_OKSTOP	BOOL	Decimal	OK to stop main oil circ pump	Read/Write
Fuel_Oil_Selected	BOOL	Decimal	Diesel Oil Selected for Use	Read/Write
Alarm_Exists	BOOL	Decimal	An Alarm Condition exists within the Heater Elements	Read/Write
New_Alarm	BOOL	Decimal	A New Alarm Exist within the Heater Elements	Read/Write
DampersPositionControlFromHX	BOOL	Decimal		Read/Write
InletDamperPosition	INT	Decimal		Read/Write
OutletDamperPosition	INT	Decimal		Read/Write
Spare01	REAL	Float		Read/Write
Spare02	REAL	Float		Read/Write
Spare03	REAL	Float		Read/Write
Spare04	DINT	Decimal		Read/Write
Spare05	DINT	Decimal		Read/Write

5.2.13 UDTLookupTable

Members: Data Type Size: 72 byte(s)

Name	Data Type	Style	Description	External Access
Waypoint	REAL[11]	Float		Read/Write
Input	REAL	Float		Read/Write
Output	REAL	Float		Read/Write
LowPoint	REAL	Float		Read/Write
HighPoint	REAL	Float		Read/Write
Rise	REAL	Float		Read/Write
Offset	REAL	Float		Read/Write
Tread	REAL	Float		Read/Write

5.2.14 UDTMisc

Members: Data Type Size: 96 byte(s)

Name	Data Type	Style	Description	External Access
SCADA_Watchdog_Timer	TIMER		Comms watchdog fault timer	Read/Write
MSG_Fryer_Timer	TIMER		Timer for triggering message instruction	Read/Write
PLC_IO_LED_Status	DINT	Decimal		Read/Write
DINT_All_On	DINT	Decimal	Double integer with All bits ON	Read/Write
DINT_All_Off	DINT	Decimal	Double integer with All bits OFF	Read/Write
Fryer_Watchdog_Mem	DINT	Decimal	Copy of comms watchdog	Read/Write
SCADA_Watchdog_Mem	DINT	Decimal	Copy of comms watchdog	Read/Write
Always_Zero	BOOL	Decimal	Always off bit	Read/Write
Always_One	BOOL	Decimal	Always On bit	Read/Write
First_Scan	BOOL	Decimal	Bit on for first program scan	Read/Write
New_Fault_Detected	BOOL	Decimal	New fault detected	Read/Write
Alarm_Horn_Req	BOOL	Decimal	Alarm horn request	Read/Write
Alarm_Beacon_Req	BOOL	Decimal	Alarm beacon request	Read/Write
Fault_Reset	BOOL	Decimal	Combined Global fault Reset.	Read/Write
Alarm_Mute	BOOL	Decimal	Alarm mute command	Read/Write
Pulse_1Sec	BOOL	Decimal	Bit is on for 1 scan every 1 seconds	Read/Write
Flash_1Sec	BOOL	Decimal	Bit is on for 1 second and off for 1 second.	Read/Write
Flash_1Sec_Mem	BOOL	Decimal	Storage bit	Read/Write
Pulse_Half_Sec	BOOL	Decimal		Read/Write
Flash_Half_Sec	BOOL	Decimal		Read/Write
Flash_Half_Sec_Men	BOOL	Decimal		Read/Write
Simulation_On	BOOL	Decimal	!!!!!!!!!!! Simulation active !!!!!!!!!!	Read/Write
A_Fault_Exists	BOOL	Decimal	Bit is on if any fault bit is on.	Read/Write
Start_Warning	BOOL	Decimal	Something is about to start	Read/Write

5.2.15 UDTMotorDOL

Members: Data Type Size: 44 byte(s)

Name	Data Type	Style	Description	External Access
MAN	BOOL	Decimal	- Manual Mode	Read/Write
MANMEM	BOOL	Decimal	- Manual State Memory	Read/Write
CHANGED	BOOL	Decimal	- Manual State Changed	Read/Write
JGMEM	BOOL	Decimal	- Jog Request Memory	Read/Write
JG	BOOL	Decimal	- Jog the Motor for X Seconds	Read/Write
K	BOOL	Decimal	- Run Request Output	Read/Write
QA	BOOL	Decimal	- Tripped Input	Read/Write
ISO	BOOL	Decimal	- Isolated Input	Read/Write
KA	BOOL	Decimal	- Running Input	Read/Write
FRU	BOOL	Decimal	- Run Fault	Read/Write
FOL	BOOL	Decimal	- Overload Fault	Read/Write
FISO	BOOL	Decimal	- Isolated Fault	Read/Write
SPD	BOOL	Decimal	- Stopped	Read/Write
STN	BOOL	Decimal	- Starting	Read/Write
ROK	BOOL	Decimal	- Running	Read/Write
FLT	BOOL	Decimal	- Faulted	Read/Write
WARN	BOOL	Decimal	- Warning of a Fault Condition	Read/Write
S_JG	BOOL	Decimal	- Scada Bit to Jog the Motor	Read/Write
S_MST	BOOL	Decimal	- Scada Bit to Start the Motor	Read/Write
S_MOF	BOOL	Decimal	- Scada Bit to Stop the Motor	Read/Write
⊞ TOM	TIMER		- Fault Timer	Read/Write
⊞ TFM	TIMER		- Fault Timer	Read/Write
⊞ TSN	TIMER		- Start Timer	Read/Write
SCADA	DINT	Decimal	- Scada Status Value	Read/Write

5.2.16 UDTPIDControl

Members: Data Type Size: 248 byte(s)

Name	Data Type	Style	Description	External Access
⊞ PID	PID		- PID Control	Read/Write
CV	REAL	Float	- Control Variable	Read/Write
Enable	BOOL	Decimal	- PID Control Enable	Read/Write
Manual	BOOL	Decimal	- PID Manual Control	Read/Write
⊞ ScaleTag	SCP		- Setpoint Scaling	Read/Write

5.2.17 UDTPLCStatus

Members: Data Type Size: 12 byte(s)

Name	Data Type	Style	Description	External Access
Battery	BOOL	Decimal		Read/Write
Checksum	BOOL	Decimal		Read/Write
ProgMode	BOOL	Decimal		Read/Write
RunMode	BOOL	Decimal		Read/Write
TestMode	BOOL	Decimal		Read/Write
DNETA	INT	Decimal		Read/Write
DNETB	INT	Decimal		Read/Write
LastScan	INT	Decimal		Read/Write
MaxScan	INT	Decimal		Read/Write

5.2.18 UDTPowerFlexINV

Name	Data Type	Style	Description	External Access
REQ	BOOL	Decimal	- Auto Request	Read/Write
MST	BOOL	Decimal	- Manual Start Request	Read/Write
MOF	BOOL	Decimal	- Manual Stop Request	Read/Write
CFR	BOOL	Decimal	- Fault Reset	Read/Write
MAN	BOOL	Decimal	- Manual Mode	Read/Write
MANMEM	BOOL	Decimal	- Manual State Memory	Read/Write
CHANGED	BOOL	Decimal	- Manual State Changed	Read/Write
JGMEM	BOOL	Decimal	- Jog Request Memory	Read/Write
JG	BOOL	Decimal	- Jog the Motor for X Seconds	Read/Write
K	BOOL	Decimal	- Run Request Output	Read/Write
PWR	BOOL	Decimal	- Power Enable	Read/Write
PWD	BOOL	Decimal	- Powered	Read/Write
QA	BOOL	Decimal	- Tripped Input	Read/Write
ISO	BOOL	Decimal	- Isolated Input	Read/Write
KA	BOOL	Decimal	- Running Input	Read/Write
FRU	BOOL	Decimal	- Run Fault	Read/Write
FOL	BOOL	Decimal	- Overload Fault	Read/Write
FISO	BOOL	Decimal	- Isolated Fault	Read/Write
SPD	BOOL	Decimal	- Stopped	Read/Write
STN	BOOL	Decimal	- Starting	Read/Write
ROK	BOOL	Decimal	- Running	Read/Write
FLT	BOOL	Decimal	- Faulted	Read/Write
INV_FLT	BOOL	Decimal	- Inverter Fault	Read/Write
WARN	BOOL	Decimal	- Warning of a Fault Condition	Read/Write
S_JG	BOOL	Decimal	- Scada Bit to Jog the Motor	Read/Write
S_MST	BOOL	Decimal	- Scada Bit to Start the Motor	Read/Write
S_MOF	BOOL	Decimal	- Scada Bit to Stop the Motor	Read/Write
⊞ TOM	TIMER		- Fault Timer	Read/Write
⊞ TFM	TIMER		- Fault Timer	Read/Write
⊞ TSN	TIMER		- Start Timer	Read/Write
SCADA	DINT	Decimal	- Scada Status Value	Read/Write
SCALED_SPEED	REAL	Float	- Inverter Scaled Actual Speed	Read/Write
SET_SPEED	INT	Decimal	- Inverter Set Speed	Read/Write
ACT_SPEED	INT	Decimal	- Inverter Actual Speed	Read/Write
SCADA_INPUT_SPEED	INT	Decimal	- Scada Input Speed	Read/Write
AT_SPEED	BOOL	Decimal	- Inverter At Speed	Read/Write
MaxSpeed	INT	Decimal	- Max Inverter Speed	Read/Write
InputImage	INT[2]	Decimal	- Input Image	Read/Write
OutputImage	INT[2]	Decimal	- Output Image	Read/Write
Run_Current_x_100	DINT	Decimal	- Run Current From Inverter x 100	Read/Write
Run_Current	REAL	Float	- Run Current From Inverter	Read/Write
Full_Load_Current	REAL	Float	- Full Load Current	Read/Write
HZ_ACTUAL	REAL	Float	- Inverter Actual Speed In Hz	Read/Write

5.2.19 UDTRecipe

Name	Data Type	Style	Description	External Access
⊞ ProductTypeDes	STRING			Read/Write
ProductTypeNum	INT	Decimal		Read/Write
Param01	REAL	Float		Read/Write
Param02	REAL	Float		Read/Write
Param03	REAL	Float		Read/Write
Param04	REAL	Float		Read/Write
Param05	REAL	Float		Read/Write
Param06	REAL	Float		Read/Write
Param07	REAL	Float		Read/Write
Param08	REAL	Float		Read/Write
Param09	REAL	Float		Read/Write
Param10	REAL	Float		Read/Write
Param11	REAL	Float		Read/Write
Param12	REAL	Float		Read/Write
Param13	REAL	Float		Read/Write
Param14	REAL	Float		Read/Write
Param15	REAL	Float		Read/Write
Param16	REAL	Float		Read/Write
Param17	REAL	Float		Read/Write
Param18	REAL	Float		Read/Write
Param19	REAL	Float		Read/Write
Param20	REAL	Float		Read/Write
Param21	REAL	Float		Read/Write
Param22	REAL	Float		Read/Write
Param23	REAL	Float		Read/Write
Param24	REAL	Float		Read/Write
Param25	REAL	Float		Read/Write
Param26	REAL	Float		Read/Write
Param27	REAL	Float		Read/Write
Param28	REAL	Float		Read/Write
Param29	REAL	Float		Read/Write
Param30	REAL	Float		Read/Write
Param31	REAL	Float		Read/Write
Param32	REAL	Float		Read/Write
Param33	REAL	Float		Read/Write
Param34	REAL	Float		Read/Write
Param35	REAL	Float		Read/Write
Param36	REAL	Float		Read/Write
Param37	REAL	Float		Read/Write
Param38	REAL	Float		Read/Write
Param39	REAL	Float		Read/Write
Param40	REAL	Float		Read/Write
Param41	REAL	Float		Read/Write
Param42	REAL	Float		Read/Write
Param43	REAL	Float		Read/Write
Param44	REAL	Float		Read/Write
Param45	REAL	Float		Read/Write
Param46	REAL	Float		Read/Write
Param47	REAL	Float		Read/Write
Param48	REAL	Float		Read/Write
Param49	REAL	Float		Read/Write

Param50	REAL	Float		Read/Write
Param51	REAL	Float		Read/Write
Param52	REAL	Float		Read/Write
Param53	REAL	Float		Read/Write
Param54	REAL	Float		Read/Write
Param55	REAL	Float		Read/Write
Param56	REAL	Float		Read/Write
Param57	REAL	Float		Read/Write
Param58	REAL	Float		Read/Write
Param59	REAL	Float		Read/Write
Param60	REAL	Float		Read/Write
Param61	REAL	Float		Read/Write
Param62	REAL	Float		Read/Write
Param63	REAL	Float		Read/Write
Param64	REAL	Float		Read/Write
Param65	REAL	Float		Read/Write

5.2.20 UDTScadaAnalogs

Members: Data Type Size: 88 byte(s)

Name	Data Type	Style	Description	External Access
FT3501_AI	REAL	Float		Read/Write
FT3502_AI	REAL	Float		Read/Write
FT3503_AI	REAL	Float		Read/Write
FT3504_AI	REAL	Float		Read/Write
FT3602_AI	REAL	Float		Read/Write
PT3501_AI	REAL	Float		Read/Write
SD3501_AI	REAL	Float		Read/Write
TT3501_AI	REAL	Float		Read/Write
TT3502_AI	REAL	Float		Read/Write
TT3503_AI	REAL	Float		Read/Write
TT3504_AI	REAL	Float		Read/Write
TT3505_AI	REAL	Float		Read/Write
TT3506_AI	REAL	Float		Read/Write
TT3507_AI	REAL	Float		Read/Write
TT3508_AI	REAL	Float		Read/Write
TT3509_AI	REAL	Float		Read/Write
TT3510_AI	REAL	Float		Read/Write
M3502_Spd_Ref	REAL	Float		Read/Write
M3501_Spd_Ref	REAL	Float		Read/Write
CV3609_AO	REAL	Float		Read/Write
CV3610_AO	REAL	Float		Read/Write
EFFICIENCY	REAL	Float		Read/Write

5.2.21 UDTScadaMisc

Members: Data Type Size: 40 byte(s)

Name	Data Type	Style	Description	External Access
BurnerStatus	INT	Decimal		Read/Write
Burner_Critical_Faults	BOOL	Decimal		Read/Write
Combust_Air_Test	INT	Decimal		Read/Write
Fault_Reset	BOOL	Decimal		Read/Write
Fans_Start	INT	Decimal		Read/Write
Flame_Start_Timer	INT	Decimal		Read/Write
Flame_In	INT	Decimal		Read/Write
Fryer_Boilout	BOOL	Decimal		Read/Write
Fryer_Frying	BOOL	Decimal		Read/Write
Fryer_Preheat	BOOL	Decimal		Read/Write
Fryer_PLC_Modulation	BOOL	Decimal		Read/Write
Fuel_Type	INT	Decimal		Read/Write
FuelOil_Command	BOOL	Decimal		Read/Write
Gas_Leak_Test	INT	Decimal		Read/Write
Heater_Auto	BOOL	Decimal		Read/Write
Heater_Running	BOOL	Decimal		Read/Write
HeaterPLC_Watchdog	INT	Decimal		Read/Write
High_Post_Purge_Timer	INT	Decimal		Read/Write
Low_Post_Purge_Timer	INT	Decimal		Read/Write
M3604_Running	BOOL	Decimal		Read/Write
NaturalGas_Command	BOOL	Decimal		Read/Write
Scada_Watchdog	INT	Decimal		Read/Write
Vents_Test	INT	Decimal		Read/Write
Damper_Test	INT	Decimal		Read/Write

5.2.22 UDTScadaStatus

Name	Data Type	Style	Description	External Access
Burner_Fault	INT	Decimal		Read/Write
CCOT	INT	Decimal		Read/Write
CV3609_AO	INT	Decimal		Read/Write
CV3610_AO	INT	Decimal		Read/Write
ES3501	INT	Decimal		Read/Write
FSSR	INT	Decimal		Read/Write
FS3501	INT	Decimal		Read/Write
FT3501_AI	INT	Decimal		Read/Write
FT3502_AI	INT	Decimal		Read/Write
FT3503_AI	INT	Decimal		Read/Write
FT3504_AI	INT	Decimal		Read/Write
FT3602_AI	INT	Decimal		Read/Write
MPSR	INT	Decimal		Read/Write
MASR	INT	Decimal		Read/Write
MESR	INT	Decimal		Read/Write
M3501	INT	Decimal		Read/Write
M3502	INT	Decimal		Read/Write
M3503	INT	Decimal		Read/Write
M3504	INT	Decimal		Read/Write
M3604	INT	Decimal		Read/Write
M3604_OKSTOP	INT	Decimal		Read/Write
OLFS	INT	Decimal		Read/Write
OLOT	INT	Decimal		Read/Write
OLTS	INT	Decimal		Read/Write
PAOT	INT	Decimal		Read/Write
PITD	INT	Decimal		Read/Write
PDPS	INT	Decimal		Read/Write
PS3502	INT	Decimal		Read/Write
PS3503	INT	Decimal		Read/Write
PS3505	INT	Decimal		Read/Write
PS3507	INT	Decimal		Read/Write
PS3508	INT	Decimal		Read/Write
PS3511_High	INT	Decimal		Read/Write
PS3511_HighHigh	INT	Decimal		Read/Write
PS3512	INT	Decimal		Read/Write
PT3501_AI	INT	Decimal		Read/Write
SDSR	INT	Decimal		Read/Write
SD3501_AI	INT	Decimal		Read/Write
STOT	INT	Decimal		Read/Write
TT3501_AI	INT	Decimal		Read/Write
TT3502_AI	INT	Decimal		Read/Write
TT3503_AI	INT	Decimal		Read/Write
TT3504_AI	INT	Decimal		Read/Write
TT3505_AI	INT	Decimal		Read/Write
TT3506_AI	INT	Decimal		Read/Write
TT3507_AI	INT	Decimal		Read/Write
TT3509_AI	INT	Decimal		Read/Write
TT3510_AI	INT	Decimal		Read/Write

	V3501	INT	Decimal		Read/Write
	V3502	INT	Decimal		Read/Write
	V3503	INT	Decimal		Read/Write
	V3504	INT	Decimal		Read/Write
	V3505	INT	Decimal		Read/Write
	V3506	INT	Decimal		Read/Write
	V3508	INT	Decimal		Read/Write
	V3512	INT	Decimal		Read/Write
	V3513	INT	Decimal		Read/Write
	V3514	INT	Decimal		Read/Write
	V3516	INT	Decimal		Read/Write
	V3517	INT	Decimal		Read/Write
	V3518	INT	Decimal		Read/Write
	ZS3501	INT	Decimal		Read/Write
	ZS3502	INT	Decimal		Read/Write
	ZS3503	INT	Decimal		Read/Write
	TT3508_AI	INT	Decimal		Read/Write

5.2.23 UDTValve

Members: Data Type Size: 44 byte(s)

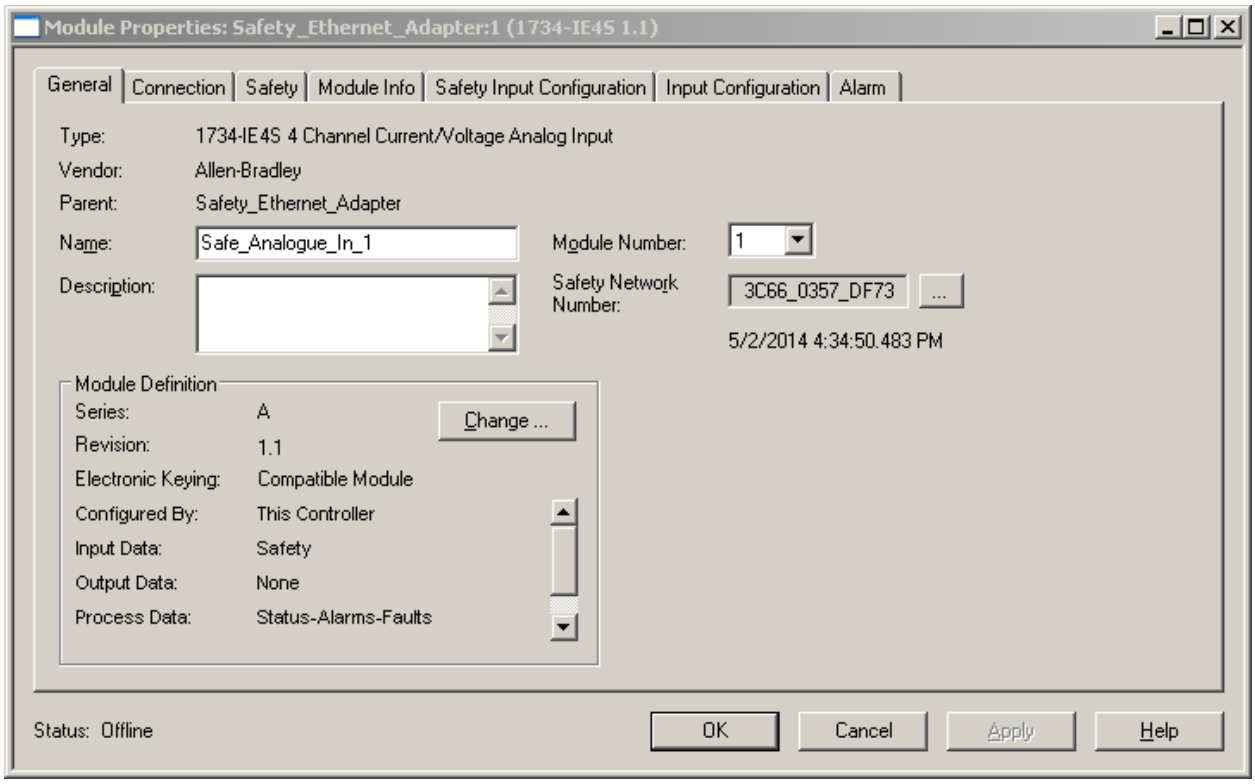
Name	Data Type	Style	Description	External Access
RQO	BOOL	Decimal	Auto Open Request	Read/Write
MOP	BOOL	Decimal	Manual Open Request	Read/Write
MCL	BOOL	Decimal	Manual Close Request	Read/Write
CFR	BOOL	Decimal	Fault Reset	Read/Write
MAN	BOOL	Decimal	Manual Mode	Read/Write
MEM	BOOL	Decimal	State Memory	Read/Write
CHANGED	BOOL	Decimal	State Changed	Read/Write
K	BOOL	Decimal	Open Output	Read/Write
OPN	BOOL	Decimal	Open Input	Read/Write
CLS	BOOL	Decimal	Closed Input	Read/Write
COK	BOOL	Decimal	Closed OK	Read/Write
OOK	BOOL	Decimal	Open OK	Read/Write
FLT	BOOL	Decimal	Fault	Read/Write
NOP	BOOL	Decimal	No Position	Read/Write
FOP	BOOL	Decimal	Open Fault	Read/Write
FCL	BOOL	Decimal	Closed Fault	Read/Write
FNP	BOOL	Decimal	No Position Fault	Read/Write
Spare1	BOOL	Decimal	Spare BOOLEAN variable	Read/Write
Spare2	BOOL	Decimal	Spare BOOLEAN variable	Read/Write
S_MCL	BOOL	Decimal	Scada Close Request	Read/Write
S_MOP	BOOL	Decimal	Scada Open Request	Read/Write
⊞ TOP	TIMER		Opening Timer	Read/Write
⊞ TCL	TIMER		Closing Timer	Read/Write
⊞ TFT	TIMER		Fault Timer	Read/Write
SCADA	DINT	Decimal	Scada Interface	Read/Write

5.3 POINT I/O Module Configuration

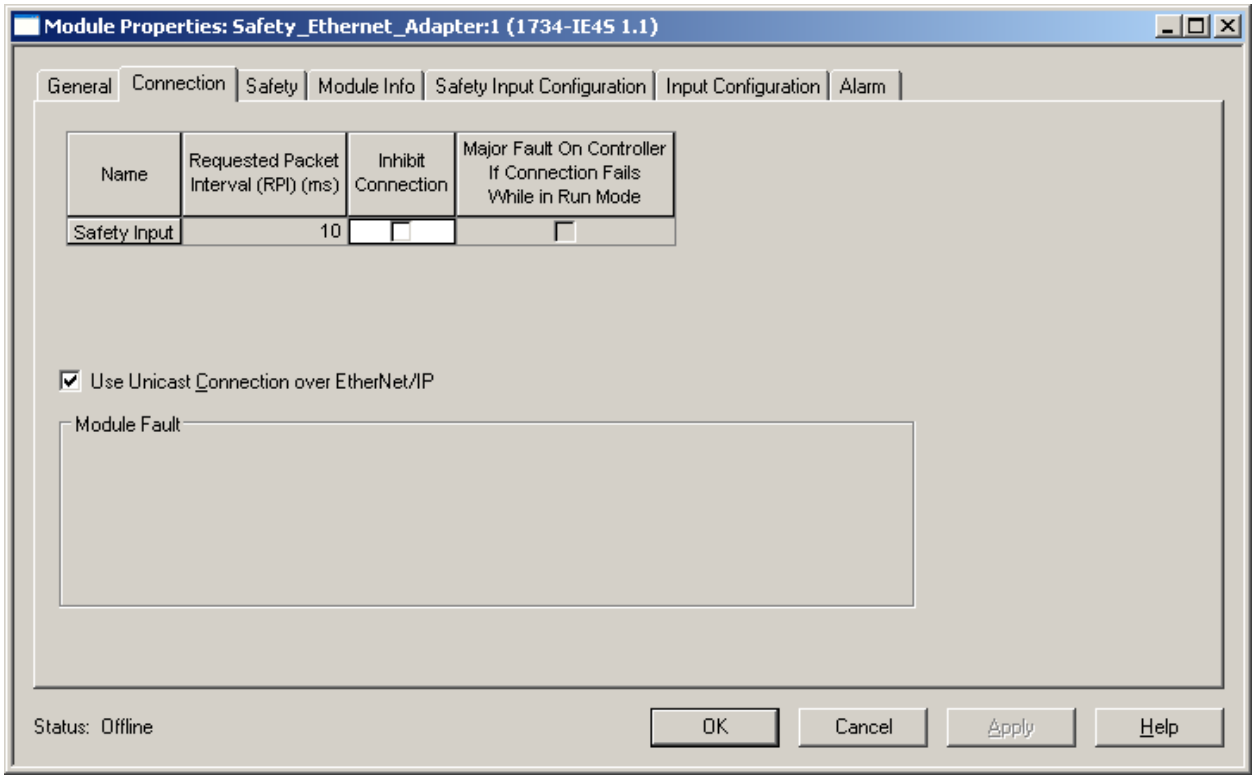
5.3.1 Slot 1 - Safe_Analogue_In_1

Module Type – 1734-IE4S (Safe 4 Channel Current/Voltage Analogue Input)

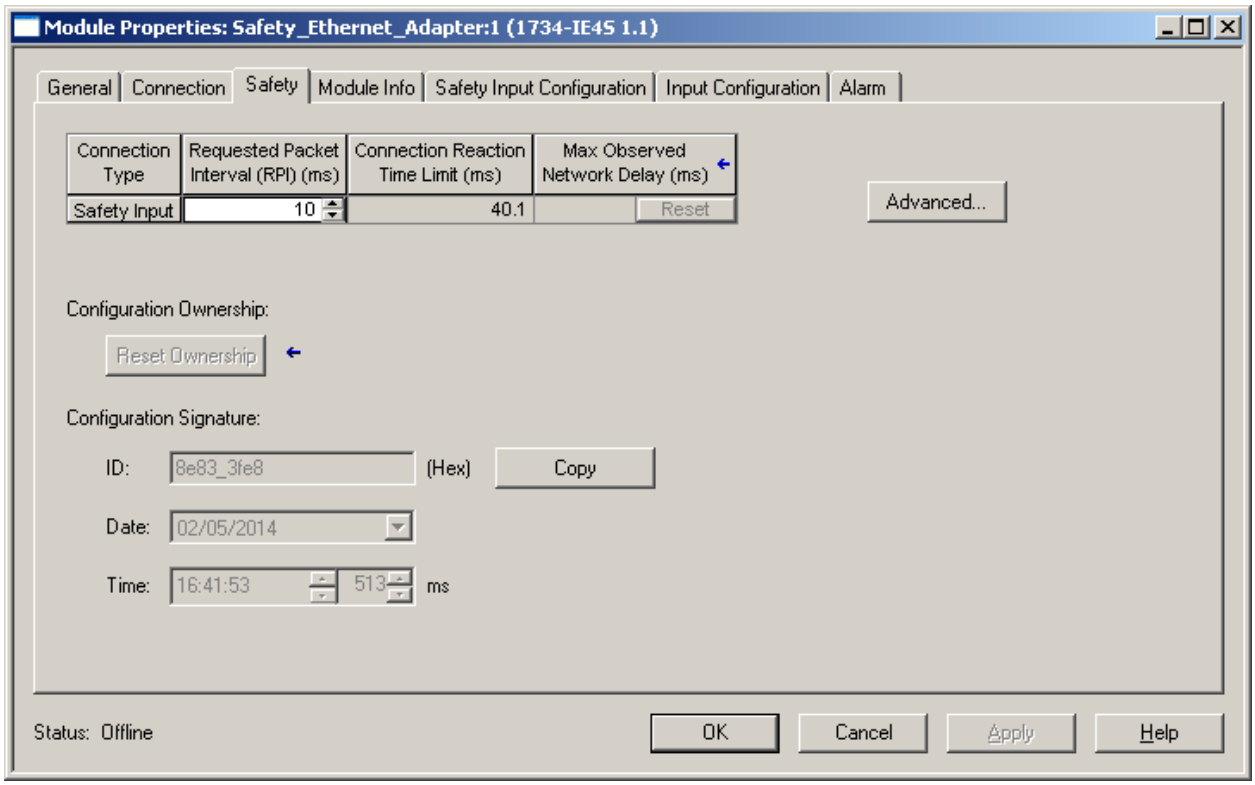
5.3.1.1 General Parameters



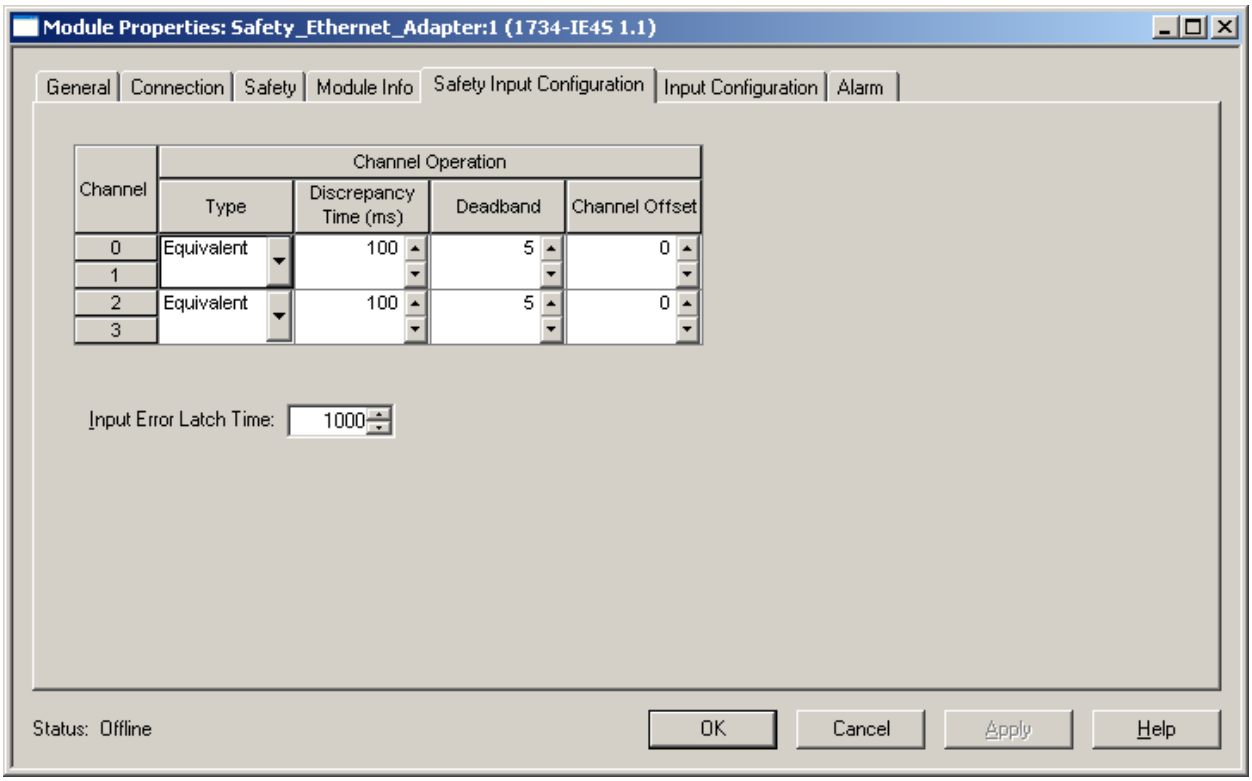
5.3.1.2 Connection Parameters



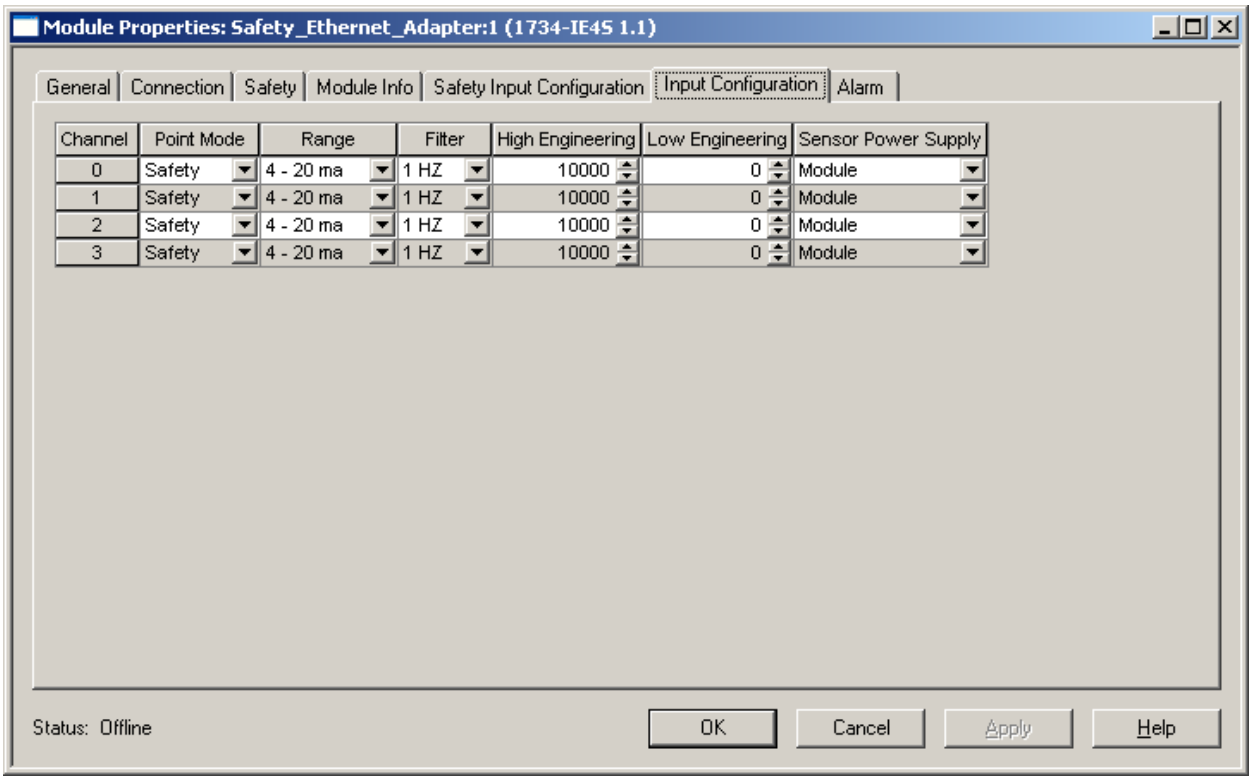
5.3.1.3 Safety Parameters



5.3.1.4 Safety Input Configuration Parameters



5.3.1.5 Input Configuration Parameters



Notes:

- The Module Information tab has no configurable parameters.
- The Alarm Tab is not used.

5.3.2 Slot 2 - Safe_Analogue_In_2

Module Type – 1734-IE4S (Safe 4 Channel Current/Voltage Analogue Input)

5.3.2.1 General Parameters

[Add finalised screen shot]

5.3.2.2 Connection Parameters

[Add finalised screen shot]

5.3.2.3 Safety Parameters

[Add finalised screen shot]

5.3.2.4 Safety Input Configuration Parameters

[Add finalised screen shot]

5.3.2.5 Input Configuration Parameters

[Add finalised screen shot]

Notes:

- The Module Information tab has no configurable parameters.
- The Alarm Tab is not used.

5.3.3 Slot 3 - Safe_Analogue_In_3

Module Type – 1734-IE4S (Safe 4 Channel Current/Voltage Analogue Input)

5.3.3.1 General Parameters

[Add finalised screen shot]

5.3.3.2 Connection Parameters

[Add finalised screen shot]

5.3.3.3 Safety Parameters

[Add finalised screen shot]

5.3.3.4 Safety Input Configuration Parameters

[Add finalised screen shot]

5.3.3.5 Input Configuration Parameters

[Add finalised screen shot]

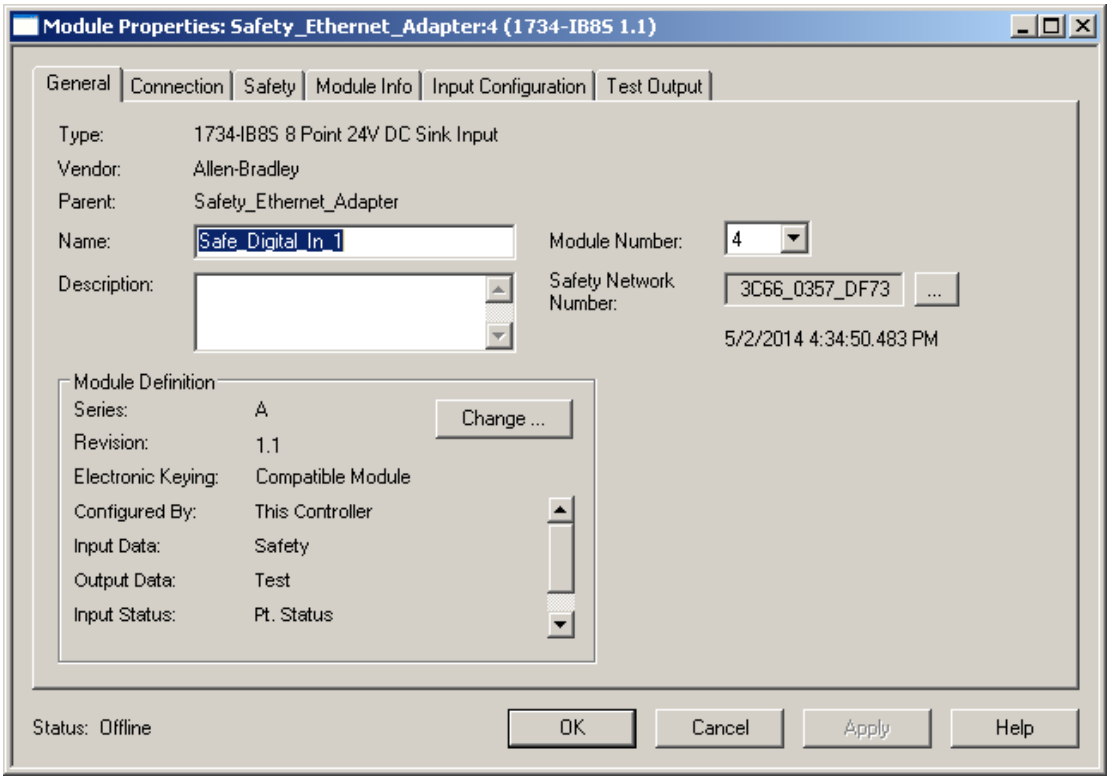
Notes:

- The Module Information tab has no configurable parameters.
- The Alarm Tab is not used.

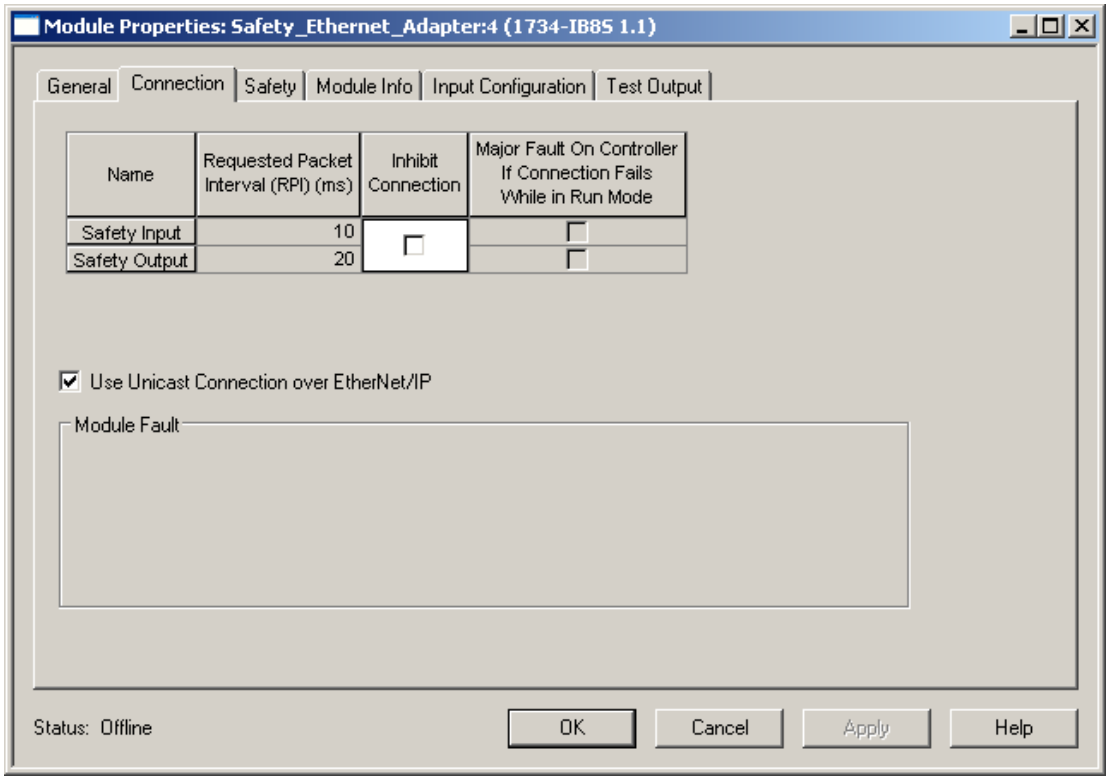
5.3.4 Slot 4 - Safe_Digital_In_1

Module Type – 1734-IB8S (Safe 8 Point 24V DC Sink Input)

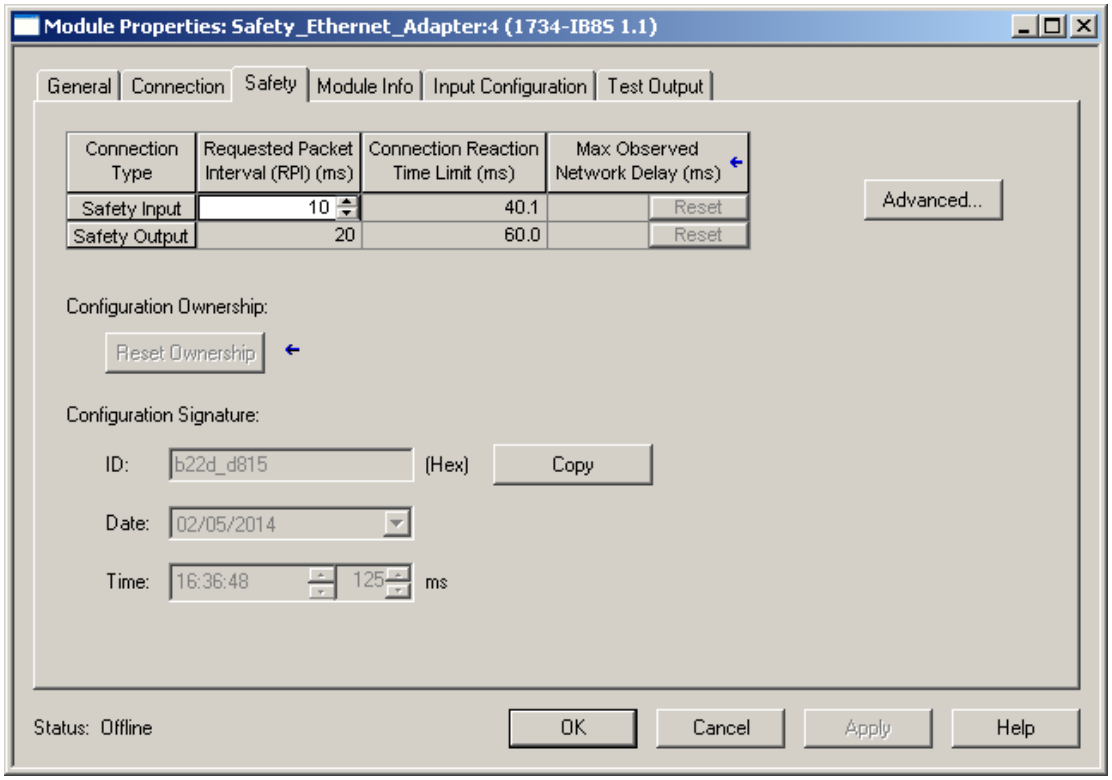
5.3.4.1 General Parameters



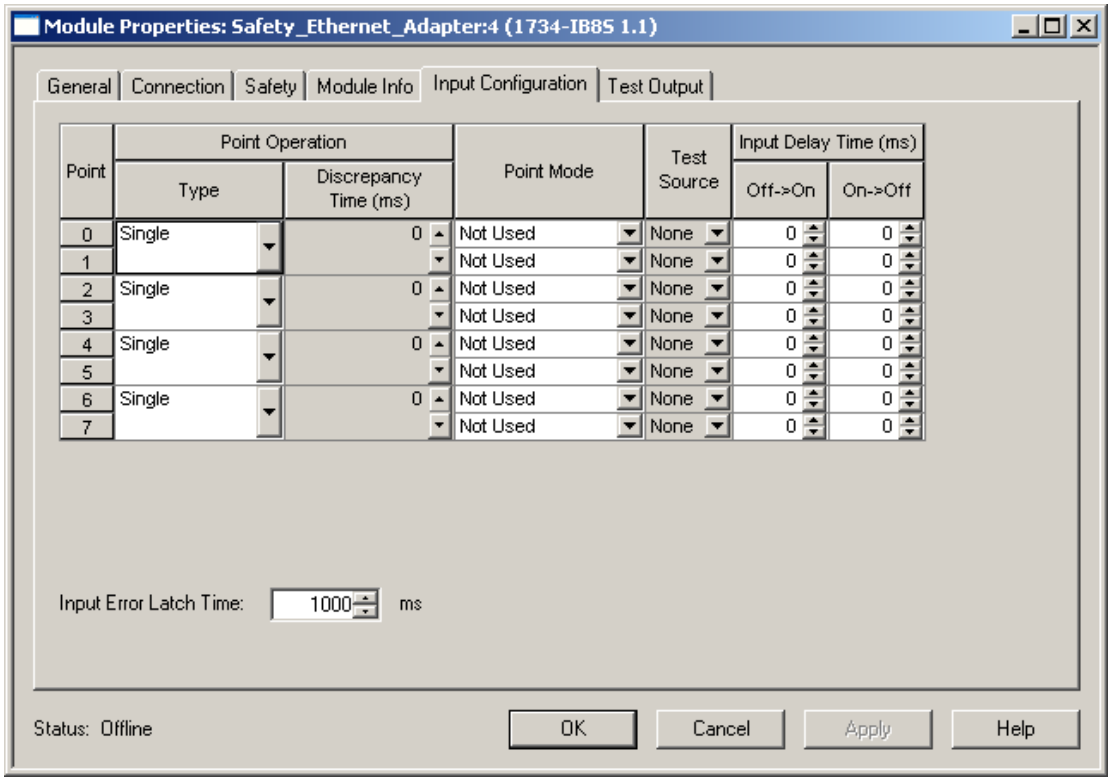
5.3.4.2 Connection Parameters



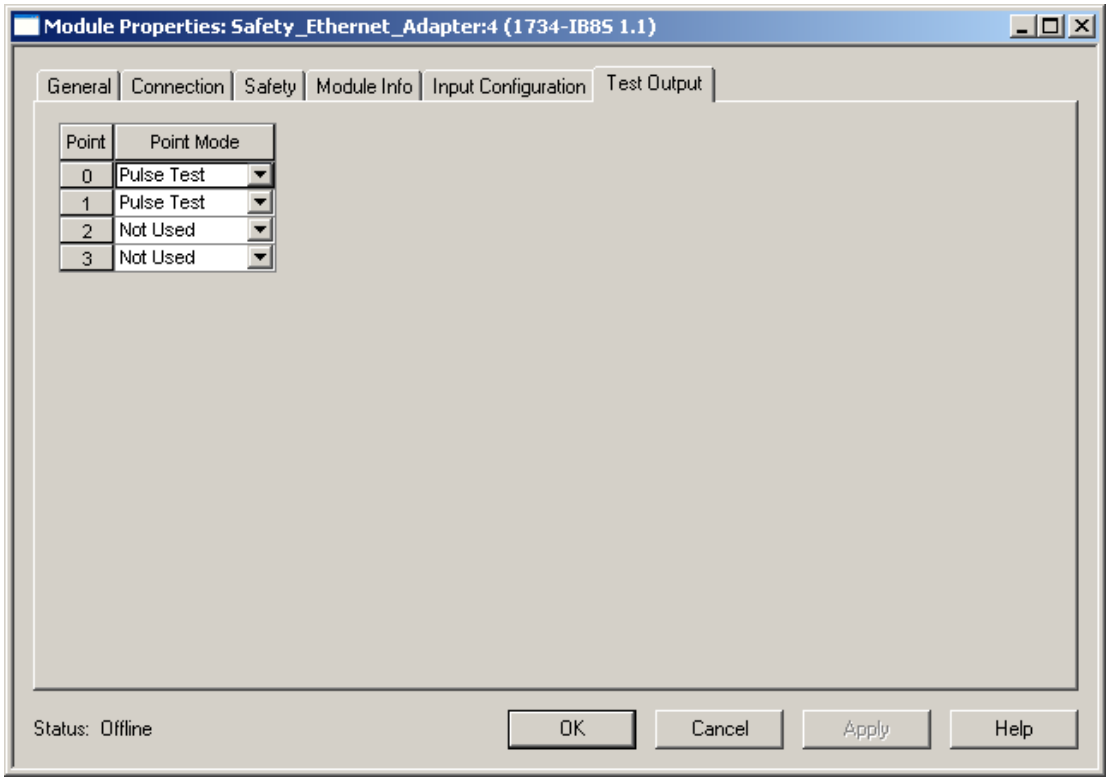
5.3.4.3 Safety Parameters



5.3.4.4 Input Configuration Parameters



5.3.4.5 Test Output Parameters



Notes:

- The Module Information tab has no configurable parameters.

5.3.5 Slot 5 - Safe_Digital_In_2

Module Type – 1734-IB8S (Safe 8 Point 24V DC Sink Input)

5.3.5.1 General Parameters

[Add finalised screen shot]

5.3.5.2 Connection Parameters

[Add finalised screen shot]

5.3.5.3 Safety Parameters

[Add finalised screen shot]

5.3.5.4 Input Configuration Parameters

[Add finalised screen shot]

5.3.5.5 Test Output Parameters

[Add finalised screen shot]

Notes:

- The Module Information tab has no configurable parameters.

5.3.6 Slot 6 - Safe_Digital_In_3

Module Type – 1734-IB8S (Safe 8 Point 24V DC Sink Input)

5.3.6.1 General Parameters

[Add finalised screen shot]

5.3.6.2 Connection Parameters

[Add finalised screen shot]

5.3.6.3 Safety Parameters

[Add finalised screen shot]

5.3.6.4 Input Configuration Parameters

[Add finalised screen shot]

5.3.6.5 Test Output Parameters

[Add finalised screen shot]

Notes:

- The Module Information tab has no configurable parameters.

5.3.7 Slot 7 - Safe_Digital_In_4

Module Type – 1734-IB8S (Safe 8 Point 24V DC Sink Input)

5.3.7.1 General Parameters

[Add finalised screen shot]

5.3.7.2 Connection Parameters

[Add finalised screen shot]

5.3.7.3 Safety Parameters

[Add finalised screen shot]

5.3.7.4 Input Configuration Parameters

[Add finalised screen shot]

5.3.7.5 Test Output Parameters

[Add finalised screen shot]

Notes:

- The Module Information tab has no configurable parameters.

5.3.8 Slot 8 - Safe_Digital_In_5

Module Type – 1734-IB8S (Safe 8 Point 24V DC Sink Input)

5.3.8.1 General Parameters

[Add finalised screen shot]

5.3.8.2 Connection Parameters

[Add finalised screen shot]

5.3.8.3 Safety Parameters

[Add finalised screen shot]

5.3.8.4 Input Configuration Parameters

[Add finalised screen shot]

5.3.8.5 Test Output Parameters

[Add finalised screen shot]

Notes:

- The Module Information tab has no configurable parameters.

5.3.9 Slot 9 - Safe_Digital_In_6

Module Type – 1734-IB8S (Safe 8 Point 24V DC Sink Input)

5.3.9.1 General Parameters

[Add finalised screen shot]

5.3.9.2 Connection Parameters

[Add finalised screen shot]

5.3.9.3 Safety Parameters

[Add finalised screen shot]

5.3.9.4 Input Configuration Parameters

[Add finalised screen shot]

5.3.9.5 Test Output Parameters

[Add finalised screen shot]

Notes:

- The Module Information tab has no configurable parameters.

5.3.10 Slot 10 - Safe_Digital_In_7

Module Type – 1734-IB8S (Safe 8 Point 24V DC Sink Input)

5.3.10.1 General Parameters

[Add finalised screen shot]

5.3.10.2 Connection Parameters

[Add finalised screen shot]

5.3.10.3 Safety Parameters

[Add finalised screen shot]

5.3.10.4 Input Configuration Parameters

[Add finalised screen shot]

5.3.10.5 Test Output Parameters

[Add finalised screen shot]

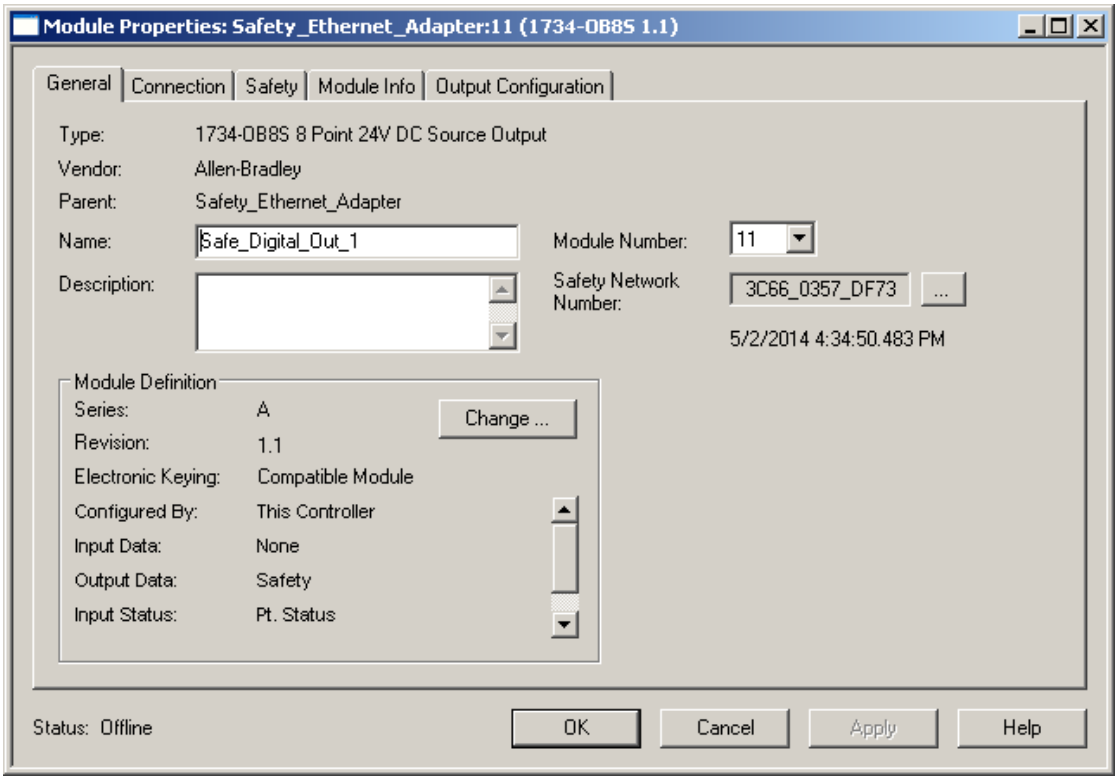
Notes:

- The Module Information tab has no configurable parameters.

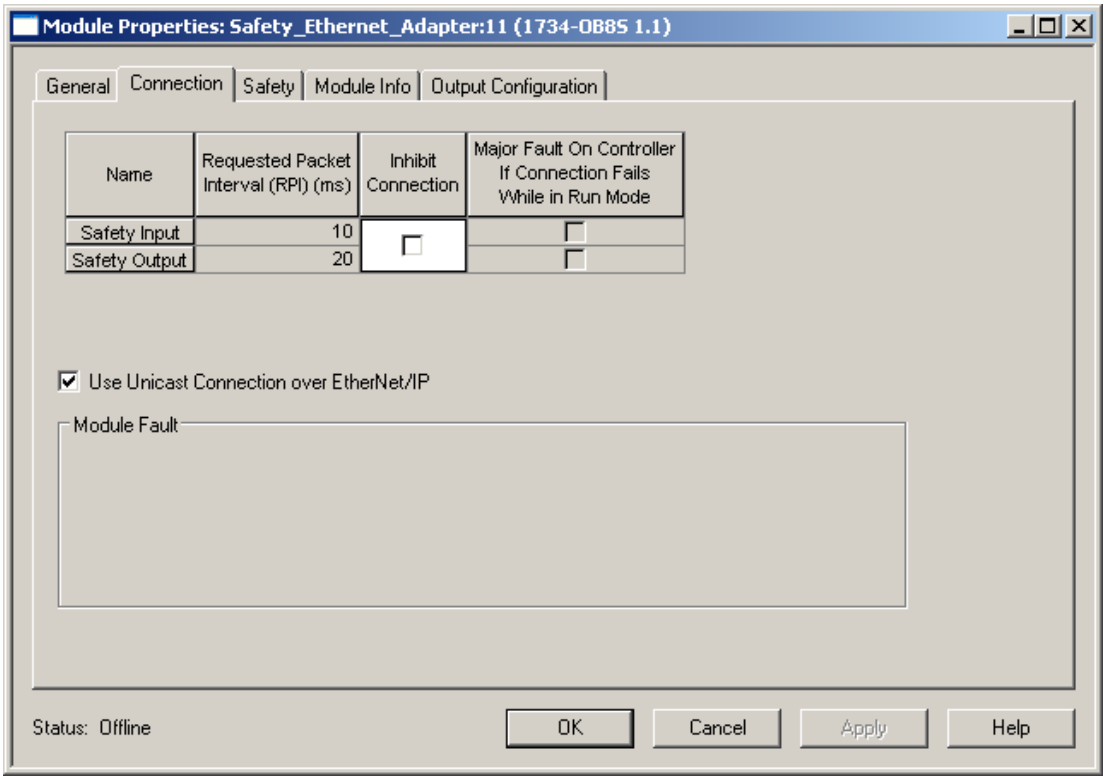
5.3.11 Slot 11 - Safe_Digital_Out_1

Module Type – 1734-OB8S (Safe 8 Point 24V DC Source Output)

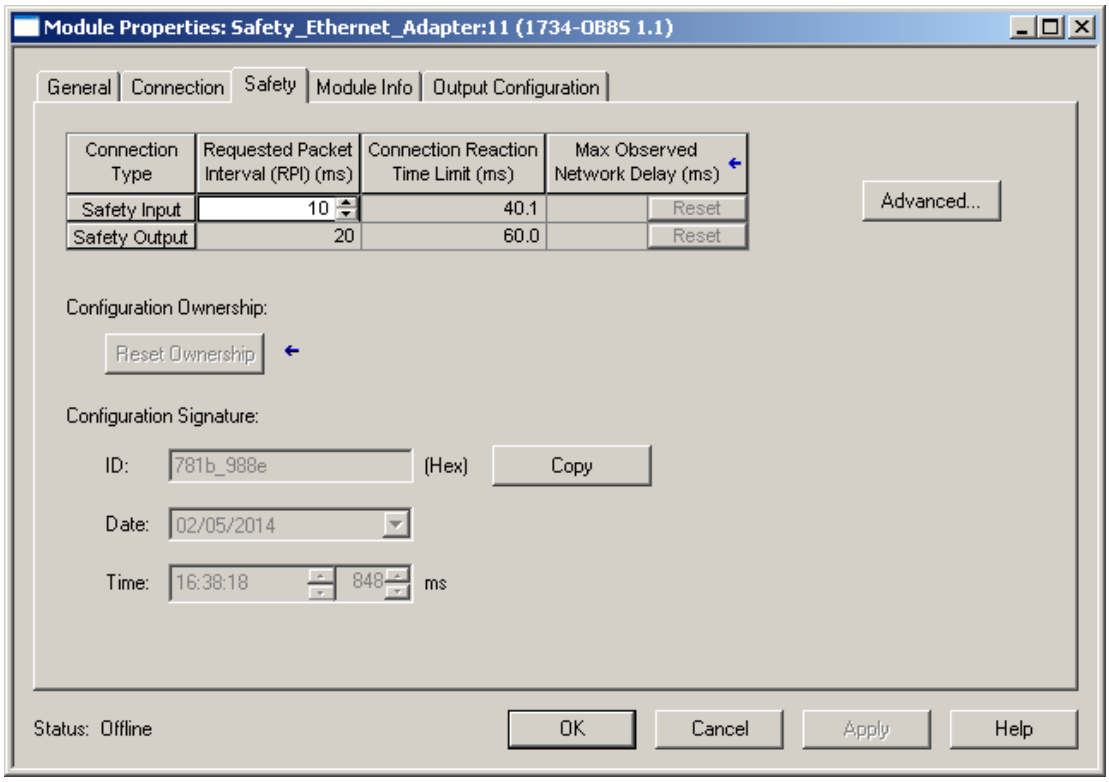
5.3.11.1 General Parameters



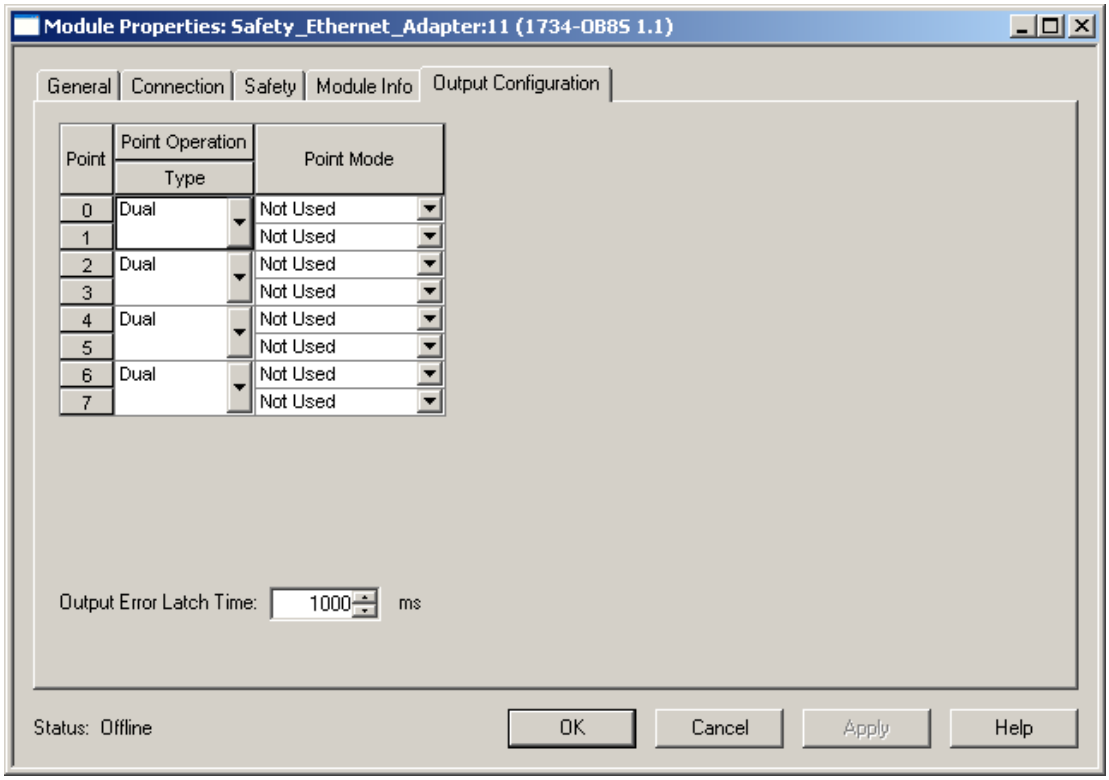
5.3.11.2 Connection Parameters



5.3.11.3 Safety Parameters



5.3.11.4 Output Configuration Parameters



Notes:

- The Module Information tab has no configurable parameters.

5.3.12 Slot 12 - Safe_Digital_Out_2

Module Type – 1734-OB8S (Safe 8 Point 24V DC Source Output)

5.3.12.1 General Parameters

[Add finalised screen shot]

5.3.12.2 Connection Parameters

[Add finalised screen shot]

5.3.12.3 Safety Parameters

[Add finalised screen shot]

5.3.12.4 Output Configuration Parameters

[Add finalised screen shot]

Notes:

- The Module Information tab has no configurable parameters.

Section 6

Appendix B

6 Appendix B

6.1 Safe I/O Test Sheets

Customer:	Walkers Snack Foods
Site:	Bursom Road
Line:	Line 4
Project:	C2381
Test Code:	
Test Name:	
Test Description:	
Test Procedure:	
Expected Result:	
Actual Result:	
Pass/Fail	
Comments:	
Signatures:	
Date:	

Example Test Sheet

6.2 Flo-Mech Sequencing Test Sheet

Customer:	CHIPSY EL OBOUR						
Subject:	G4						
Date:	12/08/2013						
Issue:	A						
Tested / Set up:	Terry Putson / Chris Ellis						
Section 2	Safety systems			Set / trip point	mA test	Checked	Date
1	Main emergency stop relay	MESR	al estop inc remote estop e suppression	NA	NA	YES	04/08/2013
2	Zone stop relay	M ZR	Pollution fan	A	NA	YES	04/08/2013
3	Zone stop relay		Combustion fan	A	NA	YES	04/08/2013
4	Zone stop relay		Flame sensor cooling fa	A	NA	YES	04/08/2013
5	Safety relay	M R	Man way access safety	A	NA	YES	04/08/2013
6	Pepprl & Fuchs analogue trip monitor / alarm	C D	Main oil flow meter saf	0	11.11	YES	04/08/2013
7	Pepprl & Fuchs analogue trip monitor / alarm	C	Combustion chamber ov temperature	870 DegC	15.6 mA	YES	04/08/2013
8	Pepprl & Fuchs analogue trip monitor / alarm	STC	Stack temperature over temperature	250 DegC	7.33 mA	YES	04/08/2013
9	Pepprl & Fuchs analogue trip monitor / alarm	P	Pump inhibit safety	130 DegC	5.73 mA	YES	04/08/2013
10	Pepprl & Fuchs analogue trip monitor / alarm	O	Oil over temperature	205 DegC	17.12 mA	YES	04/08/2013
11	Pepprl & Fuchs analogue trip monitor / alarm		Polluted air over temperat	250 DegC	7.33 mA	YES	04/08/2013
12	Vent valve sequence checks for main oil pump start		Confim operation of se ce	NA	NA	YES	04/08/2013
13	Vent valve sequence checks for main oil pump start	NA	Confim V5205 failure	NA	NA	YES	04/08/2013
14	Vent valve sequence checks for main oil pump start	NA	Confim V5206 failure	NA	NA	YES	04/08/2013
15	Vent valve sequence checks for main oil pump start		nfim V5205 OPN fai	NA	NA	YES	04/08/2013
16	Vent valve sequence checks for main oil pump start		firm V5205 CLS fail	NA	NA	YES	04/08/2013
17	Vent valve sequence checks for main oil pump start	NA	Confim V5206 OPN fai	NA	NA	YES	04/08/2013
18	Vent valve sequence checks for main oil pump start	NA	Confim V5206 CLS failu	NA	NA	YES	04/08/2013
19	Vent valve sequence checks for main oil pump start	NA	Confim LIS5201 activat	NA	NA	YES	04/08/2013
20	Pump inhibit sequence	NA	Confim operation of pu inhibit			YES	04/08/2013
21	Start pump <130 Celcius	NA	Normal start up	130 DegC	5.73 mA	YES	04/08/2013
22	Stop pump <130 Celcius	A	Normal stop	130 DegC	5.73 mA	YES	04/08/2013
23	Start pump <130 Celcius	A	mal start up	130 DegC	5.73 mA	YES	04/08/2013
24	Stop pump >130 Celcius (pump will not stop)	A	o will not stop	130 DegC	5.73 mA	YES	04/08/2013
25	Start pump >130 Celcius (pump will not start) (after e-stop)	NA	P o will not start	130 DegC	5.73 mA	YES	04/08/2013
26	Smoke detector		conds detection a s main oil pump	80%	NA	YES	04/08/2013
27	Fan start sequence for dampers		firm operation of damper sequence	NA	NA	YES	04/08/2013
	Tests with burner flame established						
28	High gas pressure fault test	NA	Confim burner sh	150 mbar	NA	YES	10/03/2013
29	Low gas pressure fault test	NA	Confim burner sh	80 mbar	NA	YES	10/03/2013
30	Burner Low Oil Pressure Switch		Confim burner shut dow	18 bar	NA	YES	10/03/2013
31	Burner High Oil Pressure Switch		Confim burner shut do	8bar	NA	YES	10/03/2013
32	Valve proving fault test	NA	Confim burner shut do	80 mbar	NA	YES	10/03/2013
33	Combustion air pressure switch fault test	NA	Confim burner shut dow	10 mbar	NA	YES	10/03/2013
34	Power loss		Confim burner shut down	NA	NA	YES	10/03/2013
35	Pollution air pressure		Confim burner shut down	5 mbar	NA	YES	10/03/2013
36	Combustion air pressure		Confim burner shut dow lockout	bar	NA	YES	10/03/2013
37	Flame sensor cooling fan pressure	P	Confim burner shut down	10 mbar	NA	YES	10/03/2013
38	Combustion fan trip		Confim burner shut down	NA	NA	YES	10/03/2013
39	Pollution fan trip		Confim burner shut dow	NA	NA	YES	10/03/2013
40	Pepprl & Fuchs analogue trip monitor / alarm		Combustion chamber ov temperature	870 DegC	15.6	YES	10/03/2013
41	Pepprl & Fuchs analogue trip monitor / alarm		Polluted air over tempe re	250 DegC	7.33	YES	10/03/2013
42	Pepprl & Fuchs analogue trip monitor / alarm	S T	Stack temperature over temperature	250 DegC	7.33	YES	10/03/2013
43	Smoke detection	SDSR	Confim burner shut down, main oil pump shut down, fire damper closed, fresh air damper open, fans running	80 % opacity	NA	YES	10/03/2013
44	Oil flow (vent line)			OK	NA	YES	10/03/2013
45	Burner lockout			OK	NA	YES	10/03/2013
46	Water temperature	NA	PLC setpoint	NA	NA	YES	10/03/2013
47	Hi Hi oil temperature	OLTS	Shut down burner, Main oil pump	220 DegC	NA	YES	10/03/2013
48	Fire suppression release	FSSR	Shut down burner, Main oil pump	Tesed Healthy	NA	YES	10/03/2013
49	Frying selected	NA		OK	NA	YES	10/03/2013
50	Boil out selected	NA		OK	NA	YES	10/03/2013
53	Pepprl & Fuchs analogue trip monitor / alarm	OLFD	Main oil flow meter safety	200 m3/hr	12 mA	YES	10/03/2013
54	Pump inhibit sequence	PITD	Confim operation of pump inhibit			YES	10/03/2013
55	Oil over temp	OLOT	Shut down burner	205 DegC	17.12mA	YES	10/03/2013
56	120 min fan shut down after burner off		Shut down fans after 120 min cool down			YES	10/03/2013

Test Sheet Example

6.3 Flo-Mech Truth Table Test Sheet

Fault	Device	ID	Alarm	Main Oil Pump Shutdown	Burner Shutdown	Combustion Air Fan Running	Pollution Fan Running	Burner Cooling Fan Running	Fuel Oil Pump (Burner) Running	Fresh Air Damper	Fire Damper	Stack Vent Damper	CIP Damper	Vent Valve Position
						M3501	M3502	M3503	M3504	V3501	V3503	V3502	V3504	[illegible]
BURNER ALARMS														
High gas pressure	TBC	HGP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Low gas pressure	TBC	LGP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Valve proving (VFM Only)	TBC	N/A	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Combustion air pressure (VFM Only)	TBC	N/A	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Cooling Fan Pressure Switch	TBC	FSP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Burner Lockout	TBC	BFR	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
OPERATIONAL ALARMS														
Complete Power Loss (Incl Gen.	TBC		NA	Y	Y	[illegible]	N	N	N	Closed	Closed	Open	Open	Open
Combustion Chamber Temp (>870°C)	TBC	CCO	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Pollution Air Pressure	TBC	PDP	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Combustion Air Fan Trip	TBC		Y	N	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Closed
Pollution Fan Trip	TBC		Y	N	Y	Y	N	Y	N	Closed	Closed	Open	Open	Closed
Burner Cooling Fan Trip	TBC		Y	N	Y	Y	Y	N	N	Open	Closed	Open	Open	Closed
Fuel Oil Pump Trip	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Pollution Header Temperature	TBC	PAO	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Stack Temperature	TBC	STO	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Oil Flow (Vent Line)	TBC	VLFS	Y	Y	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Open
Vent Valve 1 Failure	TBC	VV1	Y	Y	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Open
Vent Valve 2 Failure	TBC	VV2	Y	Y	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Open
Low Oil /Water flow	TBC	OLF	Y	N	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Closed
Hi Water Temperature (PLC Setpoint	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Hi Oil Temperature	TBC	OLO	Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Hi Hi Oil Temperature	TBC	OLT	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Main Oil Pump Failure (By Others)	TBC		Y	Y	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Open
Fire Suppression Release	TBC	FSS	Y	Y	Y	[illegible]	Y	Y	N	Open	Closed	Open*	Open	Open
Instrument Air Pressure Low (Failure)	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Complete Loss of Instrument Air	TBC		Y	Y	Y	[illegible]	Y	Y	N	Closed	Closed	Open	Open	Open
Smoke Detection	TBC	SDS	Y	Y	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Open
Strainer Differential Pressure (Hi Hi)	TBC		Y	N	Y	[illegible]	Y	Y	N	Open	Closed	Open	Open	Closed
Strainer Differential Pressure (Hi)	TBC		Y	N	N	[illegible]	Y	[illegible]	N	Closed	Open	Closed	Open	Closed
Zone Stop Combustion Air	TBC	CFZ	Y	N	Y	[illegible]	Y	[illegible]	N	Open	Closed	Open	Open	Closed
Zone Stop Pollution Air	TBC	PFZ	Y	N	Y	[illegible]	N	Y	N	Open	Closed	Open	Open	Closed
Zone Stop Burner Cooling Fan	TBC	FSZ	Y	N	Y	[illegible]	Y	N	N	Open	Closed	Open	Open	Closed
Zone Stop Burner Fuel Oil Pump	TBC	OPZ	Y	N	Y	[illegible]	Y	Y	Y	Open	Closed	Open	Open	Closed
Main Emergency Stop	TBC	MES	Y	Y	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open
Fuel Oil Supply Pressure Switch	TBC		Y	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Closed
Manway Access Safety Relay	TBC	MAS	Y	N	Y	N	N	Y	N	Closed	Closed	Open	Open	Closed
OPTIMISATION ALARMS														
Combustion Flow High/Low Flow Alarm	TBC		Y	N	N	Y	Y	Y	Y	Closed	Open	Closed	Open	Closed
Pollution Flow High/Low Flow Alarm	TBC		Y	N	N	Y	Y	Y	Y	Closed	Open	Closed	Open	Closed
Combustion Chamber Temp (>850 and	TBC		Y	N	N	Y	Y	Y	Y	Closed	Open	Closed	Open	Closed
OPERATIONAL MODES														
Frying Mode			N	N	N	[illegible]	Y	Y	Y	Closed	Open	Closed	Open	Closed
Boil Out/CIP Mode			N	N	N	Y	Y	Y	Y	Open	Closed	Open	Closed	Closed
Pre Heat Mode			N	N	N	Y	Y	Y	Y	Open	Open	Closed	Open	Closed
Shut Down Mode (Circ Pump Stopped)			N	N	Y	Y	Y	Y	N	Open	Closed	Open	Open	Open

Example Test Sheet